T2K 6.5700 $\times 10^{20}$-POT Joint 3-Flavour Oscillation Analysis: detailed study of the effect of systematics

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Abstract

This is a supplementary document for the T2K 6.5700 $\times 10^{20}$-POT Joint 3-Flavour Oscillation Analysis, presented in T2K-TN-175, covering in detailed the check of the implementation of systematic errors and the study of the effect of each systematic parameter in the predicted number of single μ-like ring and single e-like ring events and reconstructed energy spectra.

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13 Effect of BANFF 6, SK numu flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 23
14 Effect of BANFF 6, SK numu flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 23
15 Effect of BANFF 7, SK numu flux, 2.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 25
16 Effect of BANFF 7, SK numu flux, 2.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 25
17 Effect of BANFF 8, SK numu flux, 3.5 - 5.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 27

18 Effect of BANFF 8, SK numu flux, 3.5 - 5.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 27
19 Effect of BANFF 9, SK numu flux, 5.0 - 7.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 29
20 Effect of BANFF 9, SK numu flux, 5.0 - 7.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 29
21 Effect of BANFF 10, SK numu flux, 7.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 31
22 Effect of BANFF 10, SK numu flux, 7.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 31
23 Effect of BANFF 11, SK numubar flux, 0.0 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 33
24 Effect of BANFF 11, SK numubar flux, 0.0 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 33
25 Effect of BANFF 12, SK numubar flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 35
26 Effect of BANFF 12, SK numubar flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 35
27 Effect of BANFF 13, SK numubar flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 37
28 Effect of BANFF 13, SK numubar flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 37
29 Effect of BANFF 14, SK numubar flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 39
30 Effect of BANFF 14, SK numubar flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 39
31 Effect of BANFF 15, SK numubar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 41
32 Effect of BANFF 15, SK numubar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 41
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34 Effect of BANFF 16, SK nue flux, 0.0 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 43
35 Effect of BANFF 17, SK nue flux, 0.5 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 45
36 Effect of BANFF 17, SK nue flux, 0.5 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 45
37 Effect of BANFF 18, SK nue flux, 0.7 - 0.8 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 47
38 Effect of BANFF 18, SK nue flux, 0.7 - 0.8 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 47
39 Effect of BANFF 19, SK nue flux, 0.8 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 49
40 Effect of BANFF 19, SK nue flux, 0.8 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 49
41 Effect of BANFF 20, SK nue flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 51
42 Effect of BANFF 20, SK nue flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 51
43 Effect of BANFF 21, SK nue flux, 2.5 - 4.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 53
44 Effect of BANFF 21, SK nue flux, 2.5 - 4.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 53
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46 Effect of BANFF 22, SK nue flux, 4.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 55
47 Effect of BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 57
48 Effect of BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 57
49 Effect of BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 59
50 Effect of BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 59
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52 Effect of BANFF, Ma QE with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 61
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60 Effect of BANFF; Norm; CC QE; 3.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 69
61 Effect of BANFF; Norm; CC 1pi; 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 71
62 Effect of BANFF; Norm; CC 1pi; 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 71
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94 Effect of SKDet + FSI/SI 0; Ereco range 0.00 - 0.40 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 103
95 Effect of SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 105
96 Effect of SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 105
97 Effect of SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 107
98 Effect of SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 107
99 Effect of SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCnQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 109
100 Effect of SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCnQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 109
101 Effect of SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; $\nu_e/\bar{\nu_e}$/sigν_e CC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 111

102 Effect of SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; $\nu_e/\bar{\nu_e}$/sigν_e CC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 111
103 Effect of SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 113
104 Effect of SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 113
105 Effect of SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 115
106 Effect of SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 115
107 Effect of SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 117
108 Effect of SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 117
109 Effect of SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 119
110 Effect of SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 119
111 Effect of SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 121
112 Effect of SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 121
113 Effect of SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 123
114 Effect of SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 123
115 Effect of SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 125
116 Effect of SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 125
117 Effect of SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 127
118 Effect of SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 127
119 Effect of SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 129
120 Effect of SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 129
121 Effect of SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 131
122 Effect of SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 131
123 Effect of SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 133
124 Effect of SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 133
125 Effect of SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 135
126 Effect of SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 135
127 Effect of SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 137
128 Effect of SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy. 137

This document presents a detailed study of the effect of systematics on the predicted number of single μ-like ring and single e-like ring events in SuperK. This information will be presented in an ensemble of tables and plots of the reconstructed energy spectra.

Each systematic error of the 64 systematic errors considered for the 2013 VALOR joint oscillation analysis (see TN175) will be teaked independently (ignoring correlations when they could exist). For each systematic error two tables will be presented in the following pages, showing the post-BANFF tuning nominal N_{SK} for $\pm 1\sigma$ deviation of the systematic parameter treated, grouped into flavour categories (first table) or for each MC mode considered in this analysis (second table). For correlated systematics, σ is defined as the square root of the corresponding diagonal entry in the corresponding covariance matrix.

Besides, for each systematic a figure will be presented containing the nominal predicted reconstructed energy spectrum for the single μ-like ring and the single e-like ring event and the predicted reconstructed energy spectra when the systematic considered is tweaked by -3σ, -1σ, $+1\sigma$ and $+3\sigma$. In the same figure the ratios of any of these tweaks with respect to the nominal spectrum are also included.

The effect of systematics in this document has been evaluated with 3-flavour oscillations in matter, using the following values of the oscillation parameters and the normal hierarchy assumption:

- Δm^2 = 2.437$\times 10^{-3} eV^2/c^4$
- δm^2 = 7.5$\times 10^{-5} eV^2/c^4$
- $sin^2\theta_{12}$ = 0.306
- $sin^2\theta_{13}$ = 0.0251
- $sin^2\theta_{23}$ = 0.513
- δ_{CP} = 0

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.6**	**0.6629**	**123.9**	**-0.6629**	**22**	**22.18**	**0.8163**	**21.82**	**-0.8163**
ν_μ	116.2	117.1	0.7112	115.4	-0.7112	0.9435	0.9441	0.06322	0.9429	-0.06322
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2746	0.1528	0.2738	-0.1528	17.49	17.67	1.023	17.31	-1.023

Table 1: Effect of BANFF 0, SK numu flux, 0.0 - 0.4 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.6**	**0.6629**	**123.9**	**-0.6629**	**22**	**22.18**	**0.8163**	**21.82**	**-0.8163**
ν_μ CCQE	70.79	71.62	1.167	69.97	-1.167	0.04778	0.04824	0.9687	0.04731	-0.9687
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9075	0.007239	0.9073	-0.007239	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0.008242	3.229	-0.008242	0.3314	0.3315	0.04032	0.3312	-0.04032
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1952	0.2115	0.1943	-0.2115	14.79	14.96	1.174	14.62	-1.174
Osc. ν_e CC 1π	0.07266	0.07267	0.009589	0.07266	-0.009589	2.624	2.629	0.1949	2.619	-0.1949
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04409	0.5876	0.04357	-0.5876
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 2: Effect of BANFF 0, SK numu flux, 0.0 - 0.4 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

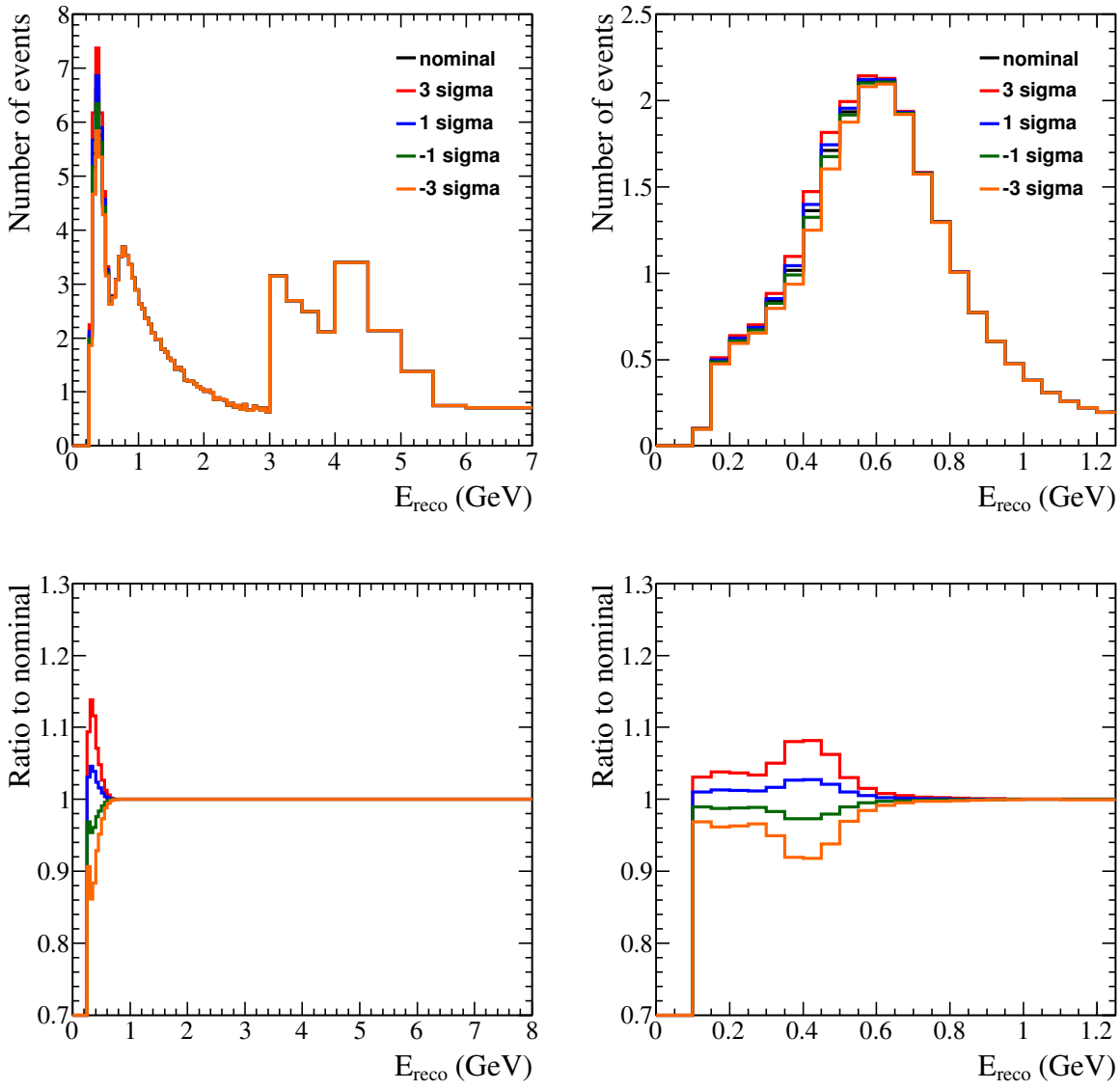


Figure 1: Effect of BANFF 0, SK numu flux, 0.0 - 0.4 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.4**	**0.5011**	**124.1**	**-0.5011**	**22**	**22**	**0.01093**	**22**	**-0.01093**
ν_μ	116.2	116.9	0.5356	115.6	-0.5356	0.9435	0.9459	0.2549	0.9411	-0.2549
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2769	0.9724	0.2716	-0.9724	17.49	17.49	0	17.49	0

Table 3: Effect of BANFF 1, SK numu flux, 0.4 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.4**	**0.5011**	**124.1**	**-0.5011**	**22**	**22**	**0.01093**	**22**	**-0.01093**
ν_μ CCQE	70.79	71.41	0.8712	70.18	-0.8712	0.04778	0.04887	2.287	0.04668	-2.287
ν_μ CC 1π	29.78	29.78	0.002494	29.77	-0.002494	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9083	0.09619	0.9065	-0.09619	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.939	0.1257	2.931	-0.1257		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3927	0.07219	0.3921	-0.07219
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1543	0.2716	0.1535	-0.2716
ν_μ/ν_τ NC other	3.229	3.229	0.01671	3.228	-0.01671	0.3314	0.332	0.1843	0.3308	-0.1843
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1971	1.221	0.1924	-1.221	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.0729	0.3229	0.07243	-0.3229	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005273	1.039	0.005165	-1.039	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 4: Effect of BANFF 1, SK numu flux, 0.4 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

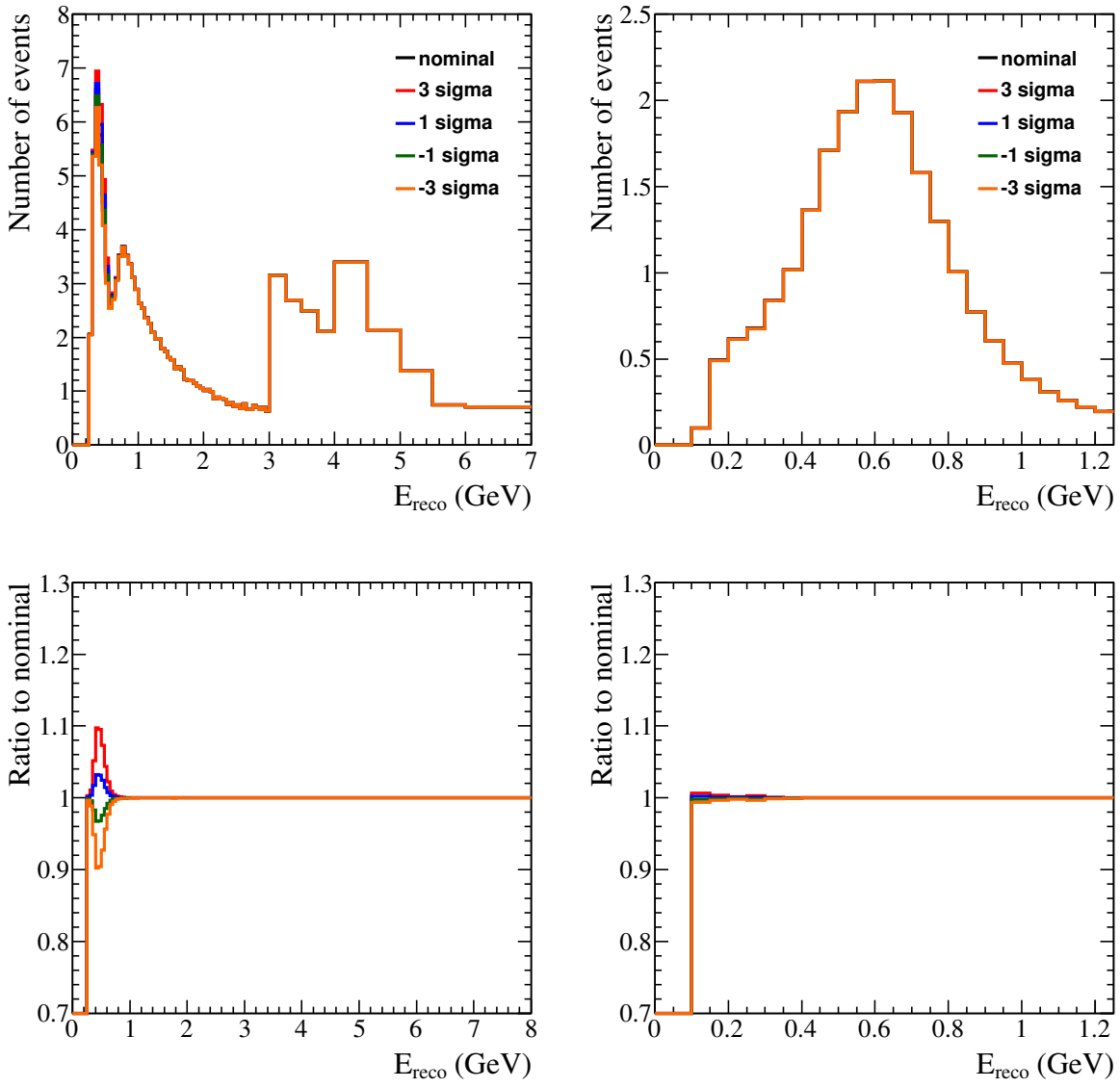


Figure 2: Effect of BANFF 1, SK numu flux, 0.4 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.9**	**0.08216**	**124.7**	**-0.08216**	**22**	**22.77**	**3.505**	**21.23**	**-3.505**
ν_μ	116.2	116.3	0.0835	116.1	-0.0835	0.9435	0.95	0.6879	0.937	-0.6879
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2797	1.988	0.2688	-1.988	17.49	18.25	4.371	16.73	-4.371

Table 5: Effect of BANFF 2, SK numu flux, 0.5 - 0.6 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.9**	**0.08216**	**124.7**	**-0.08216**	**22**	**22.77**	**3.505**	**21.23**	**-3.505**
ν_μ CCQE	70.79	70.87	0.1103	70.72	-0.1103	0.04778	0.04793	0.3199	0.04762	-0.3199
ν_μ CC 1π	29.78	29.78	0.002629	29.77	-0.002629	0.01709	0.01709	0.002437	0.01709	-0.002437
ν_μ CC coherent	0.9074	0.9076	0.02376	0.9072	-0.02376	4.366e-06	4.718e-06	8.048	4.015e-06	-8.048
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.951	0.549	2.919	-0.549		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3955	0.7881	0.3893	-0.7881
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1556	1.1	0.1522	-1.1
ν_μ/ν_τ NC other	3.229	3.231	0.05714	3.227	-0.05714	0.3314	0.3329	0.4682	0.3298	-0.4682
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1993	2.336	0.1902	-2.336	14.79	15.46	4.518	14.12	-4.518
Osc. ν_e CC 1π	0.07266	0.07346	1.097	0.07186	-1.097	2.624	2.719	3.599	2.53	-3.599
Osc. ν_e CC coherent	0.005219	0.005322	1.979	0.005116	-1.979	0.04383	0.04563	4.101	0.04203	-4.101
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03181	0.1889	0.03169	-0.1889

Table 6: Effect of BANFF 2, SK numu flux, 0.5 - 0.6 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

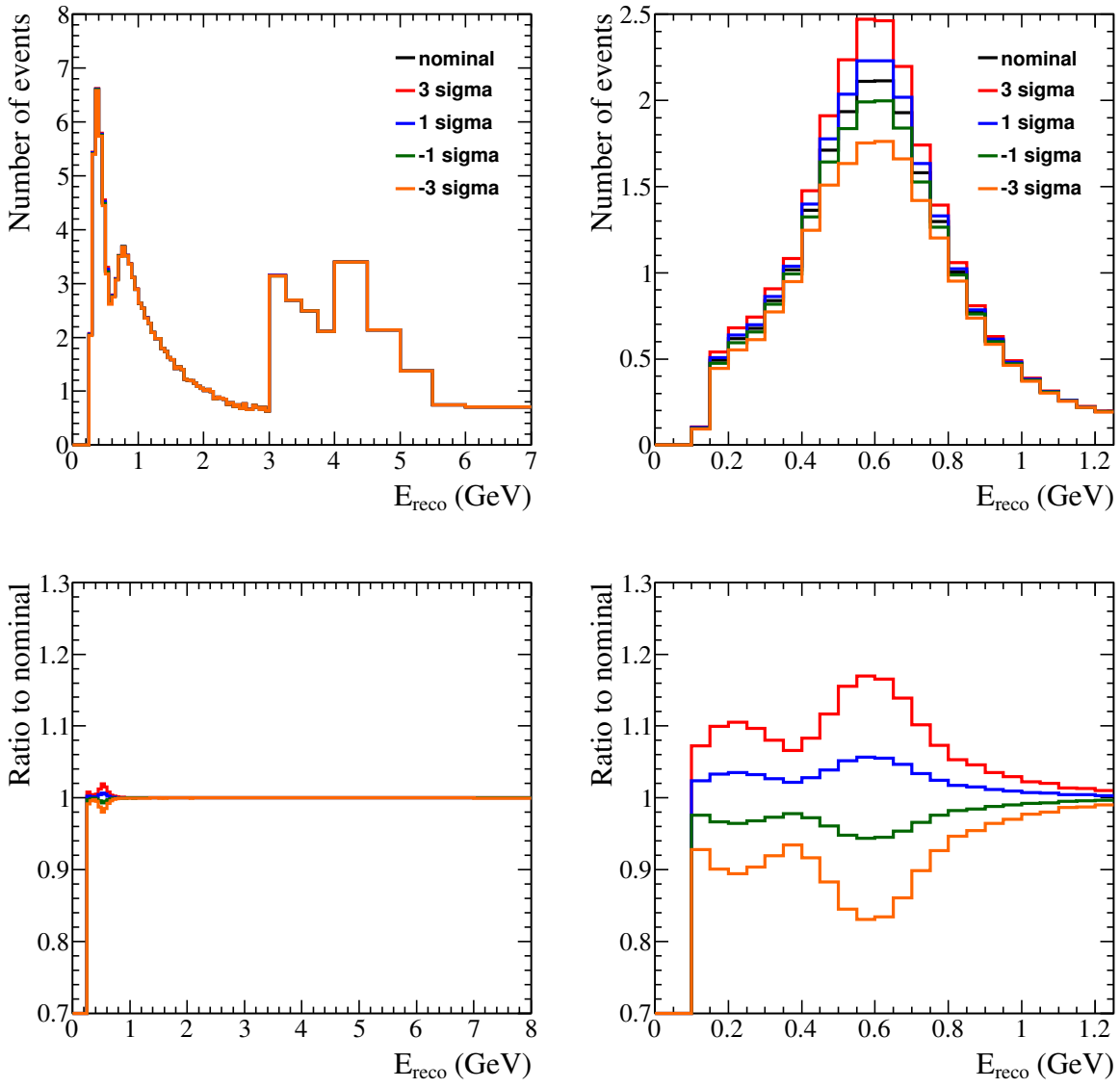


Figure 3: Effect of BANFF 2, SK numu flux, 0.5 - 0.6 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125**	**0.2116**	**124.5**	**-0.2116**	**22**	**22.01**	**0.03434**	**21.99**	**-0.03434**
ν_μ	116.2	116.5	0.222	116	-0.222	0.9435	0.951	0.8009	0.9359	-0.8009
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2802	2.176	0.2683	-2.176	17.49	17.49	0	17.49	0

Table 7: Effect of BANFF 3, SK numu flux, 0.6 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125**	**0.2116**	**124.5**	**-0.2116**	**22**	**22.01**	**0.03434**	**21.99**	**-0.03434**
ν_μ CCQE	70.79	71	0.2861	70.59	-0.2861	0.04778	0.04801	0.494	0.04754	-0.494
ν_μ CC 1π	29.78	29.8	0.07085	29.75	-0.07085	0.01709	0.01713	0.2858	0.01704	-0.2858
ν_μ CC coherent	0.9074	0.9081	0.07922	0.9067	-0.07922	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	5.517e-05	8.597	-5.517e-05	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.963	0.9604	2.907	-0.9604		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3963	0.9828	0.3885	-0.9828
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1548	0.5636	0.153	-0.5636
ν_μ/ν_τ NC other	3.229	3.234	0.169	3.223	-0.169	0.3314	0.3339	0.7687	0.3288	-0.7687
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1989	2.106	0.1907	-2.106	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07438	2.371	0.07094	-2.371	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005362	2.744	0.005076	-2.744	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 8: Effect of BANFF 3, SK numu flux, 0.6 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

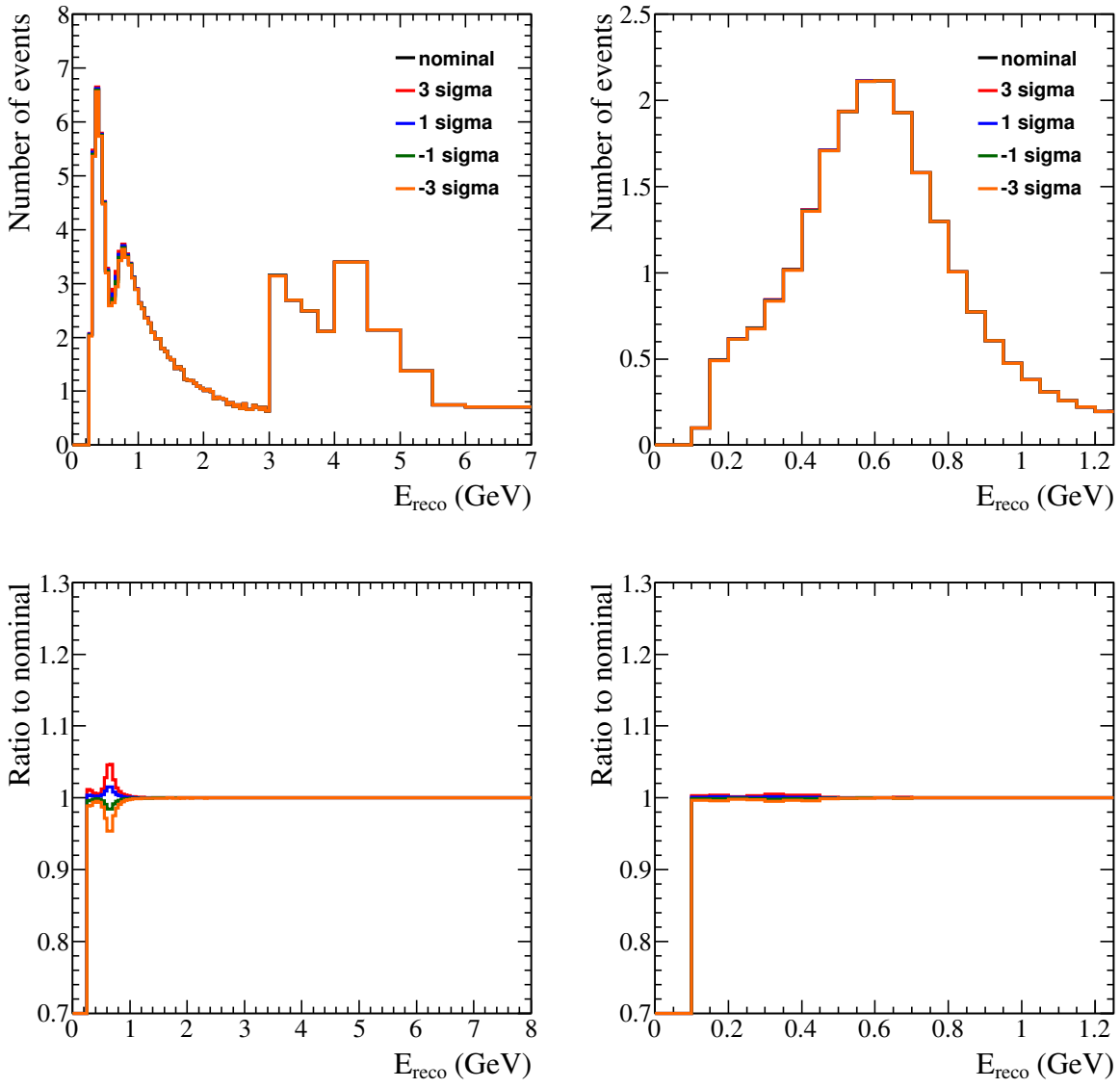


Figure 4: Effect of BANFF 3, SK numu flux, 0.6 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.7**	**1.573**	**122.8**	**-1.573**	**22**	**22.55**	**2.473**	**21.46**	**-2.473**
ν_μ	116.2	118.2	1.682	114.3	-1.682	0.9435	0.9636	2.13	0.9234	-2.13
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2818	2.773	0.2666	-2.773	17.49	18.01	2.996	16.97	-2.996

Table 9: Effect of BANFF 4, SK numu flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.7**	**1.573**	**122.8**	**-1.573**	**22**	**22.55**	**2.473**	**21.46**	**-2.473**
ν_μ CCQE	70.79	72.3	2.134	69.28	-2.134	0.04778	0.04908	2.719	0.04648	-2.719
ν_μ CC 1π	29.78	30.12	1.163	29.43	-1.163	0.01709	0.01755	2.695	0.01663	-2.695
ν_μ CC coherent	0.9074	0.9169	1.047	0.8979	-1.047	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.599	0.02304	8.595	-0.02304	0.0009536	0.0009656	1.256	0.0009416	-1.256
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	3.003	2.312	2.867	-2.312		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.4023	2.53	0.3825	-2.53
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1575	2.331	0.1503	-2.331
ν_μ/ν_τ NC other	3.229	3.248	0.5915	3.21	-0.5915	0.3314	0.3362	1.453	0.3266	-1.453
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1993	2.351	0.1902	-2.351	14.79	15.19	2.693	14.39	-2.693
Osc. ν_e CC 1π	0.07266	0.07554	3.959	0.06979	-3.959	2.624	2.746	4.652	2.502	-4.652
Osc. ν_e CC coherent	0.005219	0.00533	2.113	0.005109	-2.113	0.04383	0.04549	3.78	0.04217	-3.78
Osc. ν_e CC other	0.001592	0.001629	2.315	0.001555	-2.315	0.03175	0.03381	6.476	0.0297	-6.476

Table 10: Effect of BANFF 4, SK numu flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

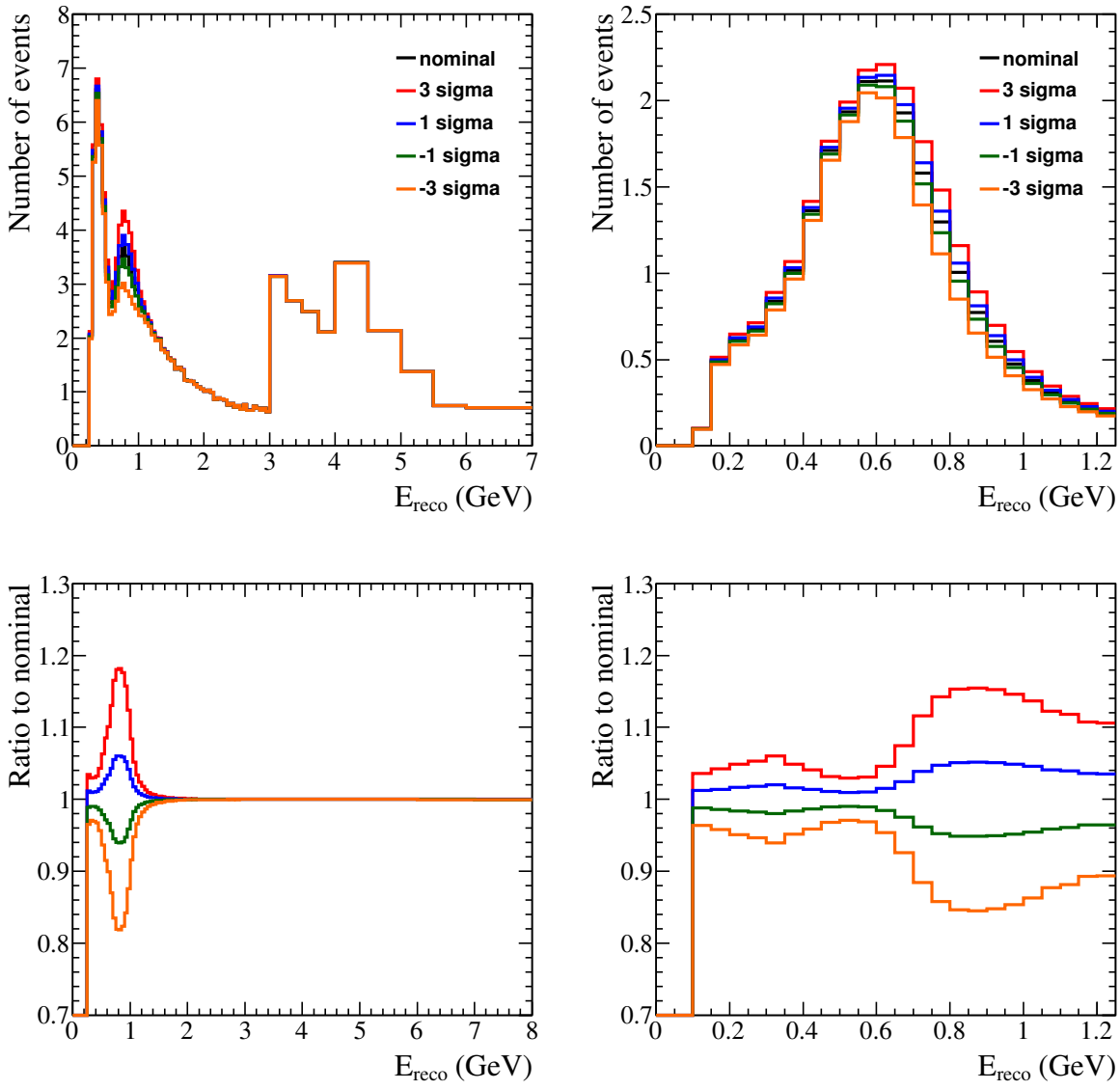


Figure 5: Effect of BANFF 4, SK numu flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.3**	**1.251**	**123.2**	**-1.251**	**22**	**22.01**	**0.05338**	**21.99**	**-0.05338**
ν_μ	116.2	117.8	1.342	114.7	-1.342	0.9435	0.9552	1.245	0.9317	-1.245
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2749	0.2392	0.2736	-0.2392	17.49	17.49	0	17.49	0

Table 11: Effect of BANFF 5, SK numu flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.3**	**1.251**	**123.2**	**-1.251**	**22**	**22.01**	**0.05338**	**21.99**	**-0.05338**
ν_μ CCQE	70.79	71.8	1.418	69.79	-1.418	0.04778	0.04839	1.279	0.04717	-1.279
ν_μ CC 1π	29.78	30.23	1.53	29.32	-1.53	0.01709	0.01771	3.676	0.01646	-3.676
ν_μ CC coherent	0.9074	0.9185	1.227	0.8963	-1.227	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.624	0.3076	8.571	-0.3076	0.0009536	0.001017	6.696	0.0008897	-6.696
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.966	1.074	2.903	-1.074		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3973	1.245	0.3875	-1.245
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1557	1.142	0.1521	-1.142
ν_μ/ν_τ NC other	3.229	3.261	0.9971	3.197	-0.9971	0.3314	0.3352	1.146	0.3276	-1.146
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1949	0.09509	0.1946	-0.09509	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07307	0.565	0.07225	-0.565	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005238	0.3533	0.005201	-0.3533	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001634	2.628	0.001551	-2.628	0.03175	0.03175	0	0.03175	0

Table 12: Effect of BANFF 5, SK numu flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

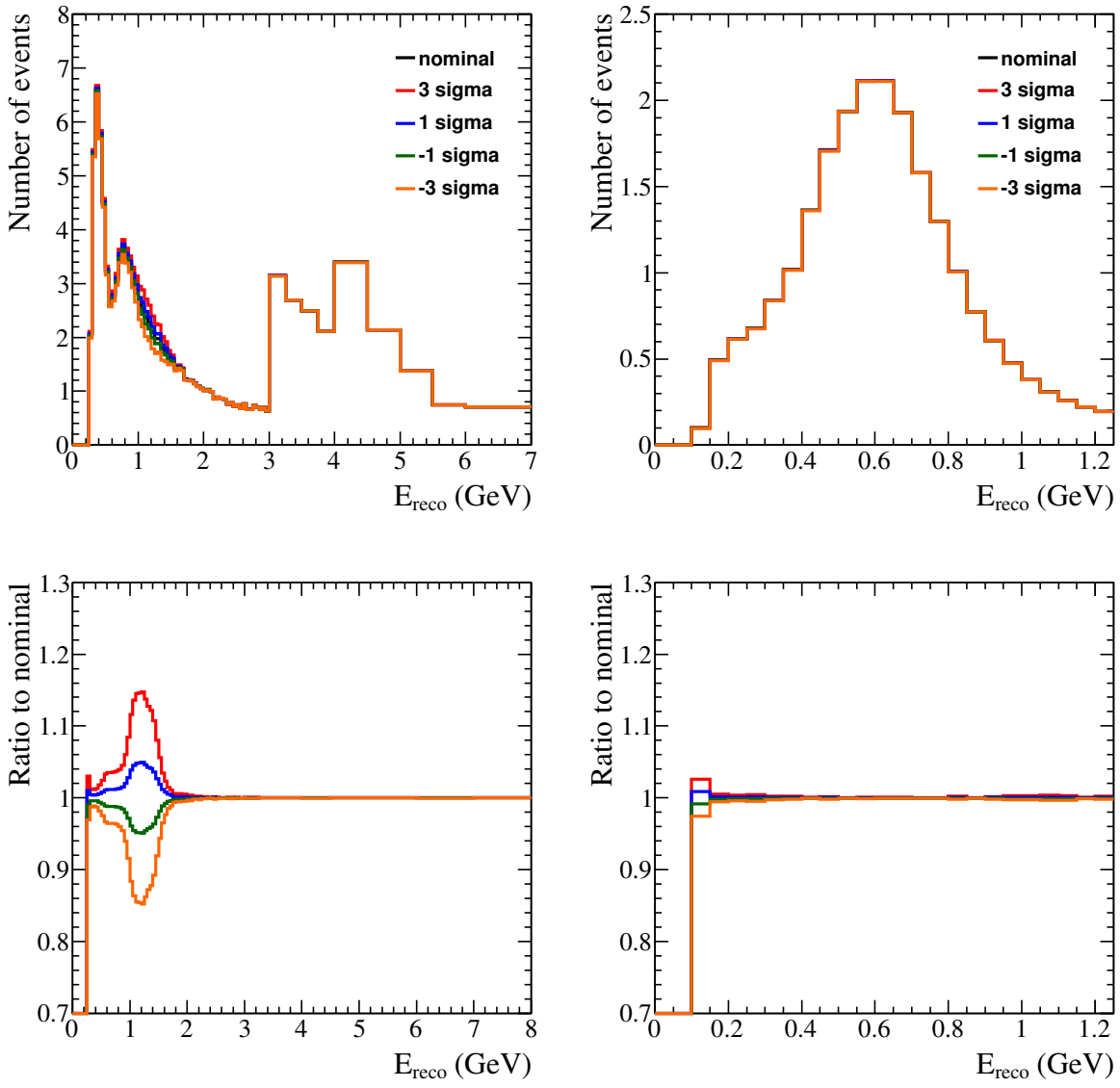


Figure 6: Effect of BANFF 5, SK numu flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.2**	**1.106**	**123.4**	**-1.106**	**22**	**22.01**	**0.06058**	**21.99**	**-0.06058**
ν_μ	116.2	117.6	1.187	114.9	-1.187	0.9435	0.9535	1.057	0.9335	-1.057
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2744	0.05728	0.2741	-0.05728	17.49	17.49	0.01919	17.49	-0.01919

Table 13: Effect of BANFF 6, SK numu flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.2**	**1.106**	**123.4**	**-1.106**	**22**	**22.01**	**0.06058**	**21.99**	**-0.06058**
ν_μ CCQE	70.79	71.49	0.9807	70.1	-0.9807	0.04778	0.04794	0.3348	0.04762	-0.3348
ν_μ CC 1π	29.78	30.29	1.722	29.26	-1.722	0.01709	0.0173	1.265	0.01687	-1.265
ν_μ CC coherent	0.9074	0.9223	1.642	0.8925	-1.642	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.683	0.9934	8.512	-0.9934	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.963	0.9446	2.907	-0.9446		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3963	0.987	0.3885	-0.987
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1554	1	0.1524	-1
ν_μ/ν_τ NC other	3.229	3.274	1.405	3.184	-1.405	0.3314	0.3356	1.262	0.3272	-1.262
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0.03252	0.1947	-0.03252	14.79	14.79	0.0003533	14.79	-0.0003533
Osc. ν_e CC 1π	0.07266	0.07273	0.09042	0.0726	-0.09042	2.624	2.627	0.1059	2.622	-0.1059
Osc. ν_e CC coherent	0.005219	0.005222	0.05778	0.005216	-0.05778	0.04383	0.04384	0.01905	0.04382	-0.01905
Osc. ν_e CC other	0.001592	0.001617	1.572	0.001567	-1.572	0.03175	0.03227	1.631	0.03124	-1.631

Table 14: Effect of BANFF 6, SK numu flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

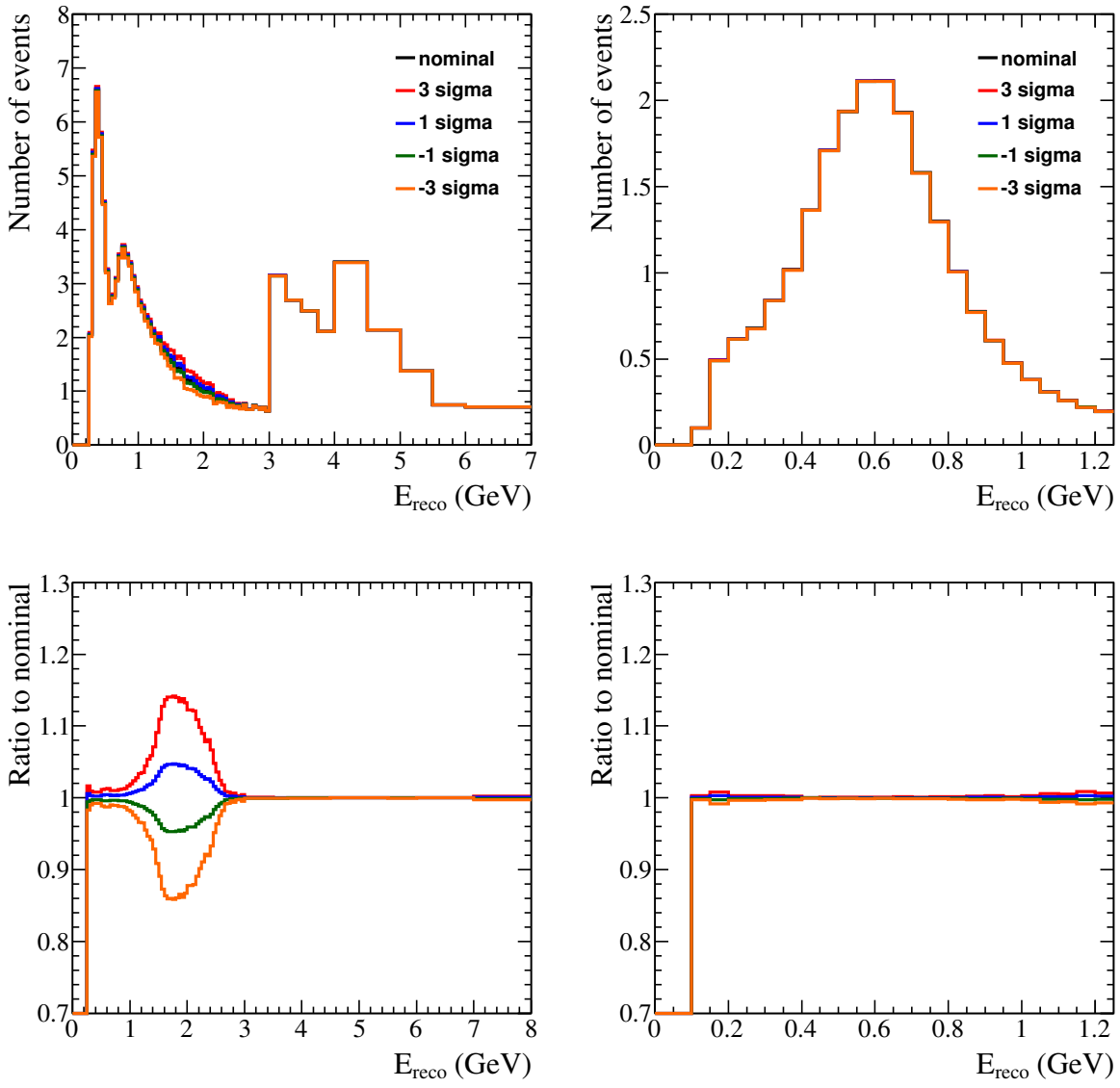


Figure 7: Effect of BANFF 6, SK numu flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.6**	**0.6483**	**124**	**-0.6483**	**22**	**22.01**	**0.02069**	**22**	**-0.02069**
ν_μ	116.2	117	0.6959	115.4	-0.6959	0.9435	0.9479	0.47	0.9391	-0.47
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2743	0.008321	0.2742	-0.008321	17.49	17.49	0.0006657	17.49	-0.0006657

Table 15: Effect of BANFF 7, SK numu flux, 2.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.6**	**0.6483**	**124**	**-0.6483**	**22**	**22.01**	**0.02069**	**22**	**-0.02069**
ν_μ CCQE	70.79	71.12	0.4568	70.47	-0.4568	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	30.09	1.058	29.46	-1.058	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9168	1.04	0.898	-1.04	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.708	1.291	8.486	-1.291	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.95	0.5196	2.92	-0.5196		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3936	0.3046	0.3912	-0.3046
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1548	0.5666	0.153	-0.5666
ν_μ/ν_τ NC other	3.229	3.264	1.085	3.194	-1.085	0.3314	0.3337	0.7145	0.329	-0.7145
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0.00128	0.1948	-0.00128	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07267	0.0155	0.07265	-0.0155	2.624	2.624	0.003466	2.624	-0.003466
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001601	0.5692	0.001583	-0.5692	0.03175	0.03178	0.0802	0.03173	-0.0802

Table 16: Effect of BANFF 7, SK numu flux, 2.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

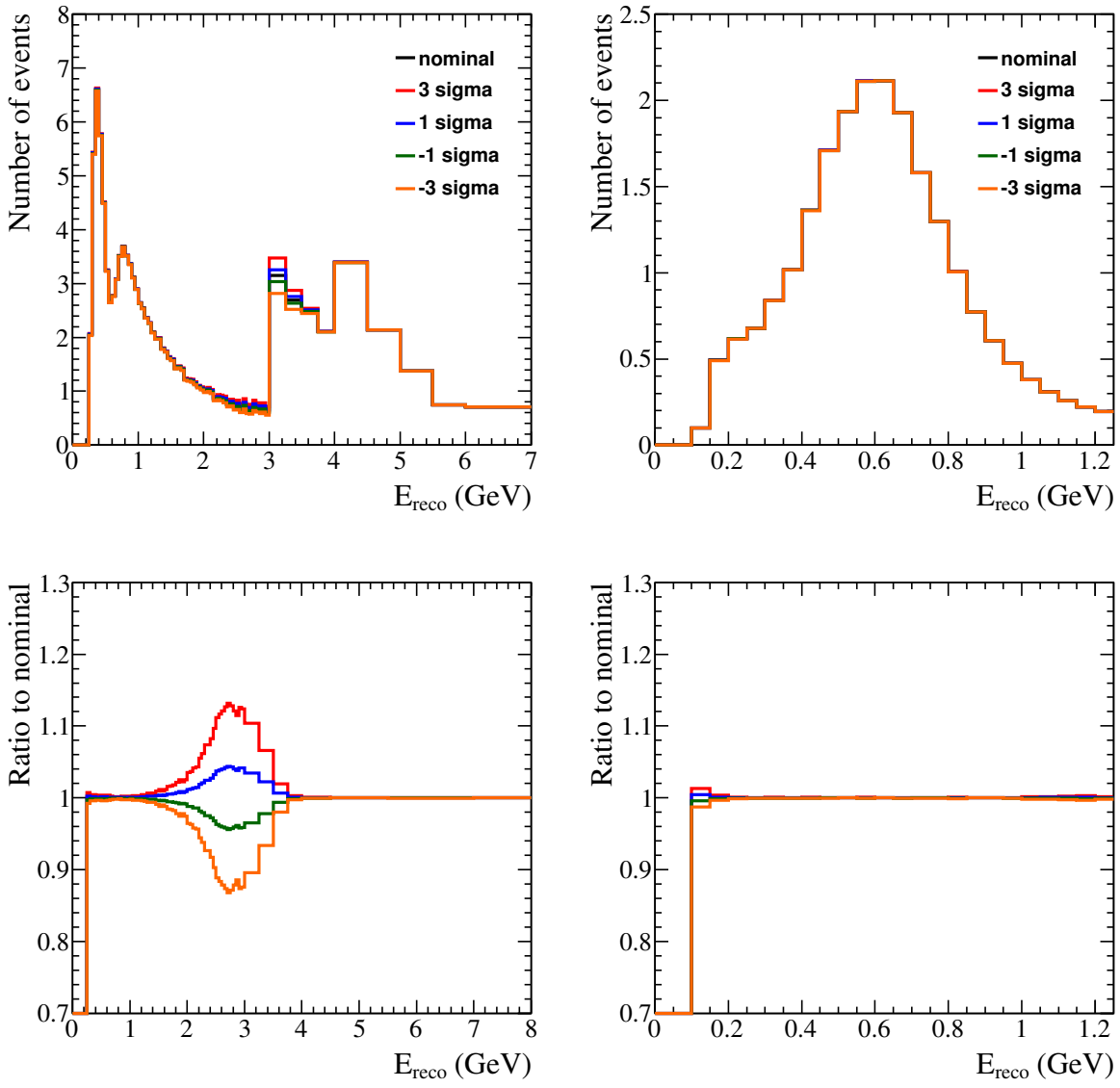


Figure 8: Effect of BANFF 7, SK numu flux, 2.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.7**	**0.7374**	**123.9**	**-0.7374**	**22**	**22.01**	**0.0255**	**22**	**-0.0255**
ν_μ	116.2	117.2	0.7915	115.3	-0.7915	0.9435	0.9491	0.5945	0.9379	-0.5945
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0.00312	0.2742	-0.00312	17.49	17.49	8.176e-06	17.49	-8.176e-06

Table 17: Effect of BANFF 8, SK numu flux, 3.5 - 5.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.7**	**0.7374**	**123.9**	**-0.7374**	**22**	**22.01**	**0.0255**	**22**	**-0.0255**
ν_μ CCQE	70.79	71.06	0.3777	70.53	-0.3777	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	30.13	1.202	29.42	-1.202	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.921	1.495	0.8938	-1.495	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.806	2.43	8.388	-2.43	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.956	0.7072	2.914	-0.7072		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3946	0.5509	0.3902	-0.5509
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1548	0.6072	0.153	-0.6072
ν_μ/ν_τ NC other	3.229	3.28	1.592	3.178	-1.592	0.3314	0.3339	0.7582	0.3289	-0.7582
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0.0006184	0.1948	-0.0006184	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0.002183	0.07266	-0.002183	2.624	2.624	2.852e-05	2.624	-2.852e-05
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001598	0.3621	0.001587	-0.3621	0.03175	0.03176	0.002147	0.03175	-0.002147

Table 18: Effect of BANFF 8, SK numu flux, 3.5 - 5.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

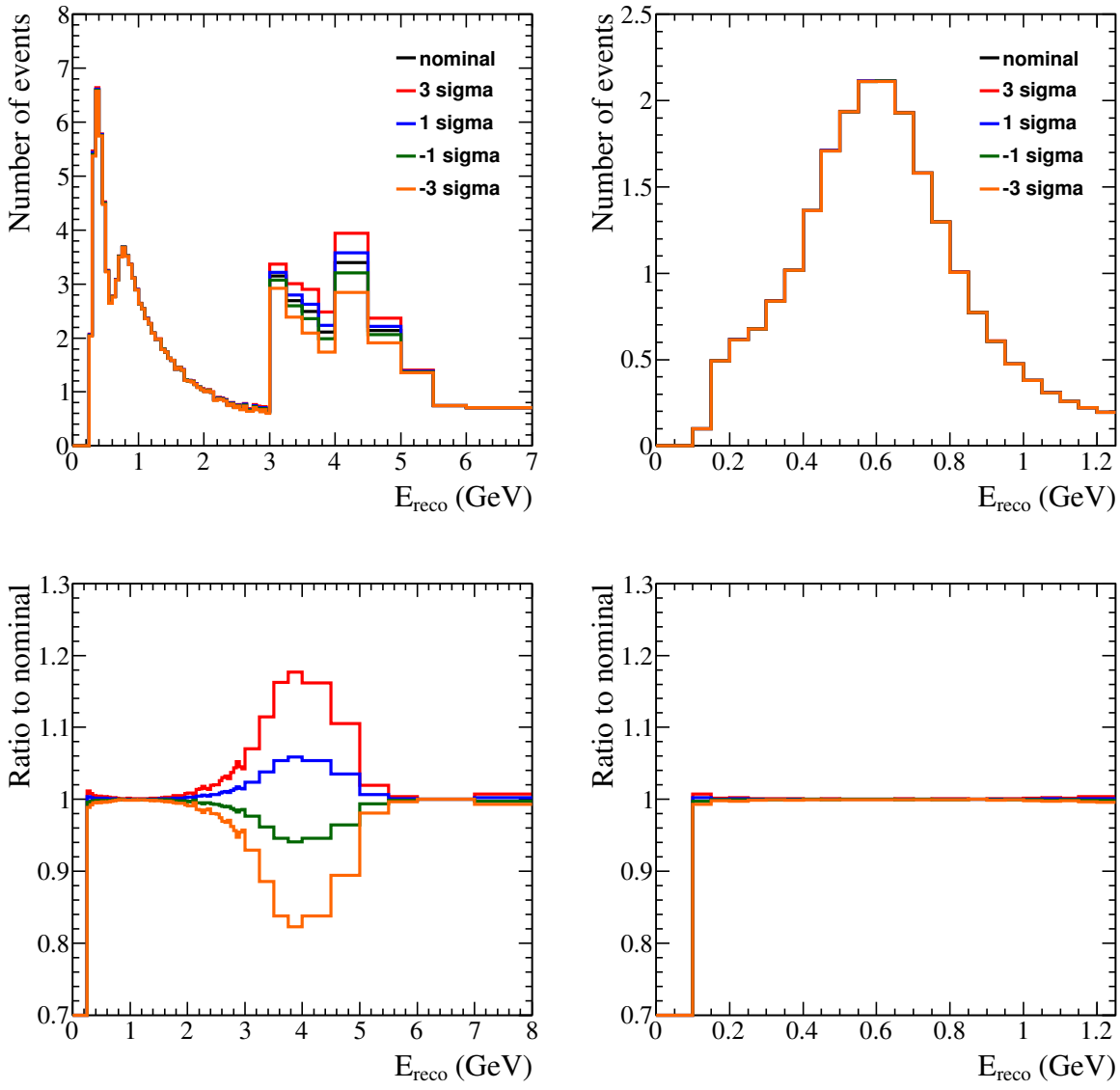


Figure 9: Effect of BANFF 8, SK numu flux, 3.5 - 5.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2692**	**124.4**	**-0.2692**	**22**	**22**	**0.01202**	**22**	**-0.01202**
ν_μ	116.2	116.6	0.289	115.9	-0.289	0.9435	0.9461	0.2803	0.9408	-0.2803
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0.0009584	0.2742	-0.0009584	17.49	17.49	0	17.49	0

Table 19: Effect of BANFF 9, SK numu flux, 5.0 - 7.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2692**	**124.4**	**-0.2692**	**22**	**22**	**0.01202**	**22**	**-0.01202**
ν_μ CCQE	70.79	70.85	0.0846	70.73	-0.0846	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.89	0.3899	29.66	-0.3899	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9115	0.4517	0.9033	-0.4517	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.72	1.431	8.474	-1.431	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.944	0.3273	2.925	-0.3273		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3936	0.3081	0.3912	-0.3081
ν_μ/ν_τ NC coherent		-		–		0.1539	0.154	0.08952	0.1538	-0.08952
ν_μ/ν_τ NC other	3.229	3.252	0.7185	3.206	-0.7185	0.3314	0.3327	0.3916	0.3301	-0.3916
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0.0001741	0.1948	-0.0001741	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0.0004046	0.07266	-0.0004046	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001594	0.1253	0.00159	-0.1253	0.03175	0.03175	0	0.03175	0

Table 20: Effect of BANFF 9, SK numu flux, 5.0 - 7.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

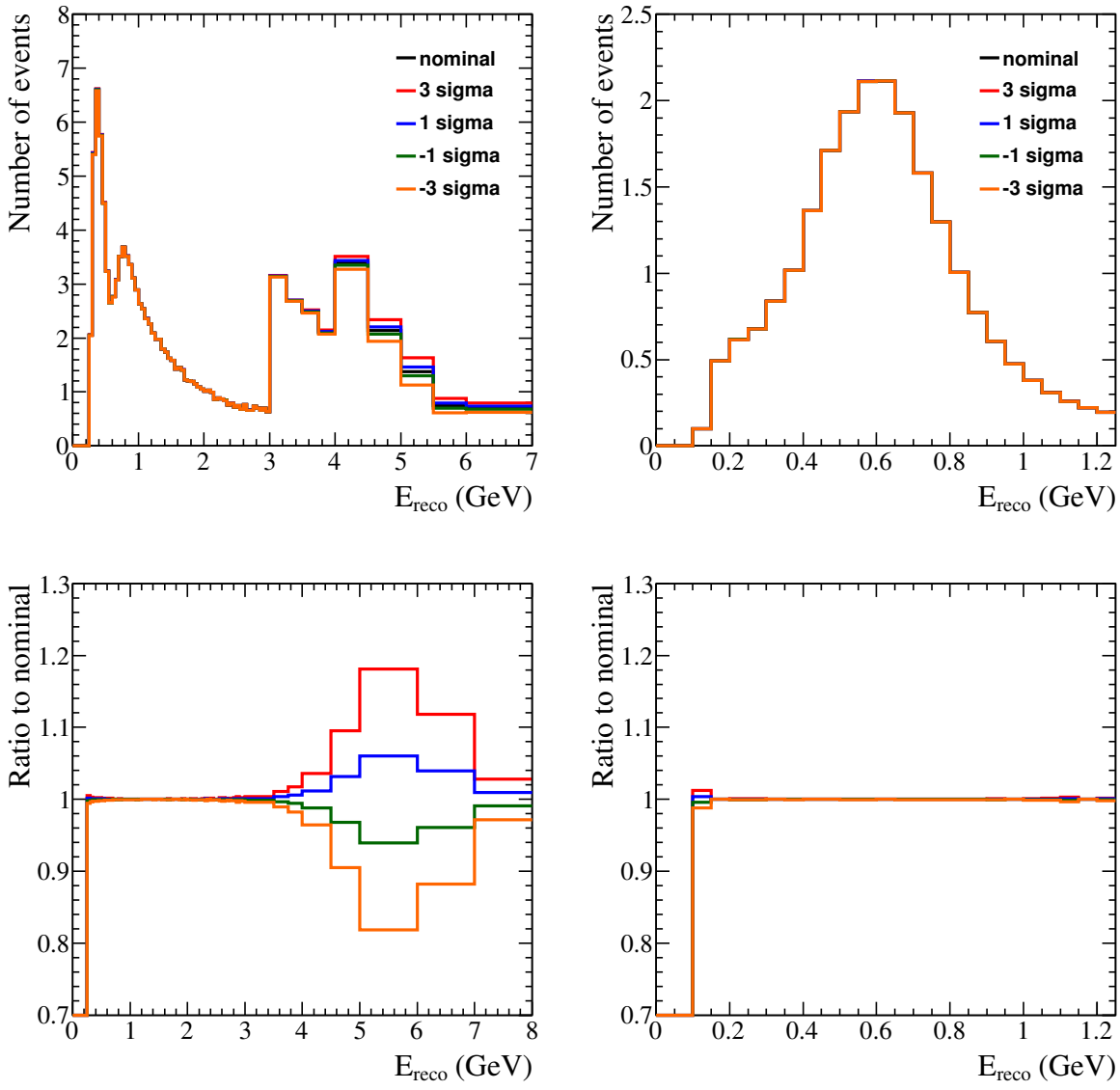


Figure 10: Effect of BANFF 9, SK numu flux, 5.0 - 7.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.9**	**0.066**	**124.7**	**-0.066**	**22**	**22**	**0.007512**	**22**	**-0.007512**
ν_μ	116.2	116.3	0.07085	116.2	-0.07085	0.9435	0.9451	0.1752	0.9418	-0.1752
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0.000245	0.2742	-0.000245	17.49	17.49	0	17.49	0

Table 21: Effect of BANFF 10, SK numu flux, 7.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.9**	**0.066**	**124.7**	**-0.066**	**22**	**22**	**0.007512**	**22**	**-0.007512**
ν_μ CCQE	70.79	70.8	0.004239	70.79	-0.004239	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.79	0.04541	29.76	-0.04541	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9079	0.05487	0.9069	-0.05487	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.639	0.4834	8.556	-0.4834	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.942	0.2477	2.928	-0.2477		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3927	0.08394	0.3921	-0.08394
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1541	0.1075	0.1537	-0.1075
ν_μ/ν_τ NC other	3.229	3.245	0.5111	3.212	-0.5111	0.3314	0.3325	0.3494	0.3302	-0.3494
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001593	0.0422	0.001592	-0.0422	0.03175	0.03175	0	0.03175	0

Table 22: Effect of BANFF 10, SK numu flux, 7.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

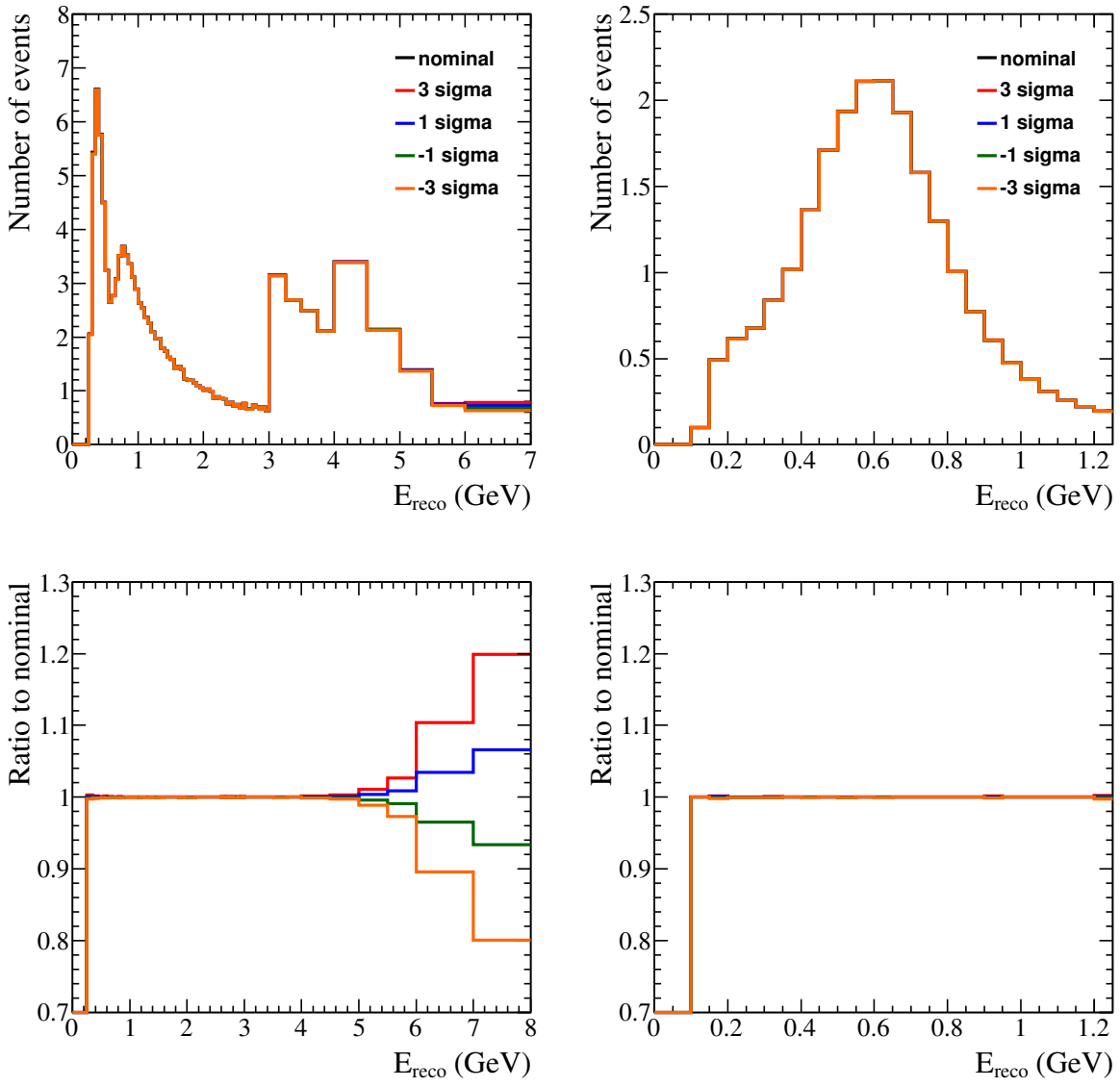


Figure 11: Effect of BANFF 10, SK numu flux, 7.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.0298**	**124.7**	**-0.0298**	**22**	**22.02**	**0.1074**	**21.98**	**-0.1074**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.841	0.4726	7.767	-0.4726	0.05456	0.05497	0.7455	0.05415	-0.7455
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003474	9.389	0.002878	-9.389	0.2339	0.2571	9.931	0.2106	-9.931
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 23: Effect of BANFF 11, SK numubar flux, 0.0 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.0298**	**124.7**	**-0.0298**	**22**	**22.02**	**0.1074**	**21.98**	**-0.1074**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.655	0.7769	4.583	-0.7769	0.0008752	0.0008975	2.546	0.0008529	-2.546
$\bar{\nu_\mu}$ CC 1π	2.132	2.133	0.01003	2.132	-0.01003	0.0002486	0.0002487	0.07758	0.0002484	-0.07758
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2484	0.1149	0.2478	-0.1149	4.264e-06	4.688e-06	9.936	3.841e-06	-9.936
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1438	0.2789	0.143	-0.2789		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.02002	0.6151	0.01978	-0.6151
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01607	1.094	0.01572	-1.094
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1752	0.05537	0.175	-0.05537	0.01752	0.0176	0.5002	0.01743	-0.5002
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002373	9.296	0.001969	-9.296	0.1808	0.1988	9.934	0.1629	-9.934
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0009326	9.683	0.0007679	-9.683	0.04064	0.04467	9.921	0.03661	-9.921
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001248	9.749	0.0001026	-9.749	0.01087	0.01195	9.936	0.009794	-9.936
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.373e-05	7.219	3.784e-05	-7.219	0.001548	0.0017	9.793	0.001396	-9.793
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 24: Effect of BANFF 11, SK numubar flux, 0.0 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

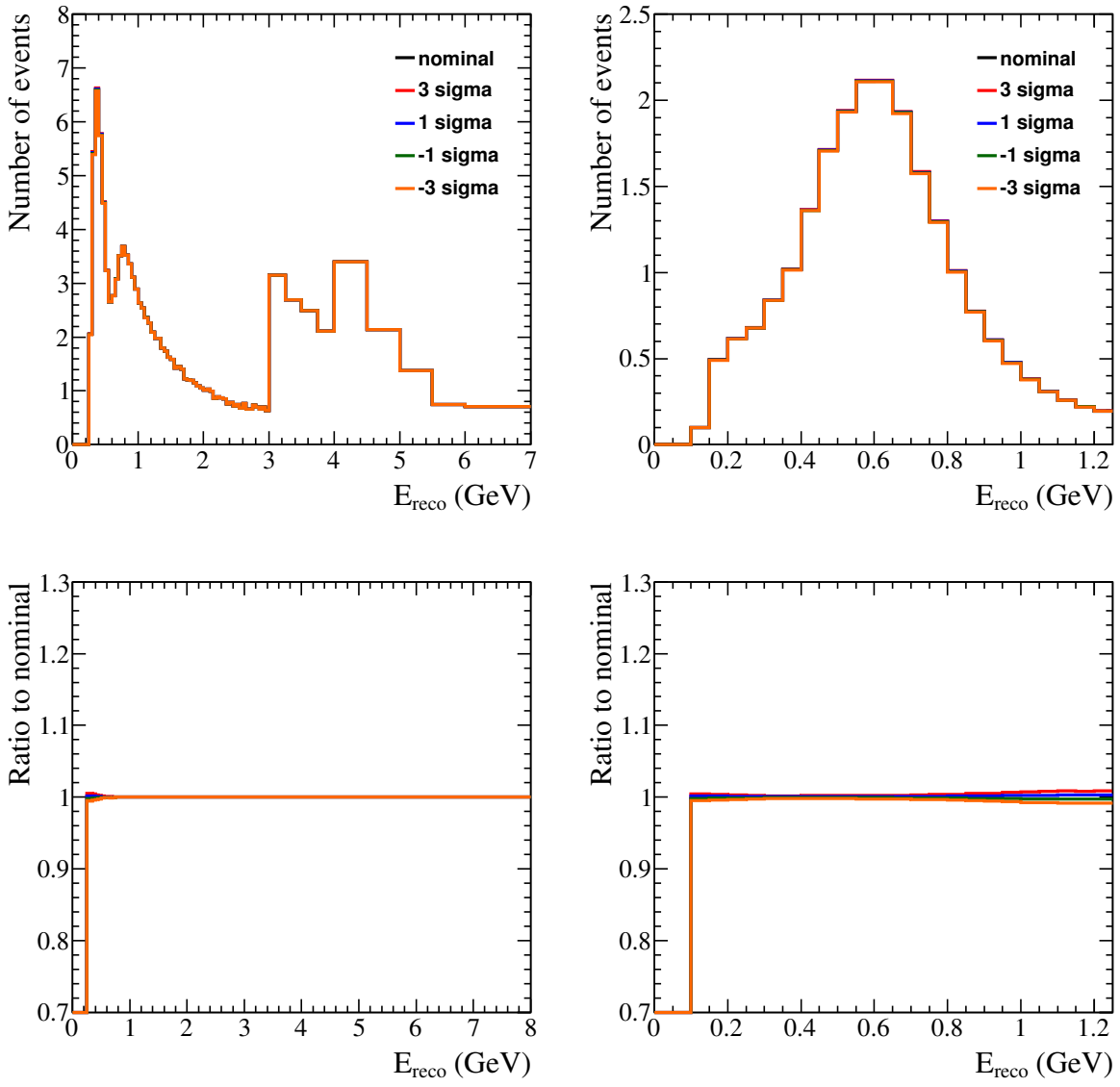


Figure 12: Effect of BANFF 11, SK numubar flux, 0.0 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.04025**	**124.7**	**-0.04025**	**22**	**22**	**0.002243**	**22**	**-0.002243**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.854	0.6436	7.754	-0.6436	0.05456	0.05505	0.9045	0.05407	-0.9045
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 25: Effect of BANFF 12, SK numubar flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.04025**	**124.7**	**-0.04025**	**22**	**22**	**0.002243**	**22**	**-0.002243**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.659	0.8563	4.58	-0.8563	0.0008752	0.0008976	2.555	0.0008529	-2.555
$\bar{\nu_\mu}$ CC 1π	2.132	2.14	0.3372	2.125	-0.3372	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2499	0.7298	0.2463	-0.7298	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4856	0.01691	0.4854	-0.01691	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1447	0.8708	0.1422	-0.8708		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.02007	0.8538	0.01973	-0.8538
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01613	1.474	0.01566	-1.474
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1754	0.1933	0.1748	-0.1933	0.01752	0.01758	0.3827	0.01745	-0.3827
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 26: Effect of BANFF 12, SK numubar flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

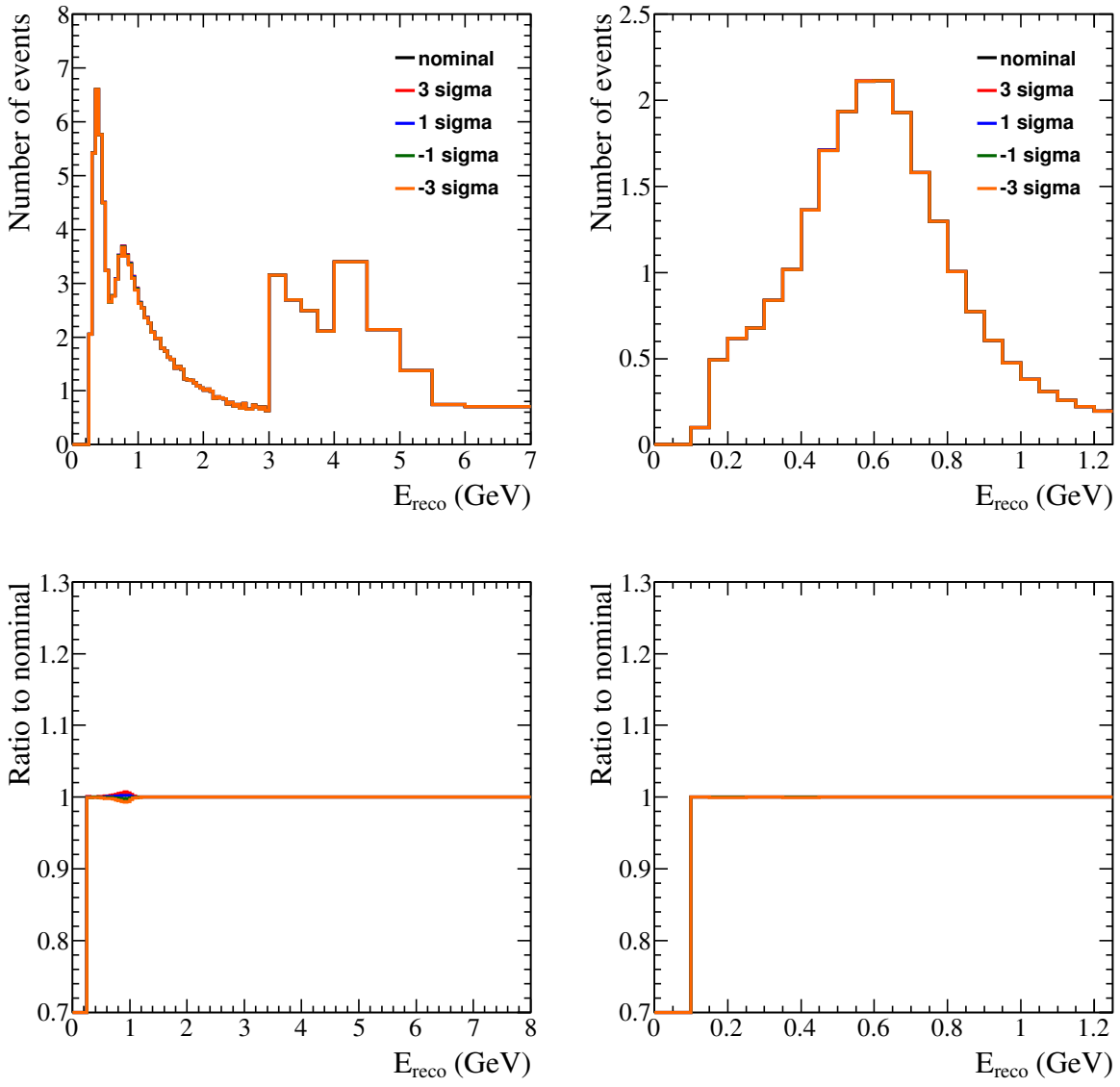


Figure 13: Effect of BANFF 12, SK numubar flux, 0.7 - 1.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.9**	**0.13**	**124.6**	**-0.13**	**22**	**22**	**0.004946**	**22**	**-0.004946**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.966	2.078	7.642	-2.078	0.05456	0.05565	1.995	0.05347	-1.995
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 27: Effect of BANFF 13, SK numubar flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.9**	**0.13**	**124.6**	**-0.13**	**22**	**22**	**0.004946**	**22**	**-0.004946**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.734	2.484	4.504	-2.484	0.0008752	0.0009036	3.247	0.0008468	-3.247
$\bar{\nu_\mu}$ CC 1π	2.132	2.168	1.653	2.097	-1.653	0.0002486	0.000262	5.416	0.0002351	-5.416
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2541	2.408	0.2422	-2.408	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4872	0.3479	0.4838	-0.3479	0.000124	0.0001279	3.196	0.00012	-3.196
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.146	1.803	0.1408	-1.803		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.02039	2.437	0.01942	-2.437
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01617	1.74	0.01562	-1.74
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.177	1.085	0.1732	-1.085	0.01752	0.0178	1.604	0.01723	-1.604
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 28: Effect of BANFF 13, SK numubar flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

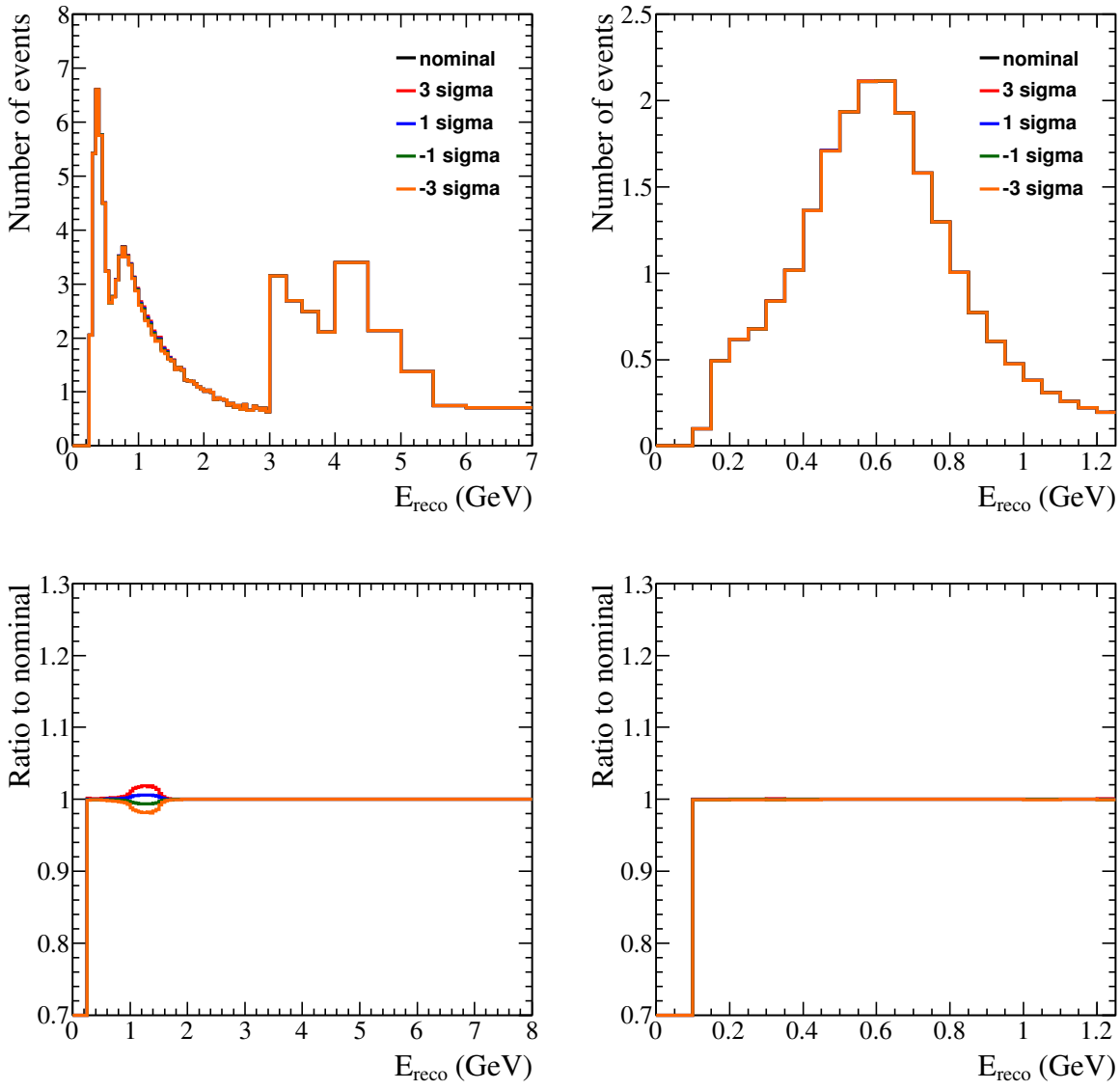


Figure 14: Effect of BANFF 13, SK numubar flux, 1.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2248**	**124.5**	**-0.2248**	**22**	**22**	**0.006598**	**22**	**-0.006598**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	8.084	3.594	7.523	-3.594	0.05456	0.05601	2.653	0.05311	-2.653
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003189	0.4159	0.003163	-0.4159	0.2339	0.2339	0.001617	0.2339	-0.001617
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 29: Effect of BANFF 14, SK numubar flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2248**	**124.5**	**-0.2248**	**22**	**22**	**0.006598**	**22**	**-0.006598**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.783	3.542	4.456	-3.542	0.0008752	0.0008848	1.094	0.0008656	-1.094
$\bar{\nu_\mu}$ CC 1π	2.132	2.22	4.095	2.045	-4.095	0.0002486	0.0002589	4.166	0.0002382	-4.166
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2576	3.799	0.2387	-3.799	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4964	2.24	0.4746	-2.24	0.000124	0.0001322	6.67	0.0001157	-6.67
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1478	3.065	0.139	-3.065		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.02048	2.914	0.01932	-2.914
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01628	2.466	0.0155	-2.466
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1799	2.769	0.1702	-2.769	0.01752	0.01796	2.557	0.01707	-2.557
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002184	0.6083	0.002158	-0.6083	0.1808	0.1808	0.002091	0.1808	-0.002091
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 30: Effect of BANFF 14, SK numubar flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

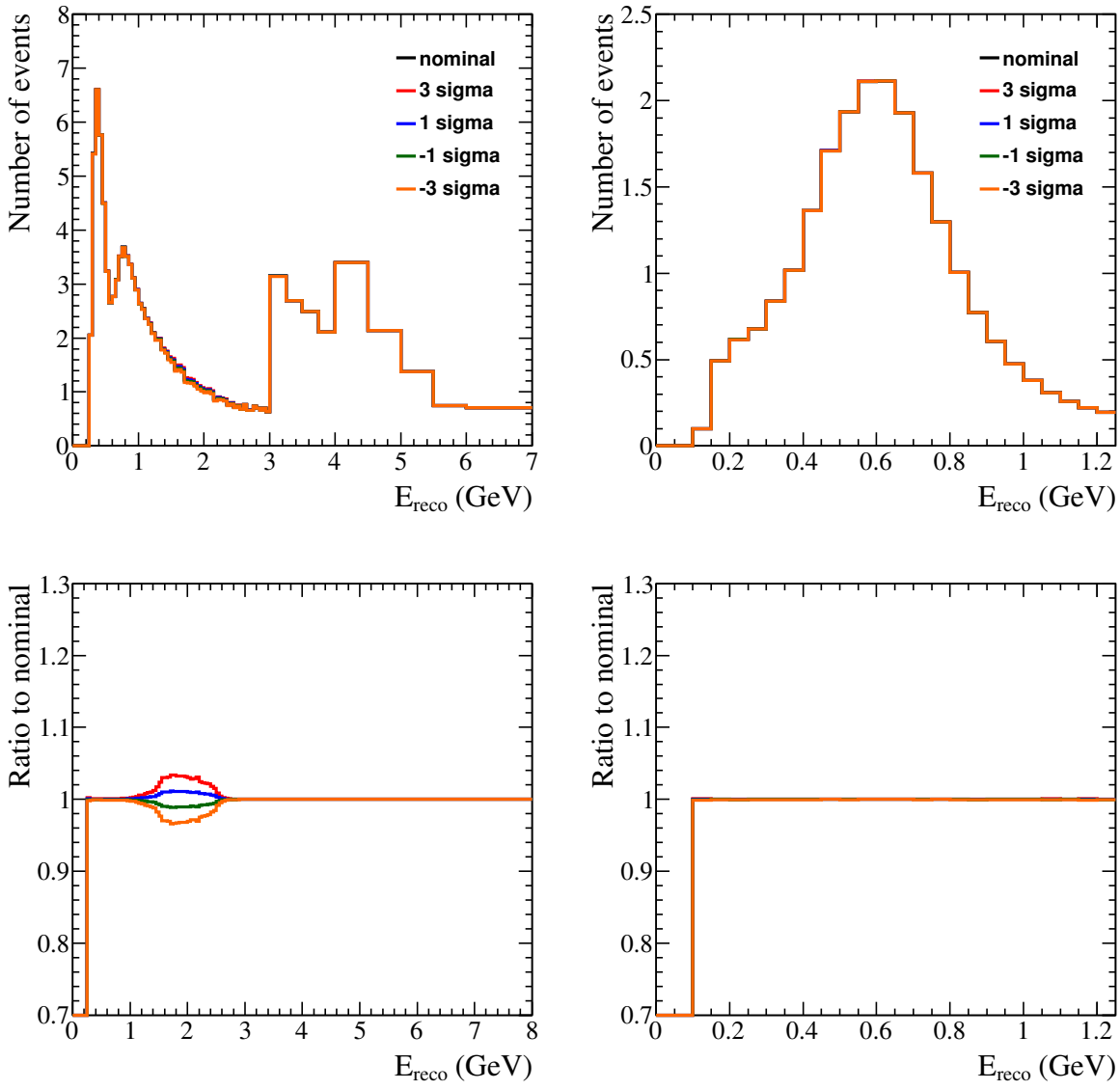


Figure 15: Effect of BANFF 14, SK numubar flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125**	**0.2003**	**124.5**	**-0.2003**	**22**	**22**	**0.009157**	**22**	**-0.009157**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	8.054	3.203	7.554	-3.203	0.05456	0.05657	3.676	0.05256	-3.676
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003181	0.1486	0.003171	-0.1486	0.2339	0.2339	0.003712	0.2339	-0.003712
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 31: Effect of BANFF 15, SK numubar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125**	**0.2003**	**124.5**	**-0.2003**	**22**	**22**	**0.009157**	**22**	**-0.009157**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.722	2.22	4.517	-2.22	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.218	4.02	2.047	-4.02	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2553	2.893	0.241	-2.893	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.5235	7.825	0.4475	-7.825	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1492	4.003	0.1377	-4.003		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.02052	3.102	0.01928	-3.102
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01639	3.115	0.0154	-3.115
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1859	6.169	0.1643	-6.169	0.01752	0.01841	5.101	0.01662	-5.101
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002172	0.04756	0.00217	-0.04756	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008526	0.2685	0.000848	-0.2685	0.04064	0.04064	0.01558	0.04063	-0.01558
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001139	0.1984	0.0001135	-0.1984	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.196e-05	2.891	3.96e-05	-2.891	0.001548	0.00155	0.1519	0.001546	-0.1519
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 32: Effect of BANFF 15, SK numubar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

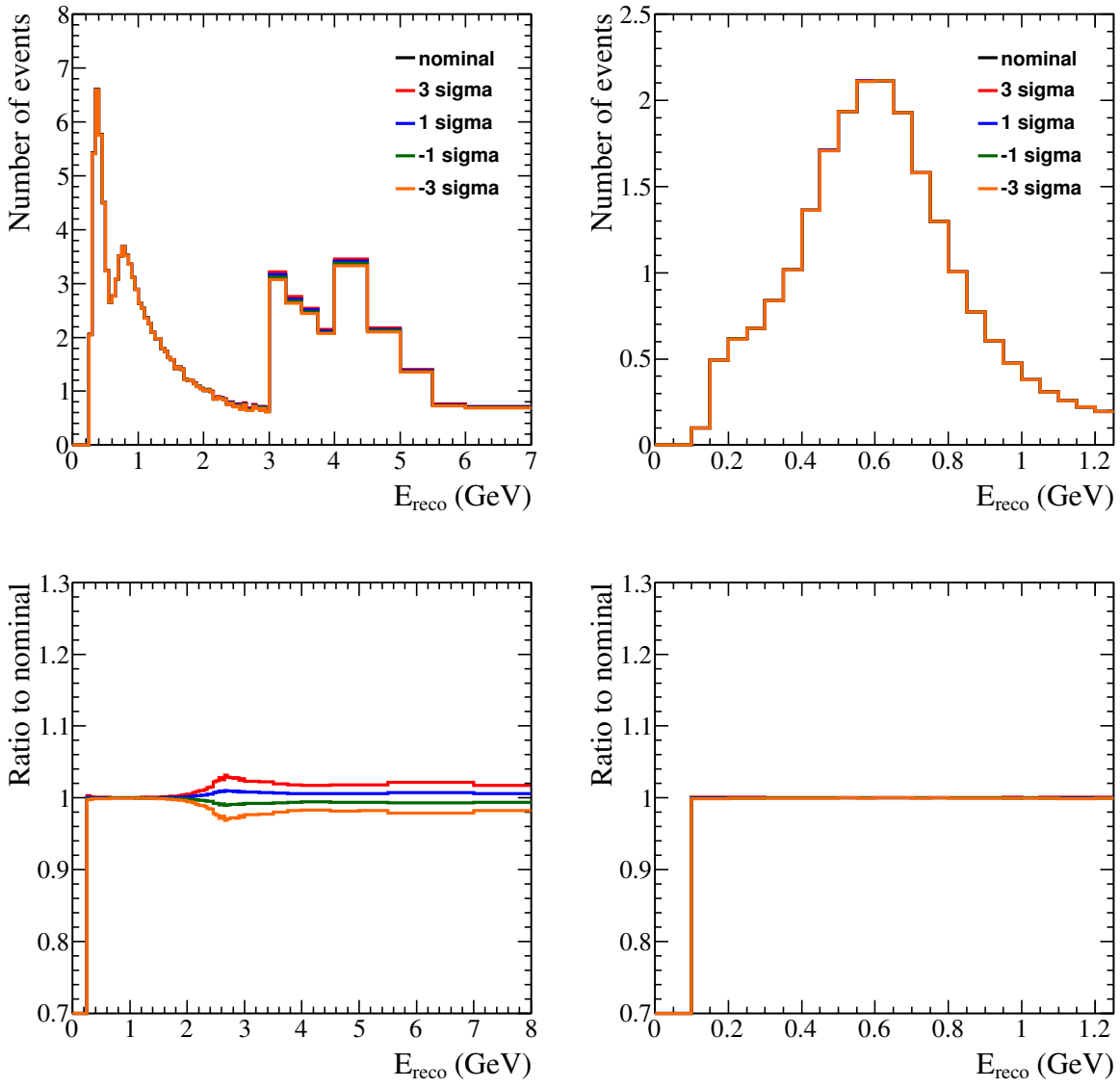


Figure 16: Effect of BANFF 15, SK numubar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.001726**	**124.8**	**-0.001726**	**22**	**22.06**	**0.2503**	**21.95**	**-0.2503**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1669	0.7348	0.1645	-0.7348	0.0001767	0.0001787	1.124	0.0001747	-1.124
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2629	0.3573	0.261	-0.3573	3.127	3.182	1.761	3.072	-1.761
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 33: Effect of BANFF 16, SK nue flux, 0.0 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.001726**	**124.8**	**-0.001726**	**22**	**22.06**	**0.2503**	**21.95**	**-0.2503**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1317	0.9299	0.1293	-0.9299	0.0001375	0.0001395	1.445	0.0001355	-1.445
Osc. ν_μ CC 1π	0.03155	0.03156	0.006068	0.03155	-0.006068	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009351	0.1976	0.0009314	-0.1976	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03593	2.444	0.03421	-2.444	2.264	2.318	2.391	2.21	-2.391
ν_e CC 1π	0.02275	0.02279	0.1827	0.02271	-0.1827	0.7705	0.7713	0.1124	0.7696	-0.1124
ν_e CC coherent	0.001091	0.001099	0.7954	0.001082	-0.7954	0.009237	0.009297	0.6474	0.009178	-0.6474
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07514	0.03817	0.07509	-0.03817		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01071	0.0286	0.0107	-0.0286
ν_e NC coherent		-		–		0.004117	0.004127	0.2283	0.004108	-0.2283
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.01011	0.0271	0.0101	-0.0271
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 34: Effect of BANFF 16, SK nue flux, 0.0 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

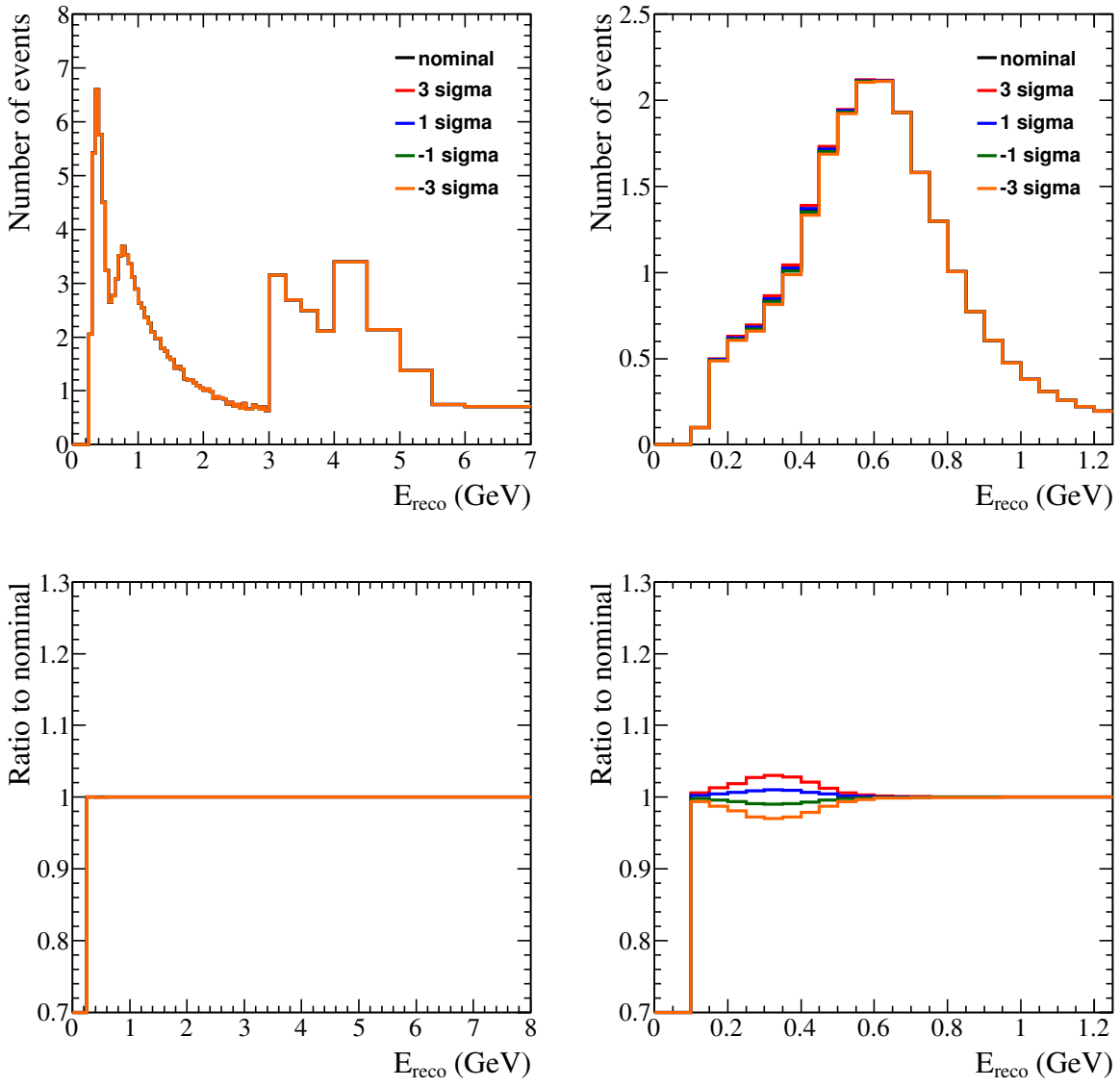


Figure 17: Effect of BANFF 16, SK nue flux, 0.0 - 0.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.004179**	**124.8**	**-0.004179**	**22**	**22.07**	**0.2984**	**21.94**	**-0.2984**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1696	2.333	0.1618	-2.333	0.0001767	0.000183	3.558	0.0001704	-3.558
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2633	0.5149	0.2606	-0.5149	3.127	3.193	2.099	3.061	-2.099
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 35: Effect of BANFF 17, SK nue flux, 0.5 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.004179**	**124.8**	**-0.004179**	**22**	**22.07**	**0.2984**	**21.94**	**-0.2984**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1342	2.792	0.1269	-2.792	0.0001375	0.0001433	4.18	0.0001318	-4.18
Osc. ν_μ CC 1π	0.03155	0.03176	0.6654	0.03134	-0.6654	3.655e-05	3.701e-05	1.269	3.608e-05	-1.269
Osc. ν_μ CC coherent	0.0009333	0.0009448	1.239	0.0009217	-1.239	8.165e-07	8.915e-07	9.183	7.415e-07	-9.183
Osc. ν_μ CC other	0.002684	0.002684	0.001381	0.002684	-0.001381	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03589	2.337	0.03425	-2.337	2.264	2.322	2.526	2.207	-2.526
ν_e CC 1π	0.02275	0.02295	0.8715	0.02255	-0.8715	0.7705	0.7786	1.063	0.7623	-1.063
ν_e CC coherent	0.001091	0.001125	3.135	0.001057	-3.135	0.009237	0.009379	1.535	0.009096	-1.535
ν_e CC other	0.00662	0.006623	0.03951	0.006618	-0.03951	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07534	0.3008	0.07489	-0.3008		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01076	0.4699	0.01065	-0.4699
ν_e NC coherent		-		–		0.004117	0.004141	0.5689	0.004094	-0.5689
ν_e NC other	0.1213	0.1213	0.05586	0.1212	-0.05586	0.0101	0.01014	0.353	0.01007	-0.353
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 36: Effect of BANFF 17, SK nue flux, 0.5 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

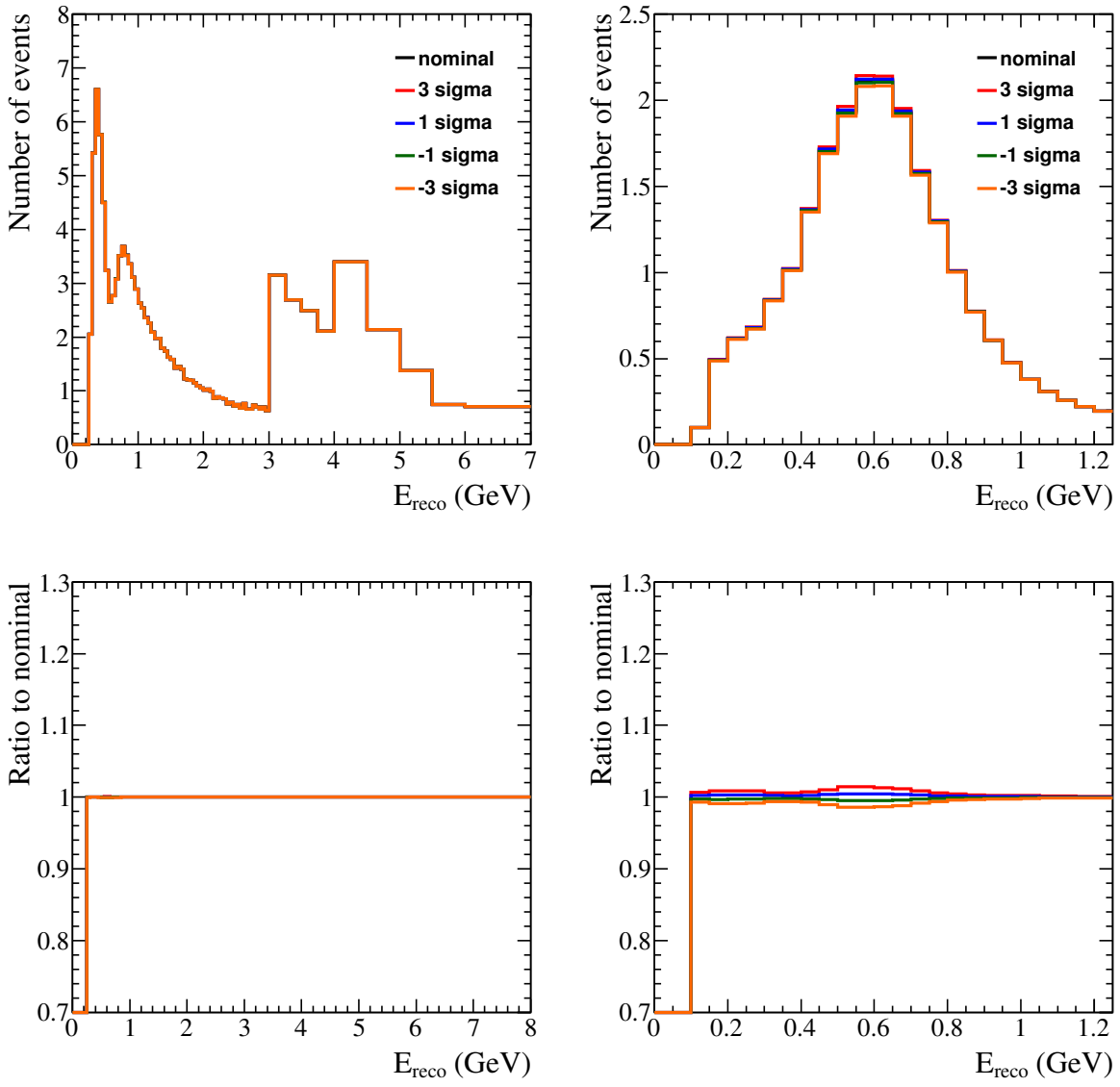


Figure 18: Effect of BANFF 17, SK nue flux, 0.5 - 0.7 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.005478**	**124.8**	**-0.005478**	**22**	**22.04**	**0.1845**	**21.96**	**-0.1845**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1717	3.612	0.1597	-3.612	0.0001767	0.0001824	3.235	0.000171	-3.235
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2628	0.3244	0.2611	-0.3244	3.127	3.168	1.298	3.086	-1.298
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 37: Effect of BANFF 18, SK nue flux, 0.7 - 0.8 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.005478**	**124.8**	**-0.005478**	**22**	**22.04**	**0.1845**	**21.96**	**-0.1845**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1354	3.718	0.1257	-3.718	0.0001375	0.0001417	3.035	0.0001333	-3.035
Osc. ν_μ CC 1π	0.03155	0.03265	3.471	0.03046	-3.471	3.655e-05	3.805e-05	4.119	3.504e-05	-4.119
Osc. ν_μ CC coherent	0.0009333	0.0009634	3.233	0.0009031	-3.233	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.00269	0.2315	0.002678	-0.2315	1.847e-06	1.886e-06	2.114	1.808e-06	-2.114
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03541	0.9629	0.03473	-0.9629	2.264	2.297	1.43	2.232	-1.43
ν_e CC 1π	0.02275	0.02297	0.9714	0.02253	-0.9714	0.7705	0.7784	1.036	0.7625	-1.036
ν_e CC coherent	0.001091	0.001097	0.5475	0.001085	-0.5475	0.009237	0.009339	1.094	0.009136	-1.094
ν_e CC other	0.00662	0.006623	0.04485	0.006617	-0.04485	0.0581	0.05813	0.04749	0.05807	-0.04749
ν_e NC 1$\pi^{+/-}$	0.07512	0.07534	0.2984	0.07489	-0.2984		-		–	
ν_e NC 1π^0		-		–		0.0107	0.01076	0.5401	0.01065	-0.5401
ν_e NC coherent		-		–		0.004117	0.004134	0.4102	0.004101	-0.4102
ν_e NC other	0.1213	0.1213	0.04778	0.1212	-0.04778	0.0101	0.01014	0.3334	0.01007	-0.3334
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 38: Effect of BANFF 18, SK nue flux, 0.7 - 0.8 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

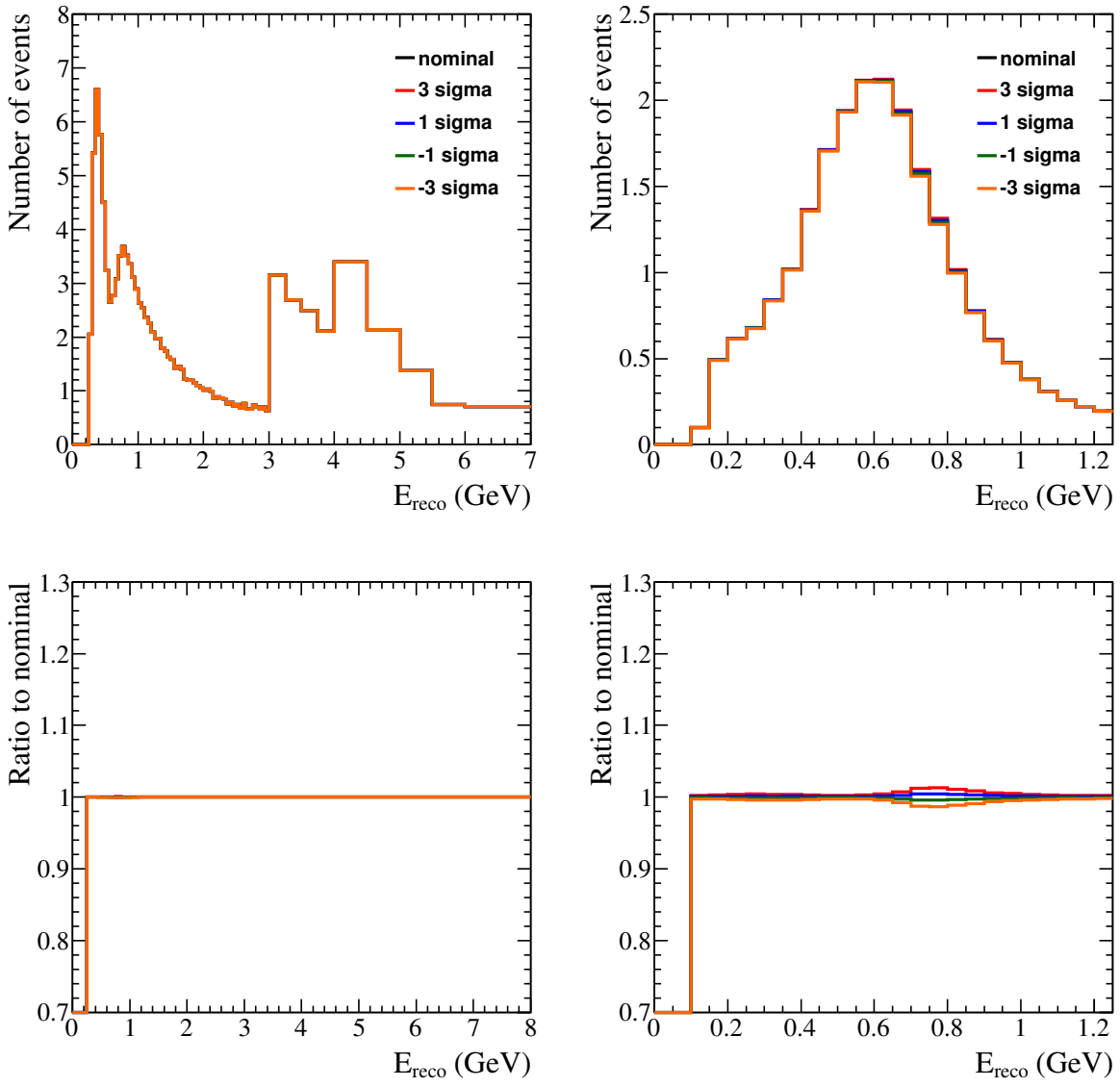


Figure 19: Effect of BANFF 18, SK nue flux, 0.7 - 0.8 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.005155**	**124.8**	**-0.005155**	**22**	**22.09**	**0.4158**	**21.91**	**-0.4158**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1682	1.484	0.1632	-1.484	0.0001767	0.000179	1.29	0.0001745	-1.29
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2659	1.517	0.2579	-1.517	3.127	3.219	2.925	3.036	-2.925
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 39: Effect of BANFF 19, SK nue flux, 0.8 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.005155**	**124.8**	**-0.005155**	**22**	**22.09**	**0.4158**	**21.91**	**-0.4158**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1322	1.277	0.1289	-1.277	0.0001375	0.0001386	0.7817	0.0001364	-0.7817
Osc. ν_μ CC 1π	0.03155	0.03229	2.331	0.03082	-2.331	3.655e-05	3.764e-05	2.997	3.545e-05	-2.997
Osc. ν_μ CC coherent	0.0009333	0.0009509	1.89	0.0009156	-1.89	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002723	1.459	0.002645	-1.459	1.847e-06	1.957e-06	5.92	1.738e-06	-5.92
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03566	1.692	0.03448	-1.692	2.264	2.321	2.52	2.207	-2.52
ν_e CC 1π	0.02275	0.02345	3.081	0.02205	-3.081	0.7705	0.8025	4.155	0.7384	-4.155
ν_e CC coherent	0.001091	0.001106	1.369	0.001076	-1.369	0.009237	0.009666	4.635	0.008809	-4.635
ν_e CC other	0.00662	0.006672	0.7793	0.006569	-0.7793	0.0581	0.05962	2.614	0.05658	-2.614
ν_e NC 1$\pi^{+/-}$	0.07512	0.07658	1.947	0.07365	-1.947		-		–	
ν_e NC 1π^0		-		–		0.0107	0.01095	2.266	0.01046	-2.266
ν_e NC coherent		-		–		0.004117	0.004188	1.71	0.004047	-1.71
ν_e NC other	0.1213	0.1224	0.9474	0.1201	-0.9474	0.0101	0.01024	1.323	0.009971	-1.323
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 40: Effect of BANFF 19, SK nue flux, 0.8 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

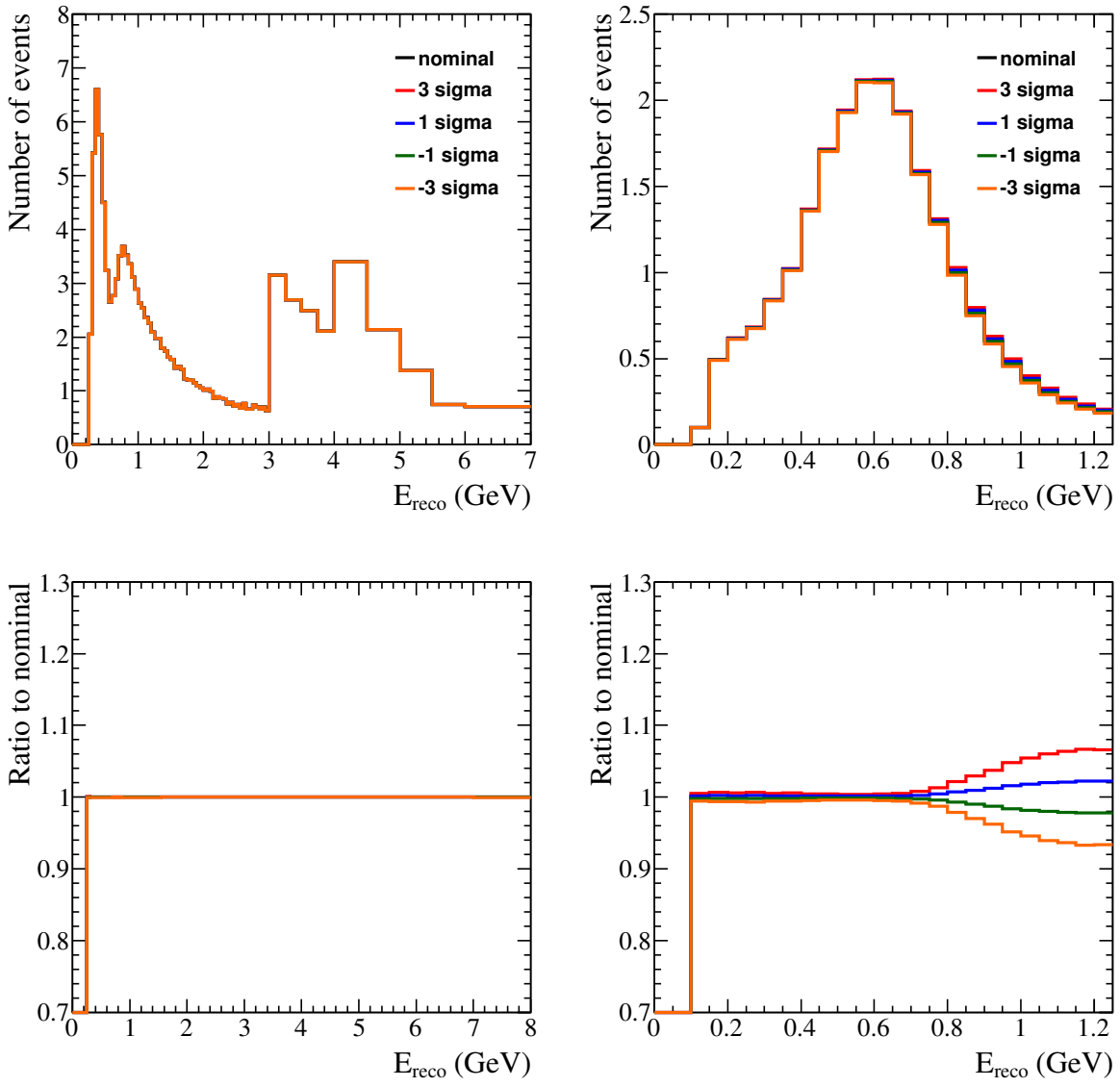


Figure 20: Effect of BANFF 19, SK nue flux, 0.8 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.005248**	**124.8**	**-0.005248**	**22**	**22.01**	**0.06027**	**21.99**	**-0.06027**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.167	0.7642	0.1644	-0.7642	0.0001767	0.0001771	0.2143	0.0001764	-0.2143
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2672	2.017	0.2566	-2.017	3.127	3.14	0.424	3.114	-0.424
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 41: Effect of BANFF 20, SK nue flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.005248**	**124.8**	**-0.005248**	**22**	**22.01**	**0.06027**	**21.99**	**-0.06027**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1312	0.5234	0.1298	-0.5234	0.0001375	0.0001377	0.1165	0.0001374	-0.1165
Osc. ν_μ CC 1π	0.03155	0.03205	1.557	0.03106	-1.557	3.655e-05	3.676e-05	0.5978	3.633e-05	-0.5978
Osc. ν_μ CC coherent	0.0009333	0.0009474	1.513	0.0009191	-1.513	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002762	2.89	0.002607	-2.89	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.0354	0.9353	0.03474	-0.9353	2.264	2.264	0.005911	2.264	-0.005911
ν_e CC 1π	0.02275	0.02312	1.613	0.02239	-1.613	0.7705	0.7808	1.344	0.7601	-1.344
ν_e CC coherent	0.001091	0.001107	1.487	0.001074	-1.487	0.009237	0.009258	0.2214	0.009217	-0.2214
ν_e CC other	0.00662	0.00677	2.261	0.00647	-2.261	0.0581	0.06032	3.817	0.05588	-3.817
ν_e NC $1\pi^{+/-}$	0.07512	0.0767	2.103	0.07354	-2.103		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01093	2.096	0.01048	-2.096
ν_e NC coherent		-		–		0.004117	0.004196	1.921	0.004038	-1.921
ν_e NC other	0.1213	0.1241	2.343	0.1184	-2.343	0.0101	0.01033	2.252	0.009877	-2.252
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 42: Effect of BANFF 20, SK nue flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

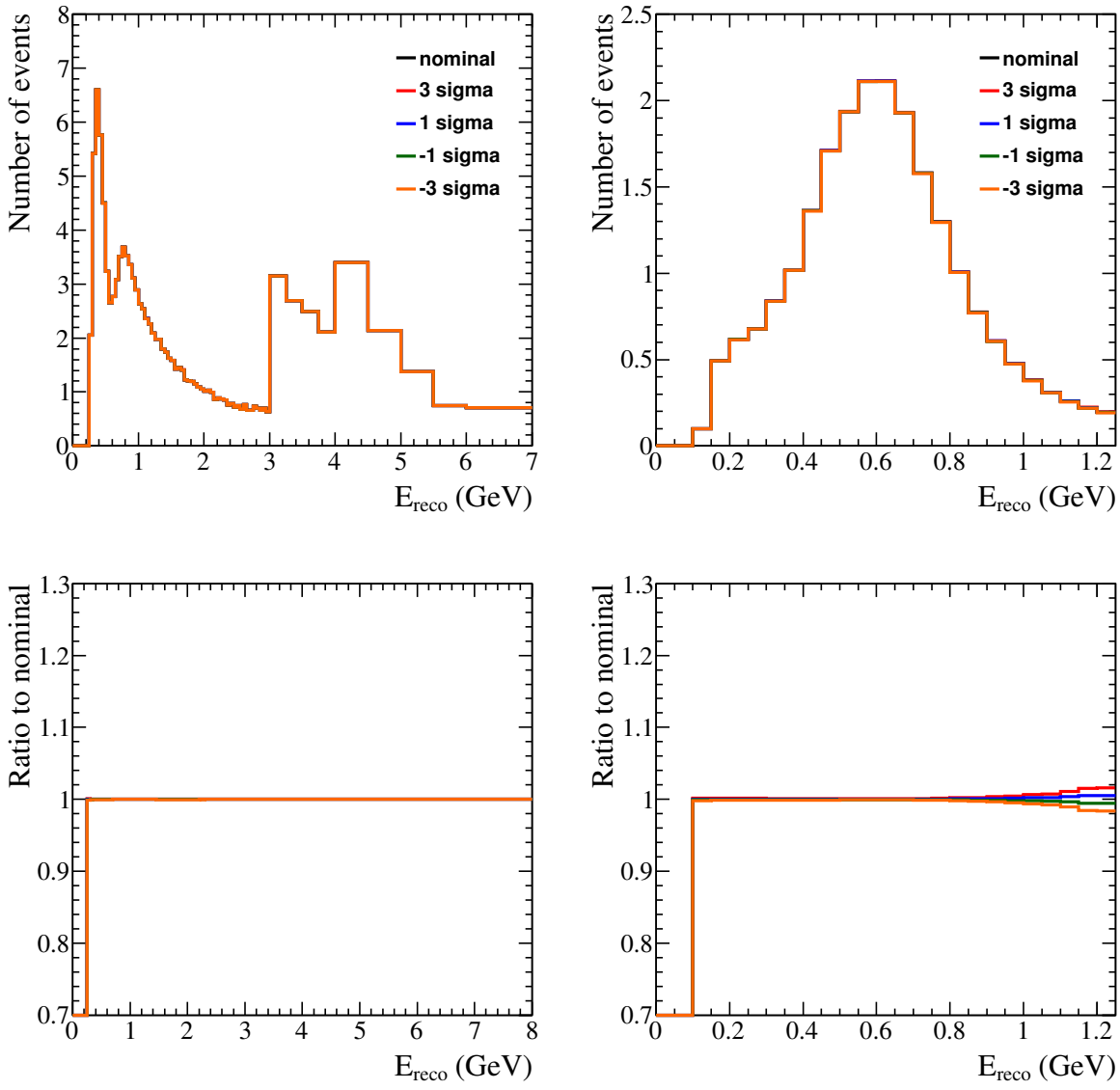


Figure 21: Effect of BANFF 20, SK nue flux, 1.5 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.003849**	**124.8**	**-0.003849**	**22**	**22**	**0.007629**	**22**	**-0.007629**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1661	0.2324	0.1653	-0.2324	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2663	1.686	0.2575	-1.686	3.127	3.129	0.05367	3.125	-0.05367
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 43: Effect of BANFF 21, SK nue flux, 2.5 - 4.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.003849**	**124.8**	**-0.003849**	**22**	**22**	**0.007629**	**22**	**-0.007629**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1307	0.123	0.1304	-0.123	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03171	0.5015	0.0314	-0.5015	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009383	0.5358	0.0009283	-0.5358	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002745	2.286	0.002623	-2.286	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03513	0.1649	0.03501	-0.1649	2.264	2.264	0.0001956	2.264	-0.0001956
ν_e CC 1π	0.02275	0.02292	0.7211	0.02259	-0.7211	0.7705	0.7714	0.1215	0.7695	-0.1215
ν_e CC coherent	0.001091	0.001095	0.4345	0.001086	-0.4345	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.006731	1.68	0.006509	-1.68	0.0581	0.05845	0.6126	0.05774	-0.6126
ν_e NC $1\pi^{+/-}$	0.07512	0.0764	1.715	0.07383	-1.715		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01084	1.268	0.01057	-1.268
ν_e NC coherent		-		–		0.004117	0.004175	1.392	0.00406	-1.392
ν_e NC other	0.1213	0.1241	2.301	0.1185	-2.301	0.0101	0.01029	1.873	0.009916	-1.873
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 44: Effect of BANFF 21, SK nue flux, 2.5 - 4.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

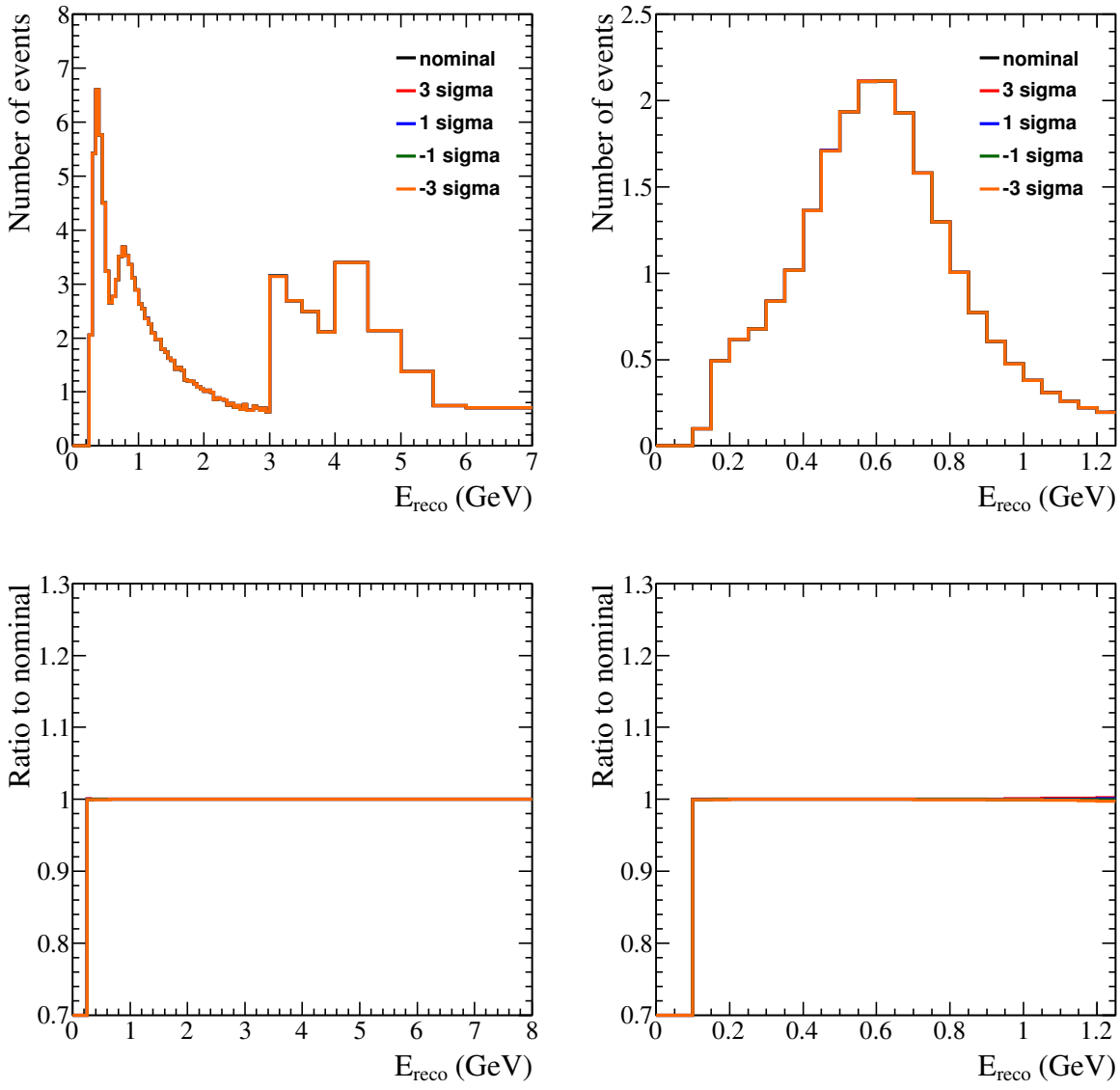


Figure 22: Effect of BANFF 21, SK nue flux, 2.5 - 4.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.002202**	**124.8**	**-0.002202**	**22**	**22**	**0.001282**	**22**	**-0.001282**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0.007293	0.1657	-0.007293	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2647	1.044	0.2592	-1.044	3.127	3.127	0.009021	3.127	-0.009021
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 45: Effect of BANFF 22, SK nue flux, 4.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.002202**	**124.8**	**-0.002202**	**22**	**22**	**0.001282**	**22**	**-0.001282**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0.001761	0.1305	-0.001761	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03156	0.01438	0.03155	-0.01438	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009334	0.0172	0.0009331	-0.0172	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002689	0.1896	0.002679	-0.1896	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03511	0.1046	0.03503	-0.1046	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02282	0.3126	0.02268	-0.3126	0.7705	0.7705	0.002846	0.7704	-0.002846
ν_e CC coherent	0.001091	0.001096	0.498	0.001085	-0.498	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.006787	2.52	0.006453	-2.52	0.0581	0.0581	0.004766	0.0581	-0.004766
ν_e NC 1$\pi^{+/-}$	0.07512	0.0758	0.9095	0.07443	-0.9095		-		–	
ν_e NC 1π^0		-		–		0.0107	0.01079	0.7917	0.01062	-0.7917
ν_e NC coherent		-		–		0.004117	0.004171	1.3	0.004064	-1.3
ν_e NC other	0.1213	0.123	1.461	0.1195	-1.461	0.0101	0.01022	1.179	0.009986	-1.179
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 46: Effect of BANFF 22, SK nue flux, 4.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

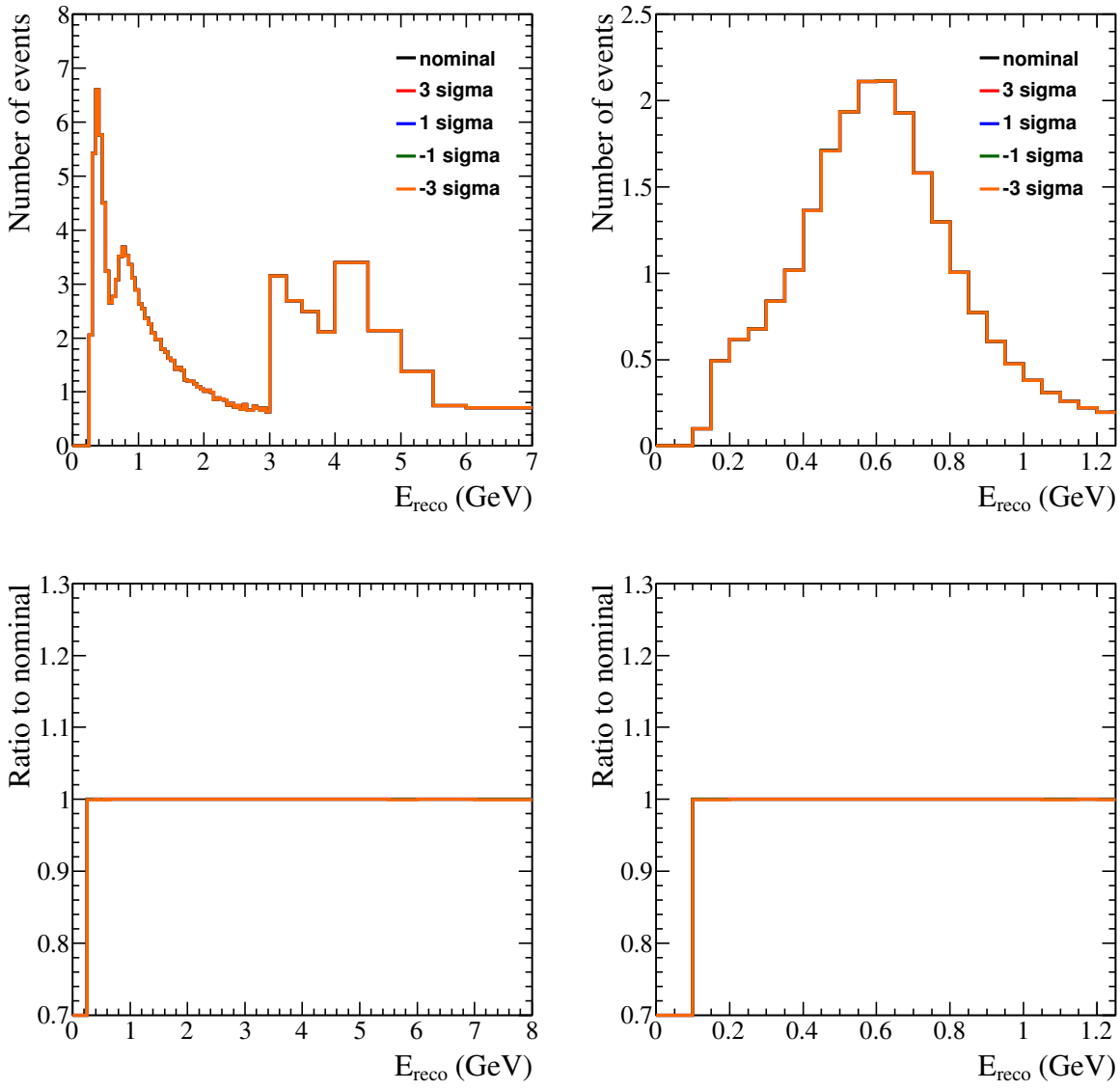


Figure 23: Effect of BANFF 22, SK nue flux, 4.0 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.001471**	**124.8**	**-0.001471**	**22**	**22.03**	**0.1208**	**21.97**	**-0.1208**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.02604	7.582	0.02237	-7.582	0.1523	0.1789	17.45	0.1257	-17.45
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 47: Effect of BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.001471**	**124.8**	**-0.001471**	**22**	**22.03**	**0.1208**	**21.97**	**-0.1208**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002337	15.35	0.001715	-15.35	0.0973	0.1146	17.74	0.08004	-17.74
$\bar{\nu_e}$ CC 1π	0.001296	0.001453	12.09	0.00114	-12.09	0.04178	0.04907	17.44	0.03449	-17.44
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001476	12.05	0.0001159	-12.05	0.006662	0.007844	17.74	0.005481	-17.74
$\bar{\nu_e}$ CC other	0.0004146	0.0004304	3.815	0.0003988	-3.815	0.003647	0.004244	16.38	0.00305	-16.38
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008912	8.197	0.007561	-8.197		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001289	8.204	0.001094	-8.204
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0008213	10.14	0.00067	-10.14
$\bar{\nu_e}$ NC other	0.0121	0.01276	5.459	0.01144	-5.459	0.0009675	0.001045	8.047	0.0008897	-8.047
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 48: Effect of BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

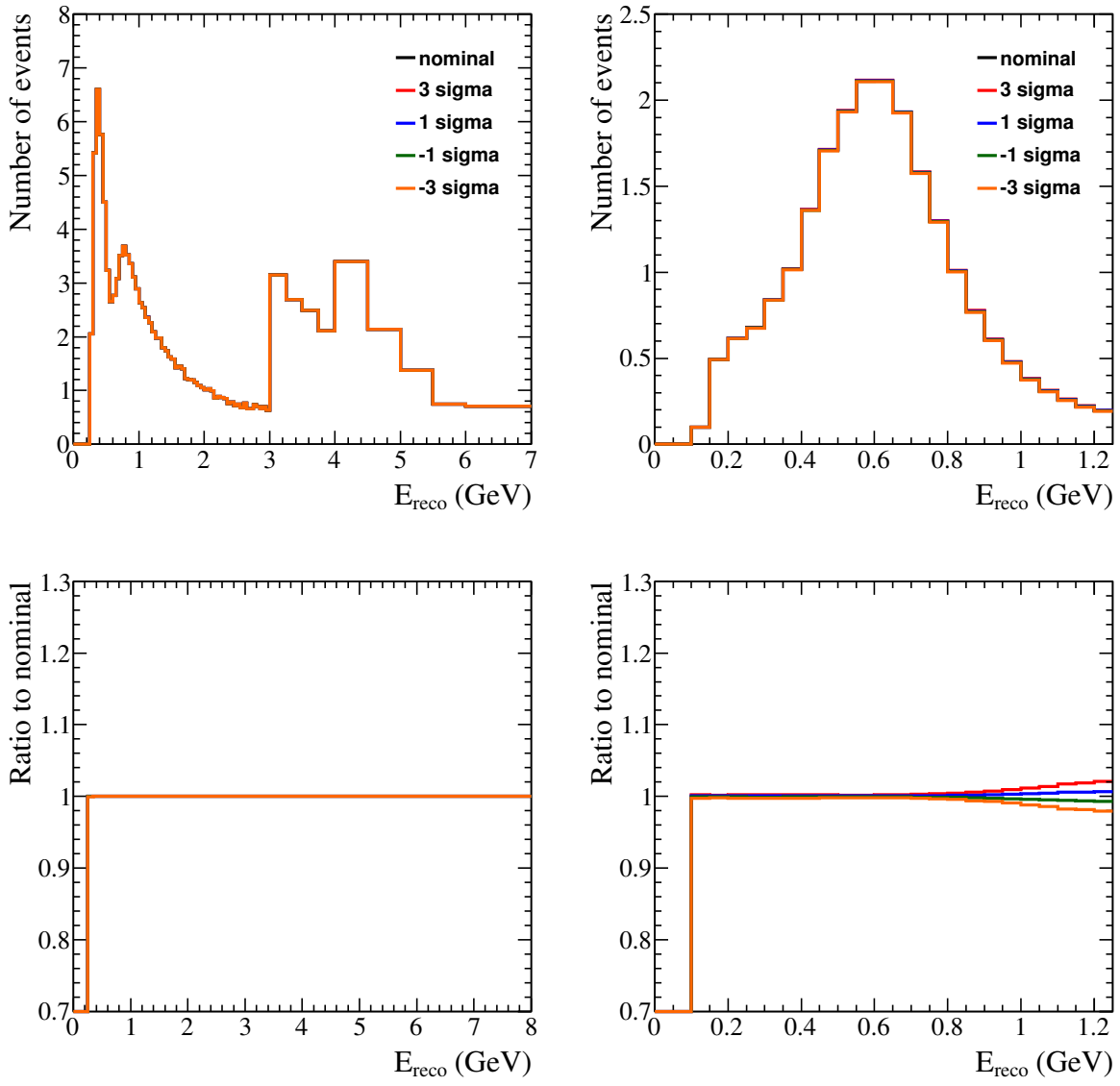


Figure 24: Effect of BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.0009283**	**124.8**	**-0.0009283**	**22**	**22**	**0.0009342**	**22**	**-0.0009342**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.02536	4.785	0.02305	-4.785	0.1523	0.1525	0.135	0.1521	-0.135
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 49: Effect of BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.0009283**	**124.8**	**-0.0009283**	**22**	**22**	**0.0009342**	**22**	**-0.0009342**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002048	1.124	0.002003	-1.124	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001331	2.659	0.001262	-2.659	0.04178	0.04184	0.1384	0.04172	-0.1384
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001353	2.677	0.0001282	-2.677	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004418	6.56	0.0003874	-6.56	0.003647	0.00367	0.6403	0.003624	-0.6403
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008607	4.495	0.007866	-4.495		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001245	4.492	0.001138	-4.492
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007723	3.578	0.000719	-3.578
$\bar{\nu_e}$ NC other	0.0121	0.0128	5.785	0.0114	-5.785	0.0009675	0.001012	4.566	0.0009233	-4.566
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 50: Effect of BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

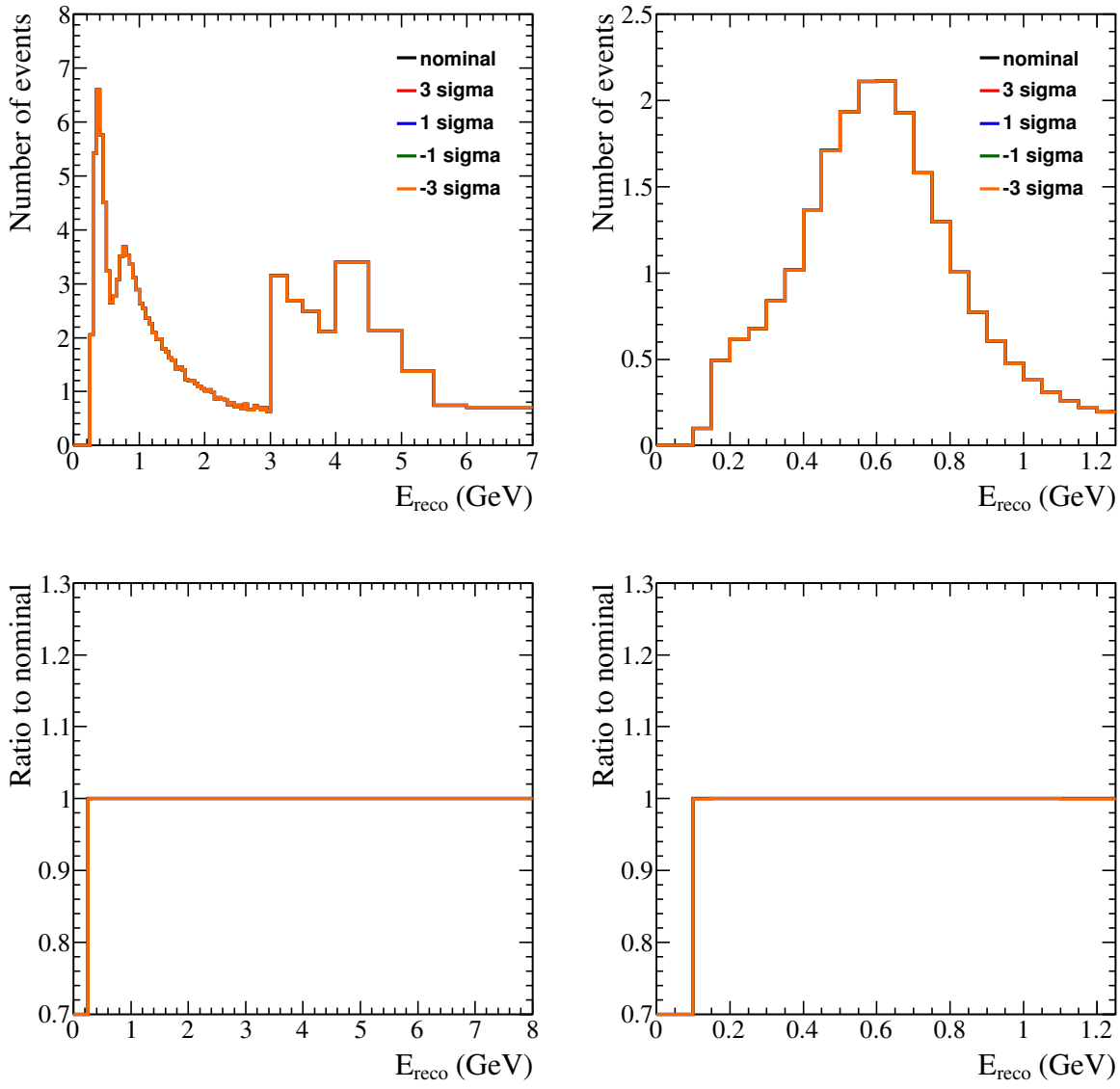


Figure 25: Effect of BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.2**	**2.738**	**121.2**	**-2.833**	**22**	**22.65**	**2.955**	**21.3**	**-3.186**
ν_μ	116.2	119.4	2.738	112.9	-2.847	0.9435	0.9453	0.1873	0.9416	-0.1994
$Osc.\nu_\mu$	0.1657	0.1713	3.388	0.1597	-3.588	0.0001767	0.0001822	3.067	0.0001709	-3.283
$\bar{\nu_\mu}$	7.804	8.022	2.798	7.595	-2.681	0.05456	0.05461	0.08274	0.05452	-0.07495
ν_e	0.2619	0.2635	0.6035	0.2603	-0.6336	3.127	3.212	2.719	3.036	-2.92
$\bar{\nu_e}$	0.0242	0.02431	0.418	0.02411	-0.4038	0.1523	0.1559	2.347	0.1488	-2.324
Osc.$\bar{\nu_e}$	0.003176	0.003253	2.437	0.003099	-2.428	0.2339	0.2403	2.737	0.2275	-2.722
$Osc.\nu_e$	0.2742	0.282	2.845	0.2658	-3.063	17.49	18.04	3.164	16.89	-3.418

Table 51: Effect of BANFF, Ma QE with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.2**	**2.738**	**121.2**	**-2.833**	**22**	**22.65**	**2.955**	**21.3**	**-3.186**
ν_μ CCQE	70.79	73.98	4.496	67.48	-4.675	0.04778	0.04954	3.699	0.0459	-3.938
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1361	4.3	0.1246	-4.554	0.0001375	0.0001429	3.941	0.0001317	-4.218
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.838	4.727	4.41	-4.529	0.0008752	0.0009204	5.158	0.0008343	-4.673
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03665	4.507	0.03341	-4.732	2.264	2.349	3.754	2.173	-4.033
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002127	4.995	0.001928	-4.824	0.0973	0.1009	3.674	0.09376	-3.637
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002249	3.565	0.002094	-3.552	0.1808	0.1872	3.541	0.1744	-3.521
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.2026	4.006	0.1864	-4.313	14.79	15.34	3.742	14.19	-4.041
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 52: Effect of BANFF, Ma QE with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

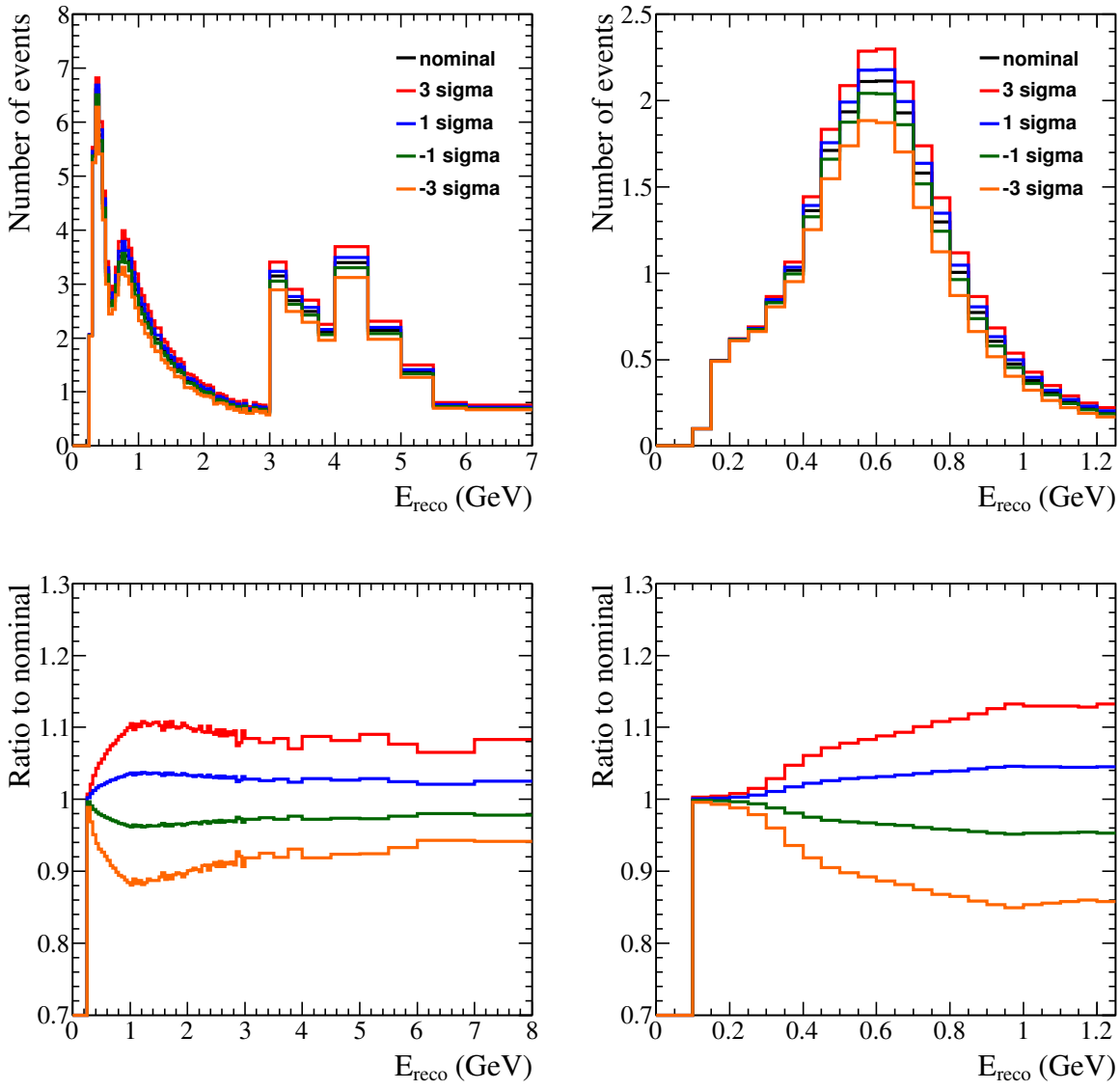


Figure 26: Effect of BANFF, Ma QE with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**127.7**	**2.384**	**121.9**	**-2.322**	**22**	**22.22**	**1.01**	**21.77**	**-1.055**
ν_μ	116.2	119	2.4	113.5	-2.348	0.9435	0.9948	5.441	0.8927	-5.387
$Osc.\nu_\mu$	0.1657	0.1678	1.278	0.1635	-1.31	0.0001767	0.0001791	1.316	0.0001743	-1.36
$\bar{\nu_\mu}$	7.804	7.969	2.123	7.654	-1.917	0.05456	0.05761	5.593	0.05169	-5.265
ν_e	0.2619	0.2738	4.55	0.2505	-4.352	3.127	3.176	1.579	3.076	-1.618
$\bar{\nu_e}$	0.0242	0.02554	5.514	0.02299	-5.031	0.1523	0.1548	1.641	0.1501	-1.464
Osc.$\bar{\nu_e}$	0.003176	0.003211	1.105	0.003145	-0.991	0.2339	0.236	0.9014	0.232	-0.807
$Osc.\nu_e$	0.2742	0.277	1.007	0.2712	-1.096	17.49	17.6	0.6505	17.37	-0.7068

Table 53: Effect of BANFF, Ma RES with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**127.7**	**2.384**	**121.9**	**-2.322**	**22**	**22.22**	**1.01**	**21.77**	**-1.055**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	32.21	8.193	27.39	-8.019	0.01709	0.01834	7.355	0.01582	-7.393
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.638	0.4766	8.558	-0.4566	0.0009536	0.0009638	1.066	0.0009429	-1.124
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	3.184	8.477	2.689	-8.371		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.4209	7.264	0.3631	-7.46
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.29	1.884	3.172	-1.753	0.3314	0.3529	6.507	0.3111	-6.118
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03366	6.677	0.0294	-6.842	3.655e-05	3.884e-05	6.29	3.417e-05	-6.499
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002695	0.4106	0.002673	-0.4117	1.847e-06	1.874e-06	1.444	1.819e-06	-1.523
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.275	6.685	2.004	-6.004	0.0002486	0.0002619	5.388	0.0002365	-4.832
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4886	0.6401	0.4824	-0.6297	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1595	11.18	0.1285	-10.44		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.02213	11.19	0.01781	-10.53
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.179	2.259	0.1716	-2.015	0.01752	0.01833	4.632	0.01675	-4.371
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02422	6.448	0.02129	-6.428	0.7705	0.818	6.169	0.7216	-6.337
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00665	0.4533	0.006592	-0.4284	0.0581	0.0583	0.3535	0.05788	-0.368
ν_e NC $1\pi^{+/-}$	0.07512	0.08305	10.56	0.06748	-10.16		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01181	10.33	0.009626	-10.08
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1238	2.055	0.119	-1.873	0.0101	0.01065	5.382	0.009623	-4.774
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001373	5.867	0.001232	-4.988	0.04178	0.04408	5.503	0.03974	-4.886
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004182	0.8808	0.000411	-0.8578	0.003647	0.003658	0.2946	0.003636	-0.3087
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.009222	11.96	0.007325	-11.07		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001331	11.71	0.001059	-11.08
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.01237	2.23	0.01186	-1.965	0.0009675	0.001018	5.187	0.0009221	-4.692
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008849	4.077	0.0008192	-3.65	0.04064	0.04274	5.176	0.03876	-4.632
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.121e-05	1.045	4.035e-05	-1.062	0.001548	0.001553	0.2967	0.001543	-0.3104
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07542	3.795	0.06966	-4.13	2.624	2.738	4.332	2.501	-4.707
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001595	0.1877	0.001589	-0.2259	0.03175	0.03184	0.2584	0.03167	-0.2719

Table 54: Effect of BANFF, Ma RES with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

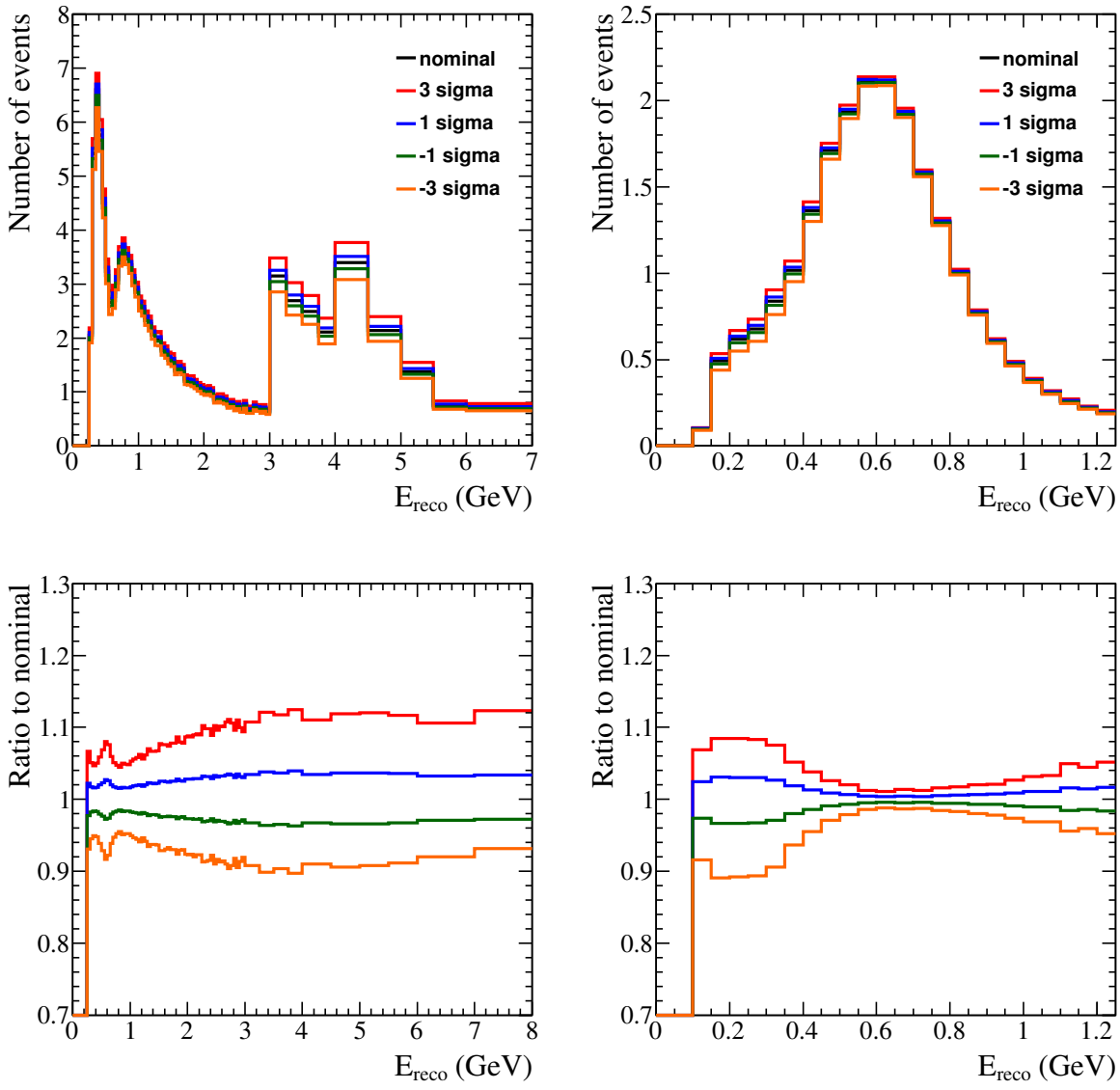


Figure 27: Effect of BANFF, Ma RES with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.9**	**3.327**	**120.6**	**-3.327**	**22**	**23.37**	**6.231**	**20.63**	**-6.231**
ν_μ	116.2	120.2	3.41	112.3	-3.41	0.9435	0.9471	0.379	0.9399	-0.379
$Osc.\nu_\mu$	0.1657	0.175	5.635	0.1564	-5.635	0.0001767	0.0001874	6.037	0.0001661	-6.037
$\bar{\nu_\mu}$	7.804	7.965	2.061	7.643	-2.061	0.05456	0.05462	0.1129	0.0545	-0.1129
ν_e	0.2619	0.2642	0.8757	0.2596	-0.8757	3.127	3.306	5.708	2.949	-5.708
$\bar{\nu_e}$	0.0242	0.02429	0.3509	0.02412	-0.3509	0.1523	0.16	5.036	0.1446	-5.036
Osc.$\bar{\nu_e}$	0.003176	0.003336	5.047	0.003016	-5.047	0.2339	0.2481	6.098	0.2196	-6.098
$Osc.\nu_e$	0.2742	0.2895	5.574	0.2589	-5.574	17.49	18.66	6.671	16.32	-6.671

Table 55: Effect of BANFF; Norm; CC QE; 0.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.9**	**3.327**	**120.6**	**-3.327**	**22**	**23.37**	**6.231**	**20.63**	**-6.231**
ν_μ CCQE	70.79	74.76	5.598	66.83	-5.598	0.04778	0.05135	7.485	0.0442	-7.485
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1399	7.153	0.1212	-7.153	0.0001375	0.0001482	7.759	0.0001269	-7.759
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.78	3.482	4.458	-3.482	0.0008752	0.0009368	7.04	0.0008136	-7.04
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03736	6.54	0.03278	-6.54	2.264	2.443	7.883	2.086	-7.883
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002111	4.192	0.001941	-4.192	0.0973	0.105	7.882	0.08963	-7.882
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002332	7.382	0.002011	-7.382	0.1808	0.1951	7.888	0.1665	-7.888
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.21	7.848	0.1795	-7.848	14.79	15.96	7.889	13.62	-7.889
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 56: Effect of BANFF; Norm; CC QE; 0.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

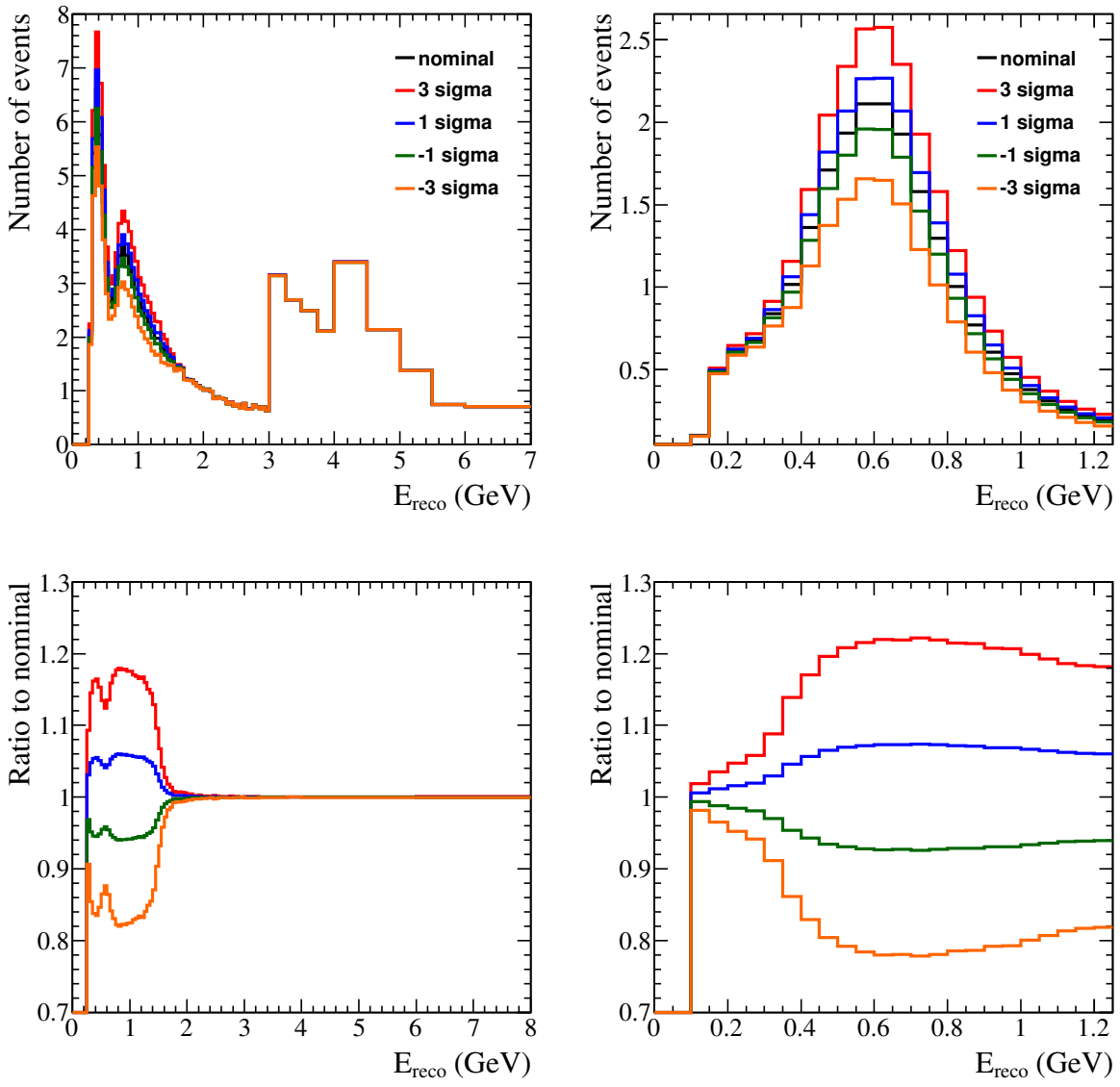


Figure 28: Effect of BANFF; Norm; CC QE; 0.0 - 1.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.8**	**1.614**	**122.8**	**-1.614**	**22**	**22**	**0.00275**	**22**	**-0.00275**
ν_μ	116.2	118	1.516	114.5	-1.516	0.9435	0.9438	0.02879	0.9432	-0.02879
$Osc.\nu_\mu$	0.1657	0.167	0.7882	0.1644	-0.7882	0.0001767	0.000177	0.1434	0.0001765	-0.1434
$\bar{\nu_\mu}$	7.804	8.053	3.191	7.555	-3.191	0.05456	0.05457	0.01917	0.05455	-0.01917
ν_e	0.2619	0.2625	0.2308	0.2613	-0.2308	3.127	3.127	0.007007	3.127	-0.007007
$\bar{\nu_e}$	0.0242	0.02429	0.3497	0.02412	-0.3497	0.1523	0.1523	0.007083	0.1523	-0.007083
Osc.$\bar{\nu_e}$	0.003176	0.003191	0.4749	0.003161	-0.4749	0.2339	0.2339	0.001765	0.2339	-0.001765
$Osc.\nu_e$	0.2742	0.2743	0.04085	0.2741	-0.04085	17.49	17.49	0.0005072	17.49	-0.0005072

Table 57: Effect of BANFF; Norm; CC QE; 1.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.8**	**1.614**	**122.8**	**-1.614**	**22**	**22**	**0.00275**	**22**	**-0.00275**
ν_μ CCQE	70.79	72.56	2.489	69.03	-2.489	0.04778	0.04805	0.5685	0.04751	-0.5685
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1318	1.001	0.1292	-1.001	0.0001375	0.0001378	0.1843	0.0001373	-0.1843
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.868	5.391	4.37	-5.391	0.0008752	0.0008857	1.195	0.0008648	-1.195
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03568	1.723	0.03447	-1.723	2.264	2.265	0.009676	2.264	-0.009676
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.00211	4.179	0.001941	-4.179	0.0973	0.09731	0.01109	0.09729	-0.01109
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002186	0.6947	0.002156	-0.6947	0.1808	0.1808	0.002283	0.1808	-0.002283
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1949	0.05752	0.1946	-0.05752	14.79	14.79	0.0005998	14.79	-0.0005998
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 58: Effect of BANFF; Norm; CC QE; 1.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

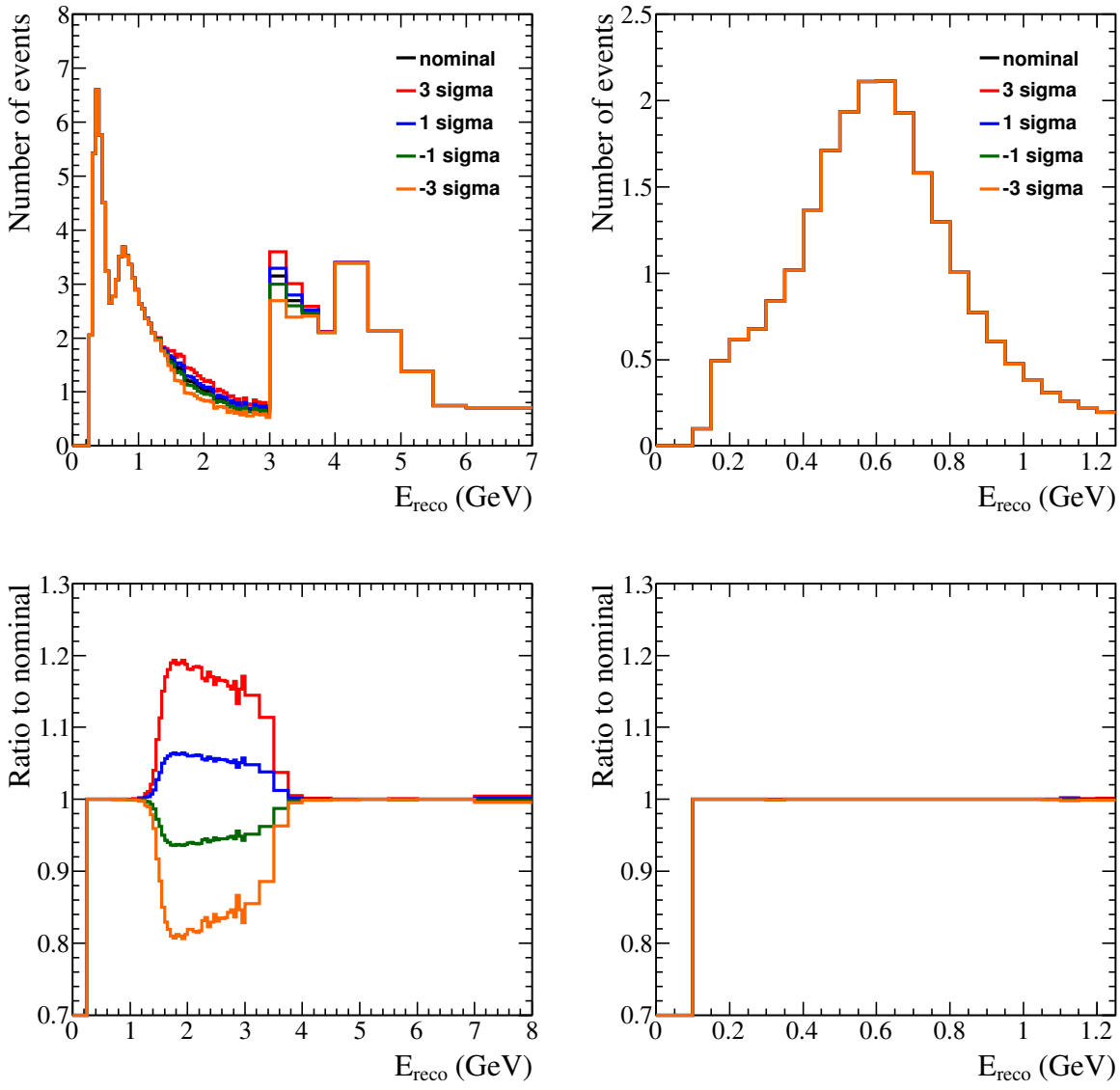


Figure 29: Effect of BANFF; Norm; CC QE; 1.5 - 3.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.4**	**0.5353**	**124.1**	**-0.5353**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	116.9	0.5361	115.6	-0.5361	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0.03284	0.1656	-0.03284	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.848	0.5724	7.759	-0.5724	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.262	0.02791	0.2618	-0.02791	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.02423	0.1025	0.02418	-0.1025	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0.01587	0.003175	-0.01587	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0.001061	0.2742	-0.001061	17.49	17.49	0	17.49	0

Table 59: Effect of BANFF; Norm; CC QE; 3.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.4**	**0.5353**	**124.1**	**-0.5353**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	71.42	0.8802	70.17	-0.8802	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1306	0.04168	0.1305	-0.04168	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.664	0.9671	4.575	-0.9671	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03514	0.2085	0.035	-0.2085	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002051	1.225	0.002001	-1.225	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002172	0.02321	0.002171	-0.02321	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0.001494	0.1948	-0.001494	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 60: Effect of BANFF; Norm; CC QE; 3.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

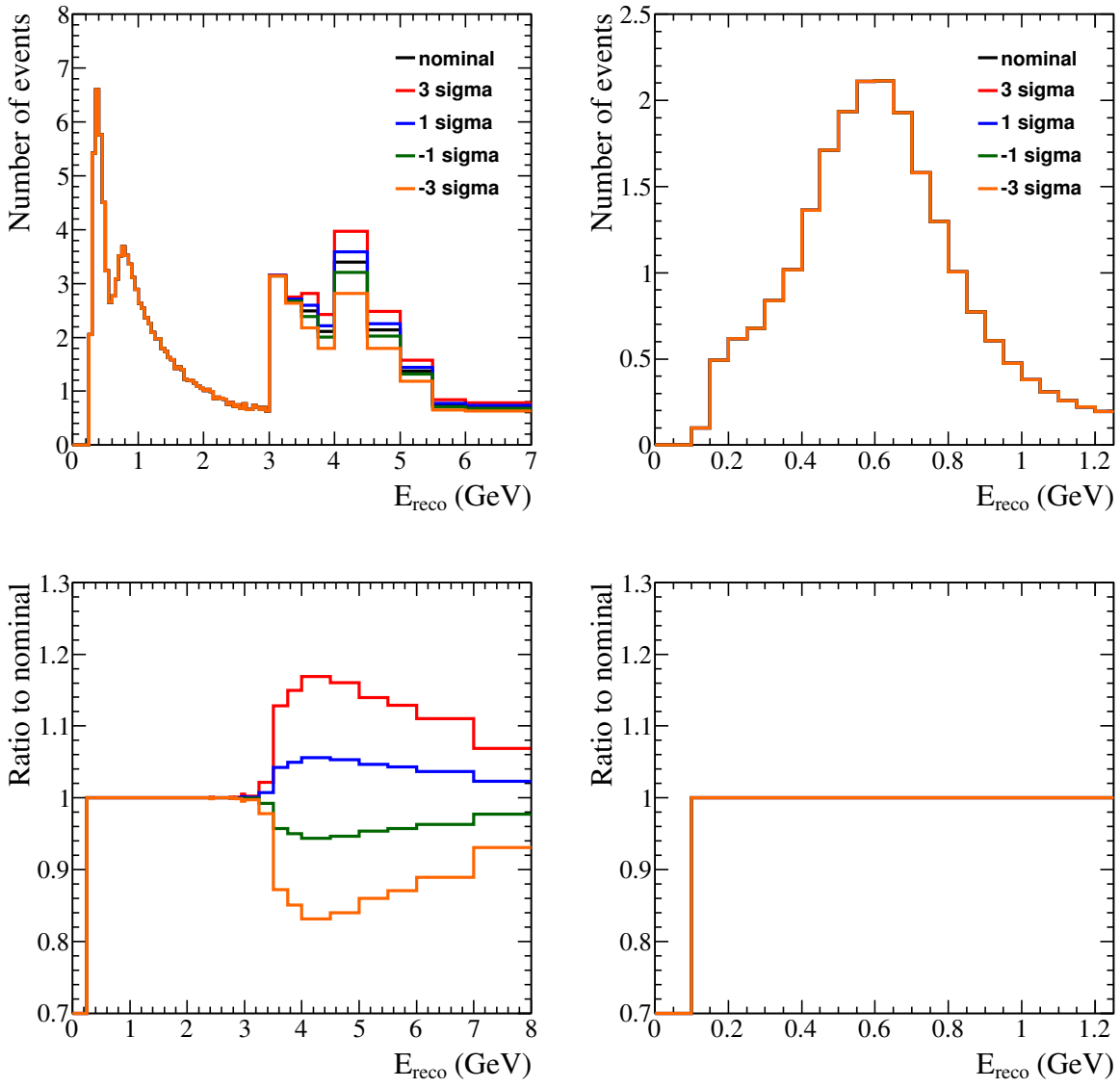


Figure 30: Effect of BANFF; Norm; CC QE; 3.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**127.3**	**1.988**	**122.3**	**-1.988**	**22**	**22.45**	**2.037**	**21.55**	**-2.037**
ν_μ	116.2	118.5	1.974	113.9	-1.974	0.9435	0.9457	0.2334	0.9413	-0.2334
$Osc.\nu_\mu$	0.1657	0.1694	2.265	0.1619	-2.265	0.0001767	0.0001814	2.665	0.000172	-2.665
$\bar{\nu_\mu}$	7.804	7.974	2.182	7.633	-2.182	0.05456	0.05459	0.0587	0.05453	-0.0587
ν_e	0.2619	0.2644	0.9551	0.2594	-0.9551	3.127	3.224	3.116	3.03	-3.116
$\bar{\nu_e}$	0.0242	0.02432	0.4706	0.02409	-0.4706	0.1523	0.1576	3.477	0.147	-3.477
Osc.$\bar{\nu_e}$	0.003176	0.003283	3.362	0.003069	-3.362	0.2339	0.2391	2.236	0.2286	-2.236
$Osc.\nu_e$	0.2742	0.2836	3.405	0.2649	-3.405	17.49	17.83	1.932	17.15	-1.932

Table 61: Effect of BANFF; Norm; CC 1pi; 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**127.3**	**1.988**	**122.3**	**-1.988**	**22**	**22.45**	**2.037**	**21.55**	**-2.037**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	32.07	7.707	27.48	-7.707	0.01709	0.01929	12.89	0.01488	-12.89
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03531	11.89	0.0278	-11.89	3.655e-05	4.126e-05	12.89	3.184e-05	-12.89
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.303	7.986	1.962	-7.986	0.0002486	0.0002806	12.89	0.0002165	-12.89
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02525	10.99	0.02025	-10.99	0.7705	0.8679	12.65	0.673	-12.65
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.00141	8.786	0.001183	-8.786	0.04178	0.04708	12.67	0.03649	-12.67
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0009571	12.56	0.0007435	-12.56	0.04064	0.04587	12.87	0.03541	-12.87
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.082	12.85	0.06333	-12.85	2.624	2.962	12.88	2.286	-12.88
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 62: Effect of BANFF; Norm; CC 1pi; 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

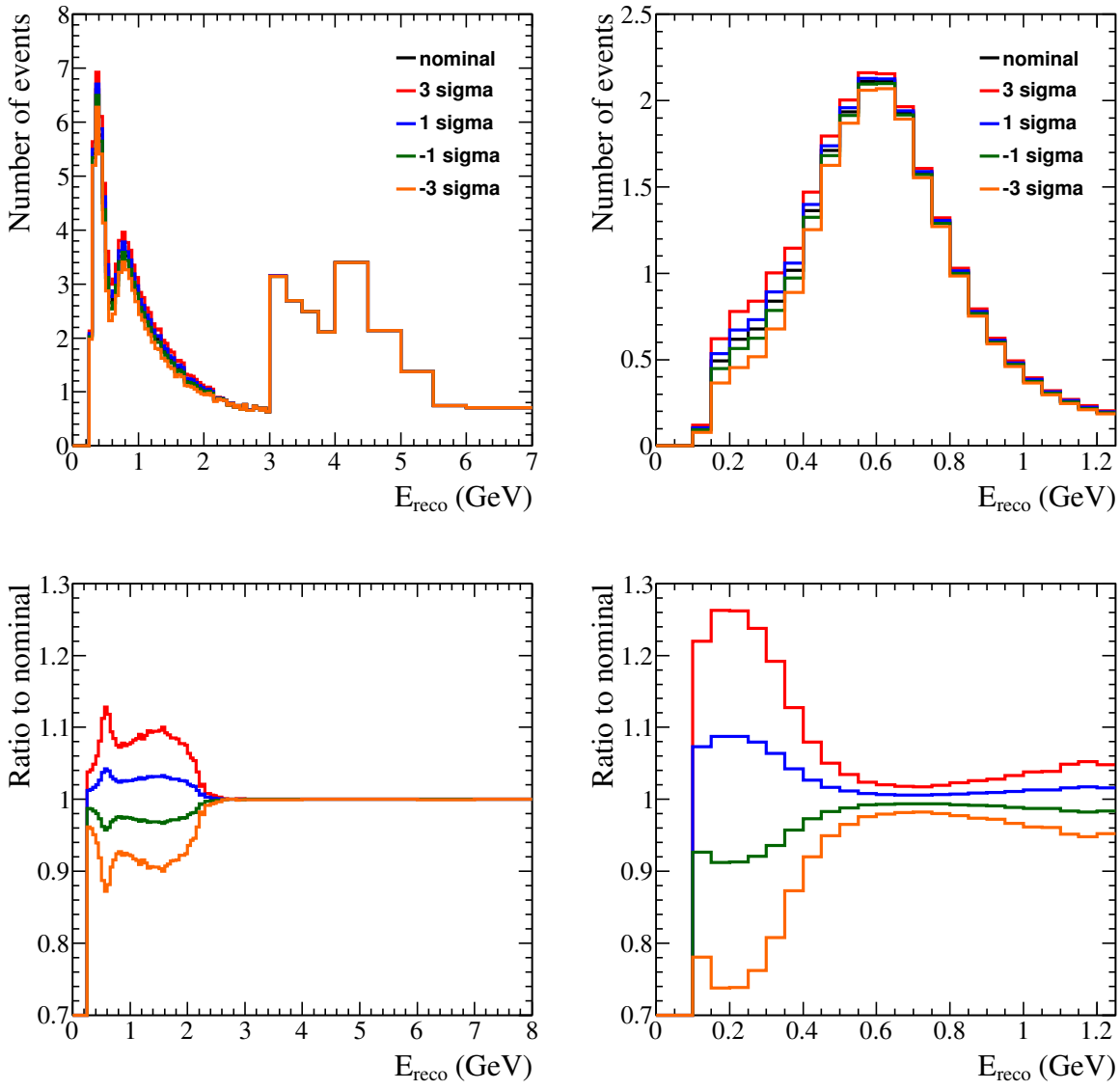


Figure 31: Effect of BANFF; Norm; CC 1pi; 0.0 - 2.5 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.7**	**1.568**	**122.8**	**-1.568**	**22**	**22**	**0.01153**	**22**	**-0.01153**
ν_μ	116.2	118.1	1.576	114.4	-1.576	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1661	0.2247	0.1653	-0.2247	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.928	1.59	7.68	-1.59	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2624	0.1952	0.2614	-0.1952	3.127	3.129	0.07012	3.125	-0.07012
$\bar{\nu_e}$	0.0242	0.02427	0.2608	0.02414	-0.2608	0.1523	0.1524	0.06953	0.1522	-0.06953
Osc.$\bar{\nu_e}$	0.003176	0.003179	0.1041	0.003173	-0.1041	0.2339	0.2339	0.00392	0.2339	-0.00392
$Osc.\nu_e$	0.2742	0.2743	0.01171	0.2742	-0.01171	17.49	17.49	0.001304	17.49	-0.001304

Table 63: Effect of BANFF; Norm; CC 1pi; 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.7**	**1.568**	**122.8**	**-1.568**	**22**	**22**	**0.01153**	**22**	**-0.01153**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	31.61	6.151	27.94	-6.151	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03193	1.18	0.03118	-1.18	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.257	5.82	2.008	-5.82	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02326	2.246	0.02224	-2.246	0.7705	0.7726	0.2846	0.7683	-0.2846
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.00136	4.869	0.001233	-4.869	0.04178	0.04189	0.2534	0.04168	-0.2534
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008536	0.3888	0.000847	-0.3888	0.04064	0.04065	0.02256	0.04063	-0.02256
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07269	0.04421	0.07263	-0.04421	2.624	2.625	0.008692	2.624	-0.008692
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 64: Effect of BANFF; Norm; CC 1pi; 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

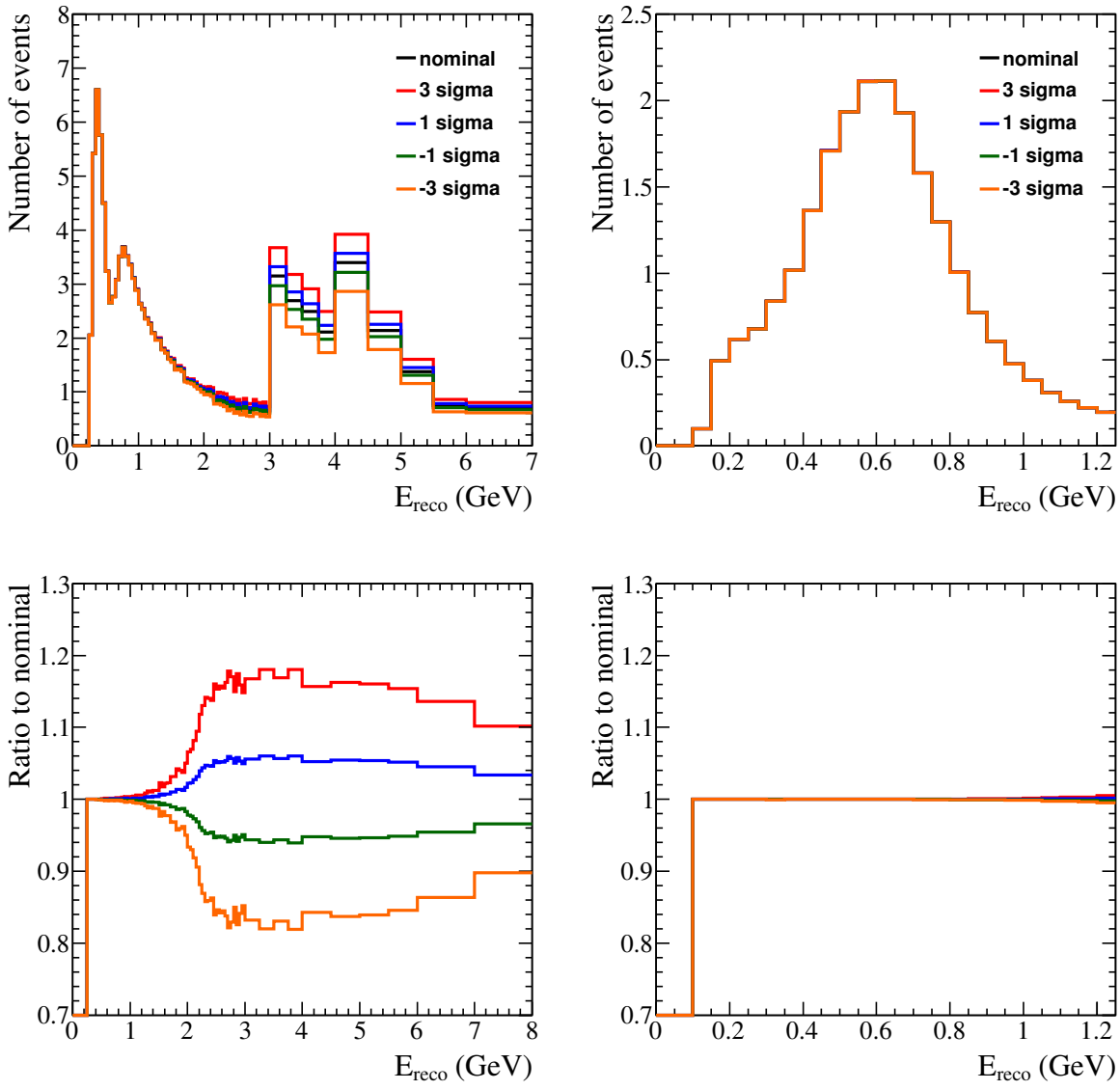


Figure 32: Effect of BANFF; Norm; CC 1pi; 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.09**	**0.4211**	**21.91**	**-0.4211**
ν_μ	116.2	116.2	0	116.2	0	0.9435	1.029	9.084	0.8578	-9.084
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05891	7.967	0.05021	-7.967
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.129	0.07477	3.125	-0.07477
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1526	0.1709	0.152	-0.1709
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 65: Effect of BANFF; Norm; NC 1pi0; 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.09**	**0.4211**	**21.91**	**-0.4211**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.4781	21.84	0.3067	-21.84
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.02425	21.84	0.01555	-21.84
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.01304	21.84	0.008367	-21.84
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001452	21.84	0.0009311	-21.84
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 66: Effect of BANFF; Norm; NC 1pi0; 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

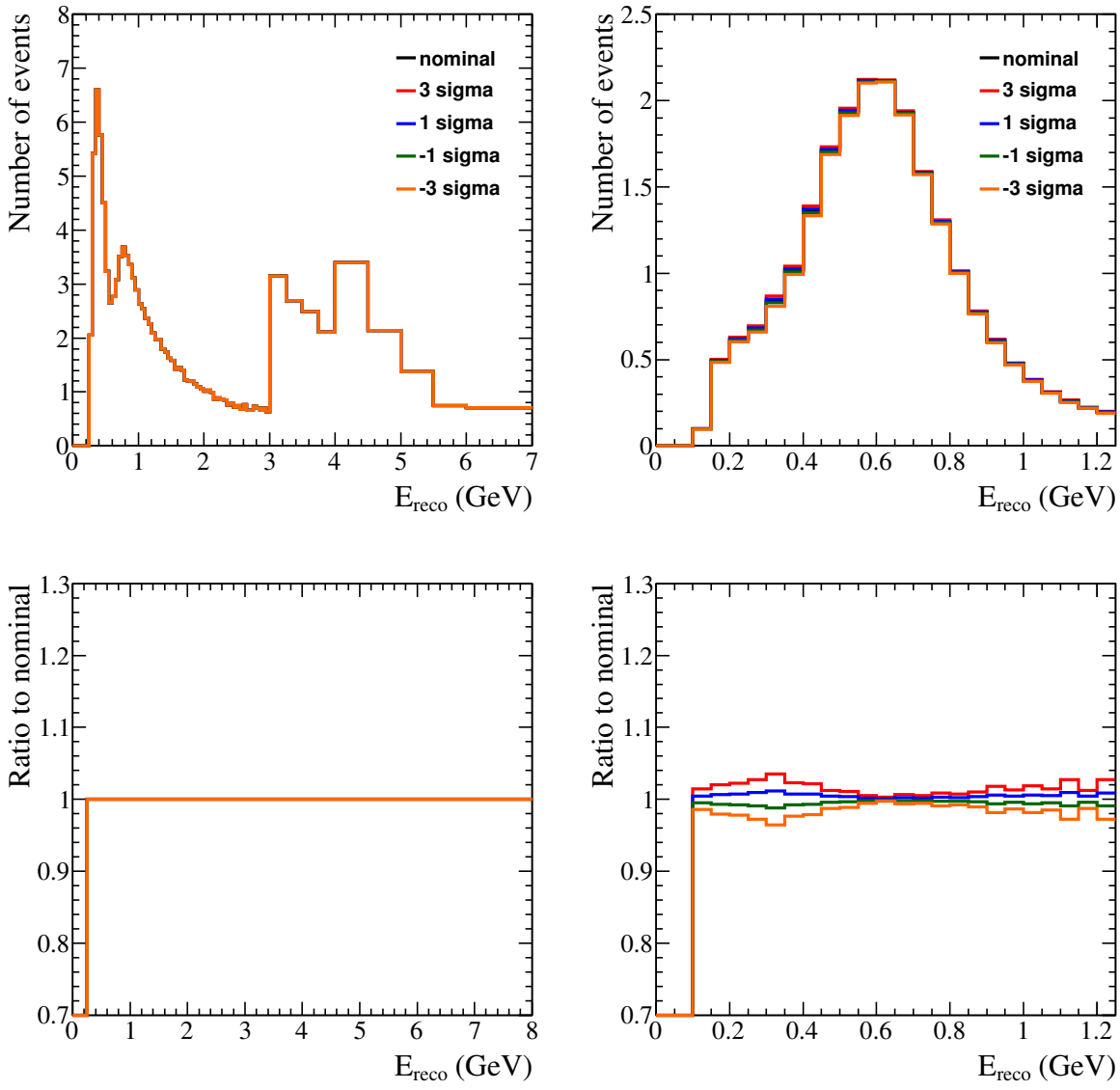


Figure 33: Effect of BANFF; Norm; NC 1pi0; 2.5 - 30.0 GeV with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125**	**0.1567**	**124.7**	**-0.02681**	**22**	**22.02**	**0.0917**	**21.98**	**-0.0945**
ν_μ	116.2	116.4	0.1661	116.2	-0.02698	0.9435	0.9429	-0.06186	0.9441	0.06973
$Osc.\nu_\mu$	0.1657	0.1663	0	0.1653	0	0.0001767	0.0001774	0	0.0001761	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2622	0.124	0.2616	-0.1072	3.127	3.125	-0.05823	3.13	0.1108
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2757	0.5288	0.2728	-0.5189	17.49	17.51	0.1291	17.47	-0.1424

Table 67: Effect of Fermi Momentum with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125**	**0.1567**	**124.7**	**-0.02681**	**22**	**22.02**	**0.0917**	**21.98**	**-0.0945**
ν_μ CCQE	70.79	70.99	0.2728	70.76	-0.0443	0.04778	0.04719	-1.222	0.04843	1.377
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1311	0.481	0.1301	-0.2932	0.0001375	0.0001382	0.51	0.0001369	-0.4486
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.0354	0.9262	0.03479	-0.8004	2.264	2.262	-0.08041	2.268	0.1531
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1962	0.7447	0.1933	-0.7307	14.79	14.81	0.1527	14.77	-0.1685
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 68: Effect of Fermi Momentum with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

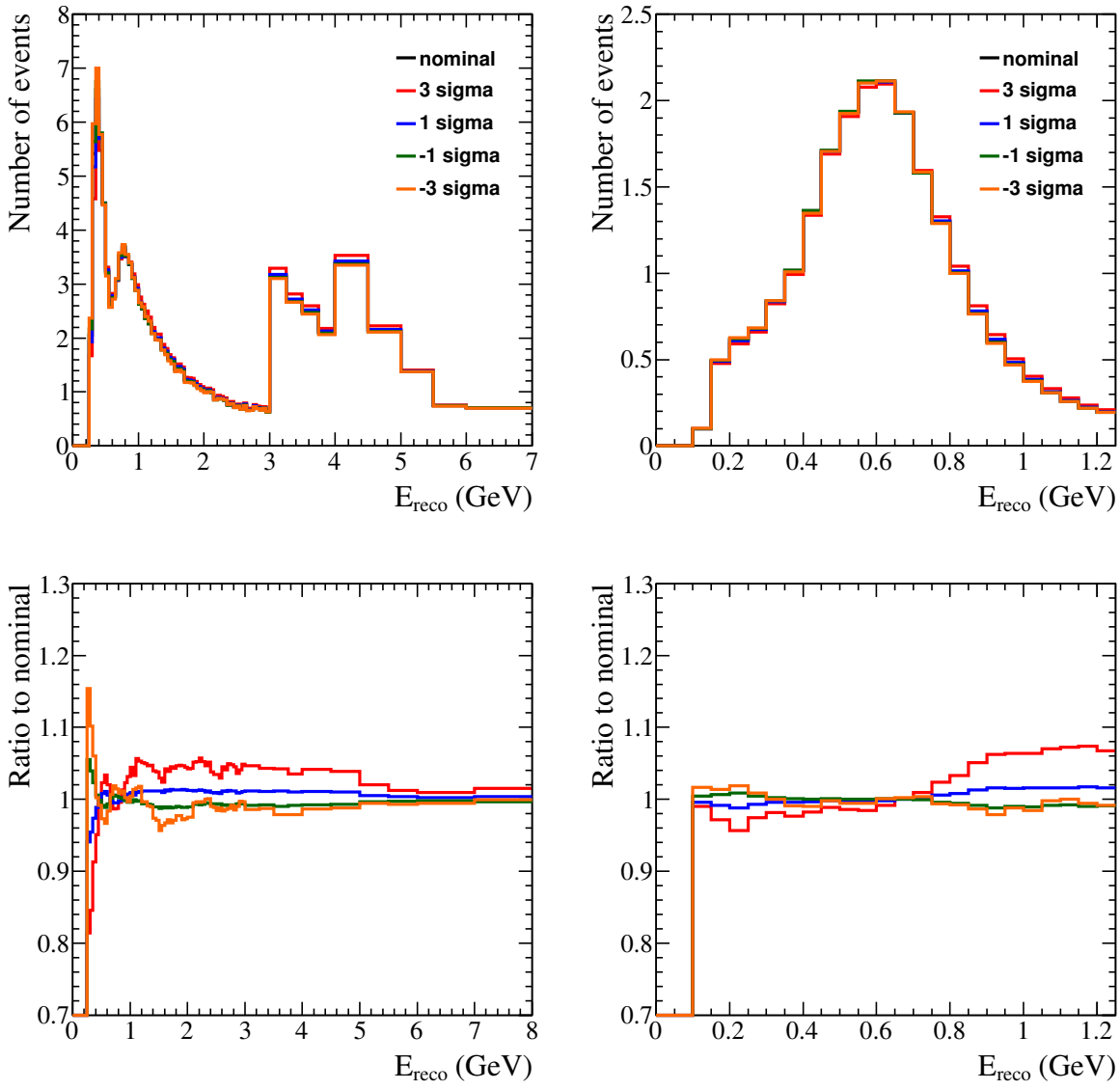


Figure 34: Effect of Fermi Momentum with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**123.7**	**-0.8974**	**125.9**	**0.8974**	**22**	**21.16**	**-3.821**	**22.84**	**3.821**
ν_μ	116.2	115.1	-0.9388	117.3	0.9388	0.9435	0.9399	-0.3754	0.947	0.3754
$Osc.\nu_\mu$	0.1657	0.1621	0	0.1693	0	0.0001767	0.0001659	0	0.0001876	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2598	-0.8047	0.264	0.8047	3.127	2.989	-4.399	3.265	4.399
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2515	-8.289	0.297	8.289	17.49	16.79	-4	18.19	4

Table 69: Effect of Spectral Func with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**123.7**	**-0.8974**	**125.9**	**0.8974**	**22**	**21.16**	**-3.821**	**22.84**	**3.821**
ν_μ CCQE	70.79	69.7	-1.541	71.88	1.541	0.04778	0.04423	-7.413	0.05132	7.413
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1269	-2.746	0.1341	2.746	0.0001375	0.0001267	-7.874	0.0001483	7.874
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03296	-6.01	0.03718	6.01	2.264	2.127	-6.076	2.402	6.076
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.172	-11.67	0.2175	11.67	14.79	14.09	-4.73	15.49	4.73
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 70: Effect of Spectral Func with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

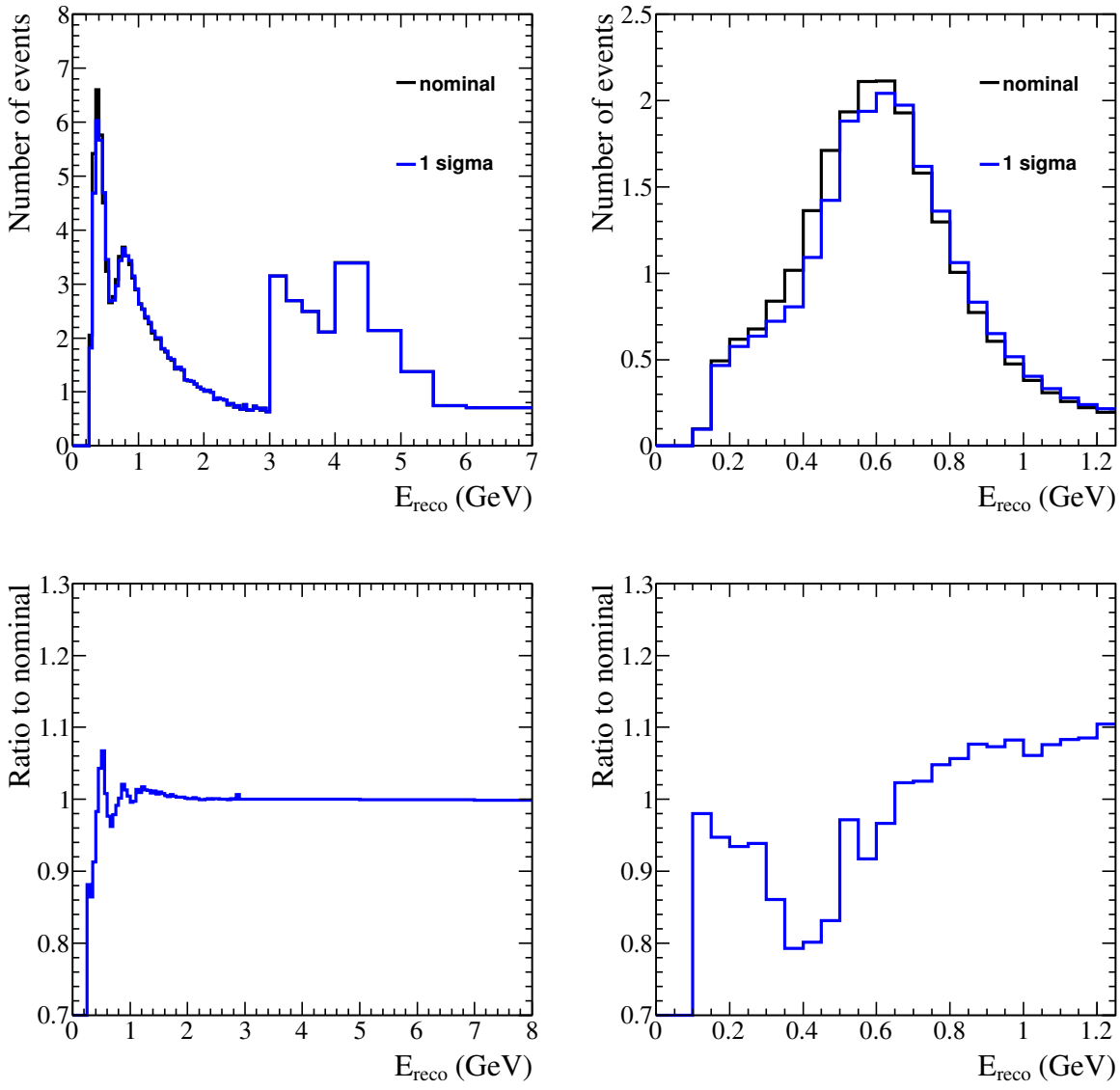


Figure 35: Effect of Spectral Func with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2438**	**124.4**	**-0.2759**	**22**	**22.04**	**0.1956**	**21.94**	**-0.2578**
ν_μ	116.2	116.5	0.224	115.9	-0.2511	0.9435	0.9807	3.946	0.8941	-5.23
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.83	0.3399	7.773	-0.3991	0.05456	0.0579	6.117	0.0502	-7.986
ν_e	0.2619	0.2771	5.81	0.2431	-7.188	3.127	3.129	0.07144	3.124	-0.08697
$\bar{\nu_e}$	0.0242	0.02619	8.19	0.02178	-10	0.1523	0.1525	0.1542	0.152	-0.1976
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 71: Effect of W-shape with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2438**	**124.4**	**-0.2759**	**22**	**22.04**	**0.1956**	**21.94**	**-0.2578**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	3.103	5.742	2.752	-6.241		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.4163	6.089	0.3584	-8.669
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.321	2.846	3.12	-3.366	0.3314	0.3447	4.025	0.316	-4.626
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1622	13.07	0.1222	-14.82		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.02172	9.15	0.01751	-12.03
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1829	4.437	0.1652	-5.643	0.01752	0.01903	8.658	0.01555	-11.21
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.08574	14.14	0.06211	-17.31		-		–	
ν_e NC 1π^0		-		–		0.0107	0.01208	12.85	0.008992	-16.01
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1259	3.792	0.1154	-4.801	0.0101	0.01096	8.499	0.009099	-9.958
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.009662	17.3	0.006501	-21.07		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.00135	13.33	0.0009735	-18.29
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.01266	4.606	0.01141	-5.672	0.0009675	0.001044	7.86	0.0008845	-8.586
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 72: Effect of W-shape with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

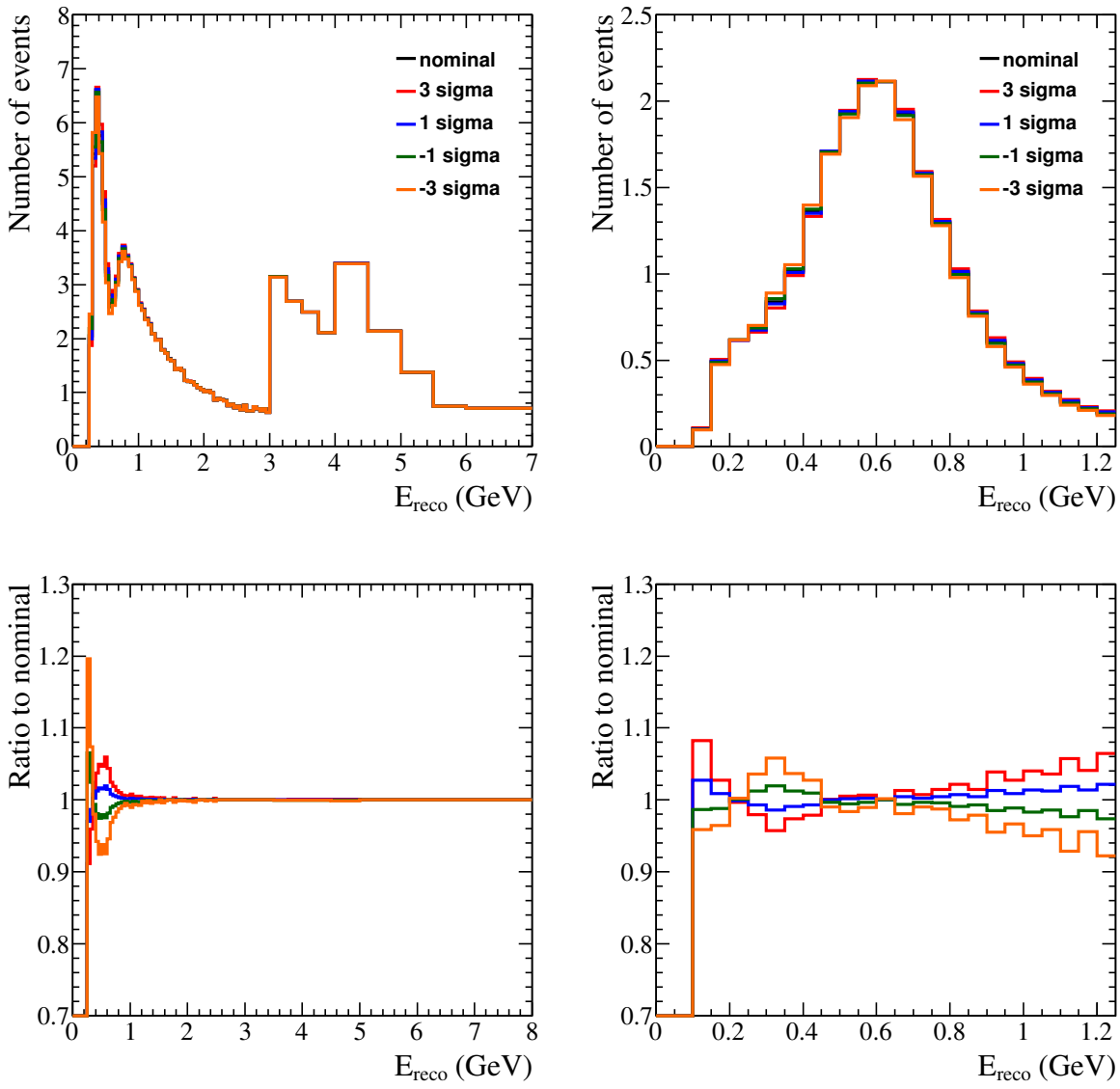


Figure 36: Effect of W-shape with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.9**	**0.8889**	**123.7**	**-0.8889**	**22**	**22.03**	**0.1229**	**21.97**	**-0.1229**
ν_μ	116.2	117.3	0.8981	115.2	-0.8981	0.9435	0.9438	0.03743	0.9431	-0.03743
$Osc.\nu_\mu$	0.1657	0.1662	0	0.1652	0	0.0001767	0.0001774	0	0.000176	0
$\bar{\nu_\mu}$	7.804	7.867	0.8084	7.741	-0.8084	0.05456	0.0546	0.06368	0.05453	-0.06368
ν_e	0.2619	0.263	0.4013	0.2609	-0.4013	3.127	3.142	0.4645	3.113	-0.4645
$\bar{\nu_e}$	0.0242	0.02425	0.206	0.02415	-0.206	0.1523	0.1532	0.5849	0.1514	-0.5849
Osc.$\bar{\nu_e}$	0.003176	0.003185	0	0.003167	0	0.2339	0.2343	0	0.2335	0
$Osc.\nu_e$	0.2742	0.2747	0.174	0.2738	-0.174	17.49	17.5	0.06188	17.48	-0.06188

Table 73: Effect of Shape; CC Oth with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.9**	**0.8889**	**123.7**	**-0.8889**	**22**	**22.03**	**0.1229**	**21.97**	**-0.1229**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	9.641	12.14	7.553	-12.14	0.0009536	0.001307	37.03	0.0006004	-37.03
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.003219	19.91	0.00215	-19.91	1.847e-06	2.55e-06	38.01	1.145e-06	-38.01
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.5486	12.99	0.4224	-13	0.000124	0.0001587	28.03	8.922e-05	-28.03
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.007671	15.88	0.005569	-15.88	0.0581	0.07262	25	0.04357	-25
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004645	12.03	0.0003647	-12.03	0.003647	0.004538	24.42	0.002756	-24.42
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.941e-05	21.16	3.215e-05	-21.16	0.001548	0.001965	26.92	0.001131	-26.92
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.00207	29.97	0.001115	-29.97	0.03175	0.04258	34.08	0.02093	-34.08

Table 74: Effect of Shape; CC Oth with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

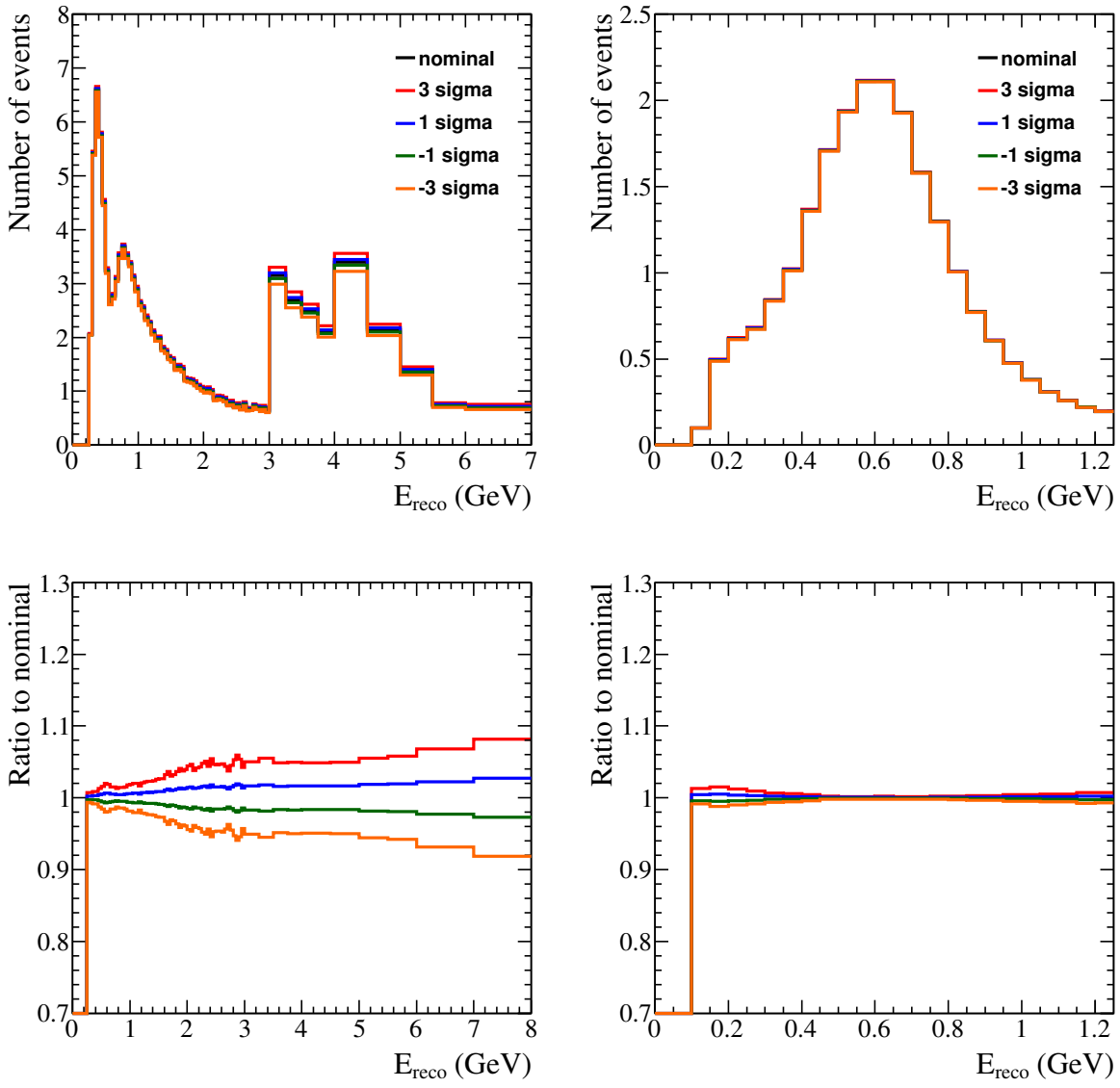


Figure 37: Effect of Shape; CC Oth with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**131.1**	**5.068**	**118.4**	**-5.068**	**22**	**22.81**	**3.675**	**21.19**	**-3.675**
ν_μ	116.2	122.4	5.264	110.1	-5.264	0.9435	0.8278	-12.27	1.059	12.27
$Osc.\nu_\mu$	0.1657	0.1728	0	0.1586	0	0.0001767	0.0001883	0	0.0001652	0
$\bar{\nu_\mu}$	7.804	8.018	2.745	7.59	-2.745	0.05456	0.04799	-12.04	0.06113	12.04
ν_e	0.2619	0.2442	-6.76	0.2796	6.76	3.127	3.388	8.349	2.866	-8.349
$\bar{\nu_e}$	0.0242	0.02204	-8.924	0.02636	8.924	0.1523	0.1617	6.185	0.1429	-6.185
Osc.$\bar{\nu_e}$	0.003176	0.003266	0	0.003086	0	0.2339	0.2434	0	0.2243	0
$Osc.\nu_e$	0.2742	0.277	1.025	0.2714	-1.025	17.49	18.14	3.72	16.84	-3.72

Table 75: Effect of Pi-less delta decay with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**131.1**	**5.068**	**118.4**	**-5.068**	**22**	**22.81**	**3.675**	**21.19**	**-3.675**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	36.63	23.01	22.93	-23.01	0.01709	0.02423	41.82	0.00994	-41.82
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.385	-18.74	3.485	18.74		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.2943	-25	0.4905	25
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.048	-5.611	3.41	5.611	0.3314	0.3066	-7.474	0.3561	7.474
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.0387	22.64	0.02441	-22.64	3.655e-05	4.807e-05	31.53	2.502e-05	-31.53
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.384	11.79	1.881	-11.79	0.0002486	0.0001888	-24.02	0.0003083	24.02
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1156	-19.42	0.1713	19.42		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.01493	-25	0.02488	25
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1657	-5.359	0.1845	5.359	0.01752	0.01598	-8.751	0.01905	8.751
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02566	12.78	0.01984	-12.78	0.7705	1.035	34.33	0.5059	-34.33
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.06054	-19.4	0.08969	19.4		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.008029	-25	0.01338	25
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1152	-4.981	0.1273	4.981	0.0101	0.009359	-7.384	0.01085	7.384
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001381	6.522	0.001212	-6.522	0.04178	0.05157	23.43	0.03199	-23.43
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.006636	-19.44	0.009838	19.44		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.0008935	-25	0.001489	25
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.01146	-5.318	0.01274	5.318	0.0009675	0.0008942	-7.579	0.001041	7.579
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0009404	10.59	0.0007602	-10.59	0.04064	0.05017	23.45	0.03111	-23.45
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07547	3.868	0.06985	-3.868	2.624	3.275	24.79	1.974	-24.79
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 76: Effect of Pi-less delta decay with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

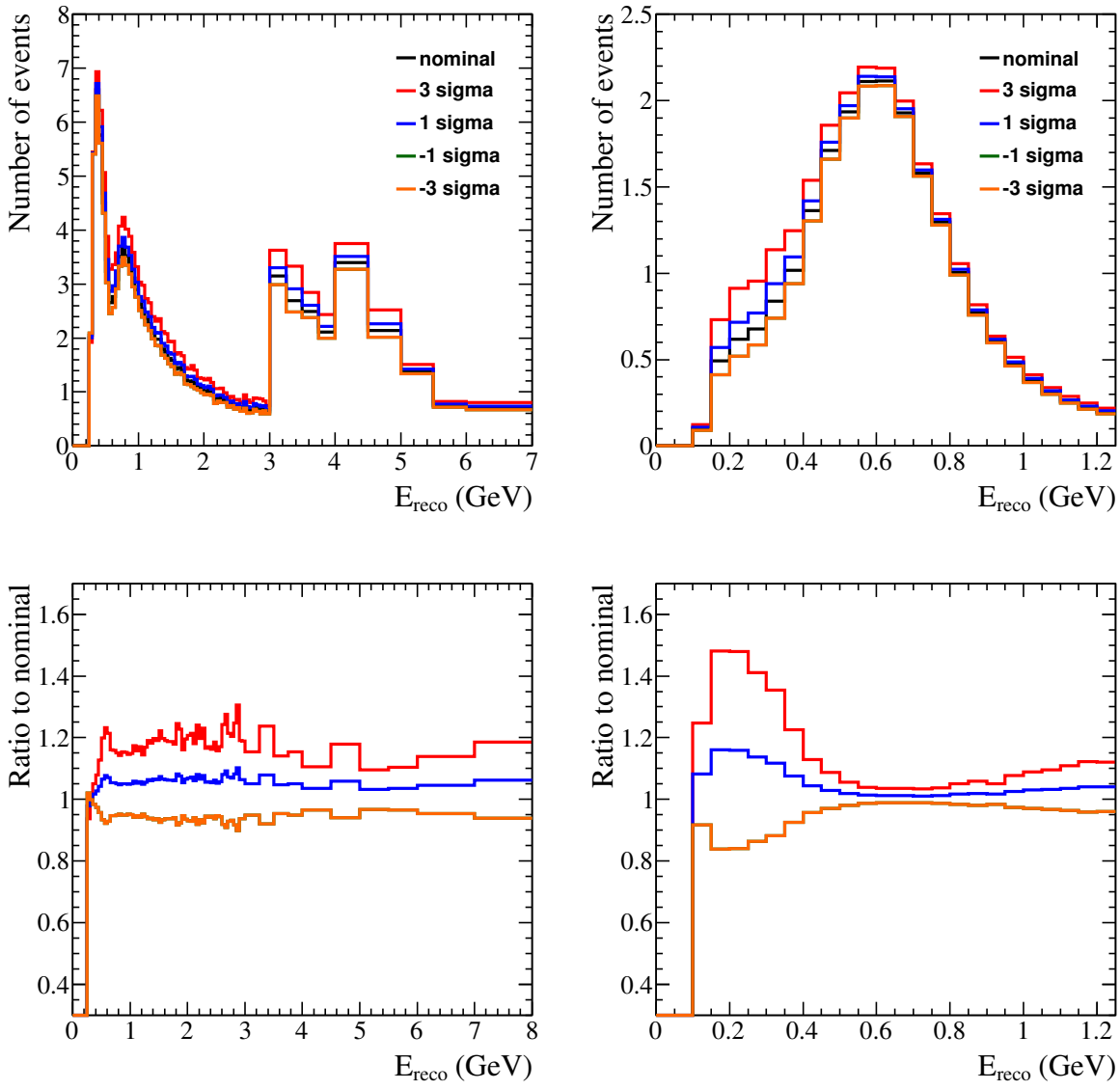


Figure 38: Effect of Pi-less delta decay with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.7**	**-0.05983**	**125.1**	**0.2441**	**22**	**21.83**	**-0.8003**	**22.15**	**0.6883**
ν_μ	116.2	116.2	-0.06074	116.5	0.2589	0.9435	0.9442	0.07883	0.943	-0.05343
$Osc.\nu_\mu$	0.1657	0.1629	0	0.1685	0	0.0001767	0.0001765	0	0.000177	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2618	-0.04902	0.2621	0.06154	3.127	3.115	-0.3802	3.143	0.4976
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2731	-0.4222	0.2749	0.2575	17.49	17.32	-0.943	17.63	0.7798

Table 77: Effect of Binding energy with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.7**	**-0.05983**	**125.1**	**0.2441**	**22**	**21.83**	**-0.8003**	**22.15**	**0.6883**
ν_μ CCQE	70.79	70.72	-0.09972	71.09	0.4251	0.04778	0.04852	1.557	0.04727	-1.055
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1278	-2.118	0.1334	2.175	0.0001375	0.0001372	-0.2003	0.0001378	0.2262
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03494	-0.3661	0.03523	0.4596	2.264	2.252	-0.5251	2.28	0.6872
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1936	-0.5945	0.1955	0.3625	14.79	14.63	-1.115	14.93	0.9221
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 78: Effect of Binding energy with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

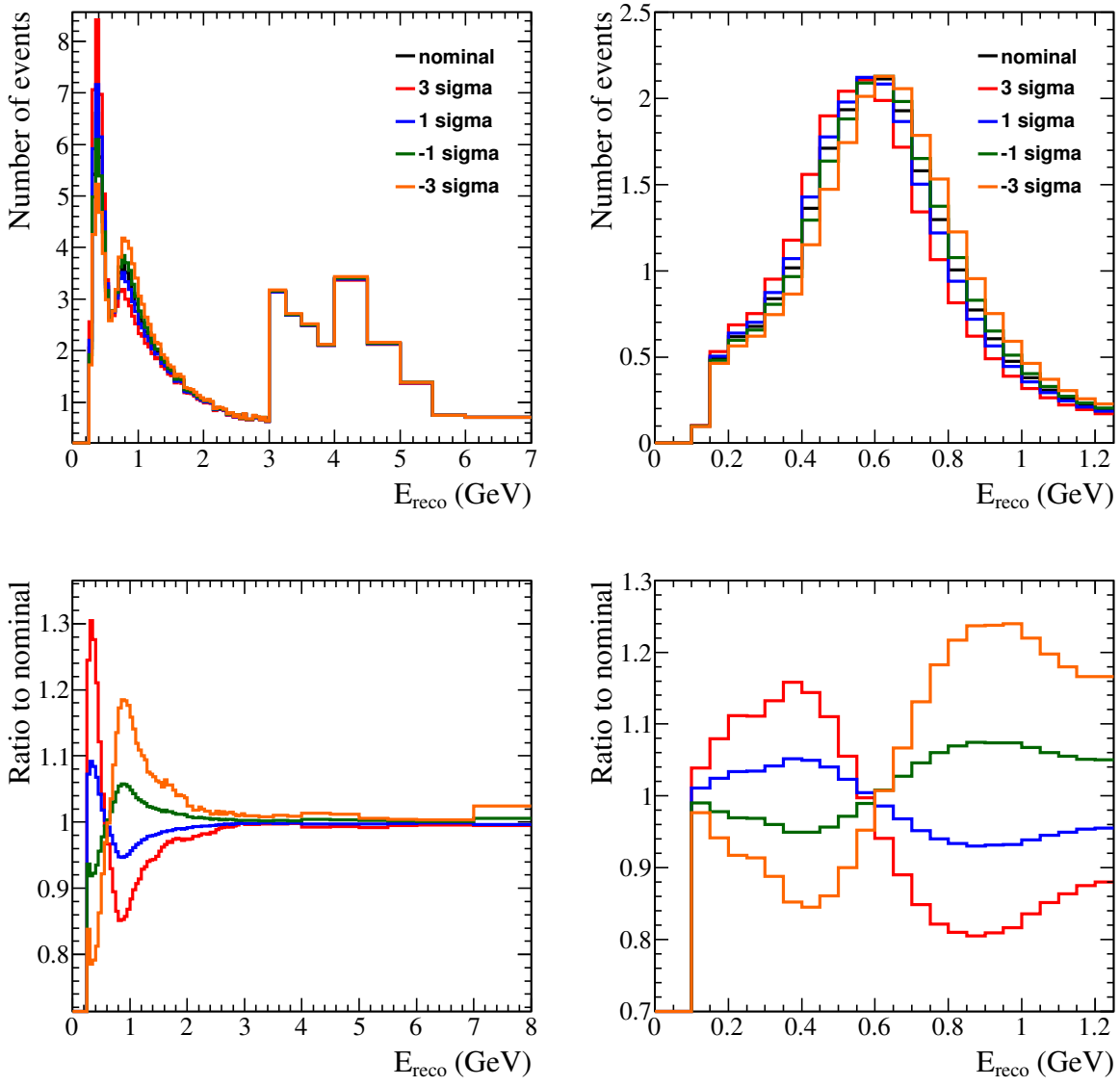


Figure 39: Effect of Binding energy with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.9**	**0.9321**	**123.6**	**-0.9321**	**22**	**22.07**	**0.3209**	**21.93**	**-0.3209**
ν_μ	116.2	117.1	0.7806	115.3	-0.7806	0.9435	0.9435	0.0004628	0.9435	-0.0004628
$Osc.\nu_\mu$	0.1657	0.1666	0	0.1648	0	0.0001767	0.0001775	0	0.0001759	0
$\bar{\nu_\mu}$	7.804	8.052	3.18	7.556	-3.18	0.05456	0.05457	0.007816	0.05456	-0.007816
ν_e	0.2619	0.263	0.4164	0.2608	-0.4164	3.127	3.136	0.2954	3.118	-0.2954
$\bar{\nu_e}$	0.0242	0.02434	0.5442	0.02407	-0.5442	0.1523	0.159	4.375	0.1456	-4.375
Osc.$\bar{\nu_e}$	0.003176	0.00329	0	0.003062	0	0.2339	0.2447	0	0.223	0
$Osc.\nu_e$	0.2742	0.2794	1.903	0.269	-1.903	17.49	17.53	0.2506	17.45	-0.2506

Table 79: Effect of Norm, CC Coh with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.9**	**0.9321**	**123.6**	**-0.9321**	**22**	**22.07**	**0.3209**	**21.93**	**-0.3209**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	1.815	100	0	-100	4.366e-06	8.733e-06	100	0	-100
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.001867	100	0	-100	8.165e-07	1.633e-06	100	0	-100
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.4963	100	0	-100	4.264e-06	8.529e-06	100	0	-100
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.002181	100	0	-100	0.009237	0.01847	100	0	-100
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0002635	100	0	-100	0.006662	0.01332	100	0	-100
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0002274	100	0	-100	0.01087	0.02175	100	0	-100
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.01044	100	0	-100	0.04383	0.08766	100	0	-100
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 80: Effect of Norm, CC Coh with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

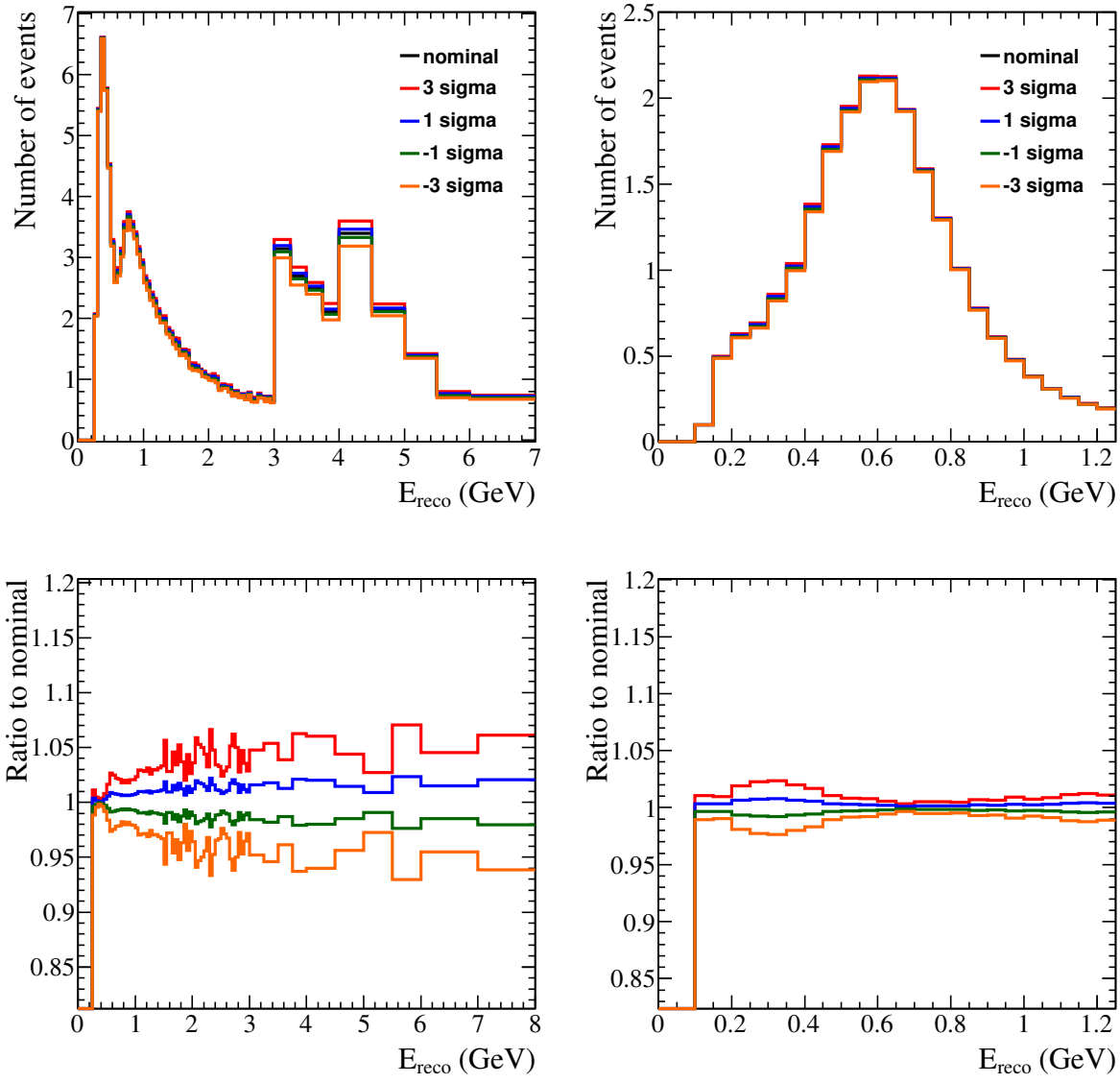


Figure 40: Effect of Norm, CC Coh with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.7**	**0.7602**	**123.8**	**-0.7602**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	117.1	0.7575	115.4	-0.7575	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.847	0.5514	7.761	-0.5514	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2845	8.604	0.2394	-8.604	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.02668	10.21	0.02173	-10.21	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 81: Effect of Norm, NC 1piC with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.7**	**0.7602**	**123.8**	**-0.7602**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	3.815	30	2.054	-30		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1865	30	0.1004	-30		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.09765	30	0.05258	-30		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.01071	30	0.005766	-30		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 82: Effect of Norm, NC 1piC with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

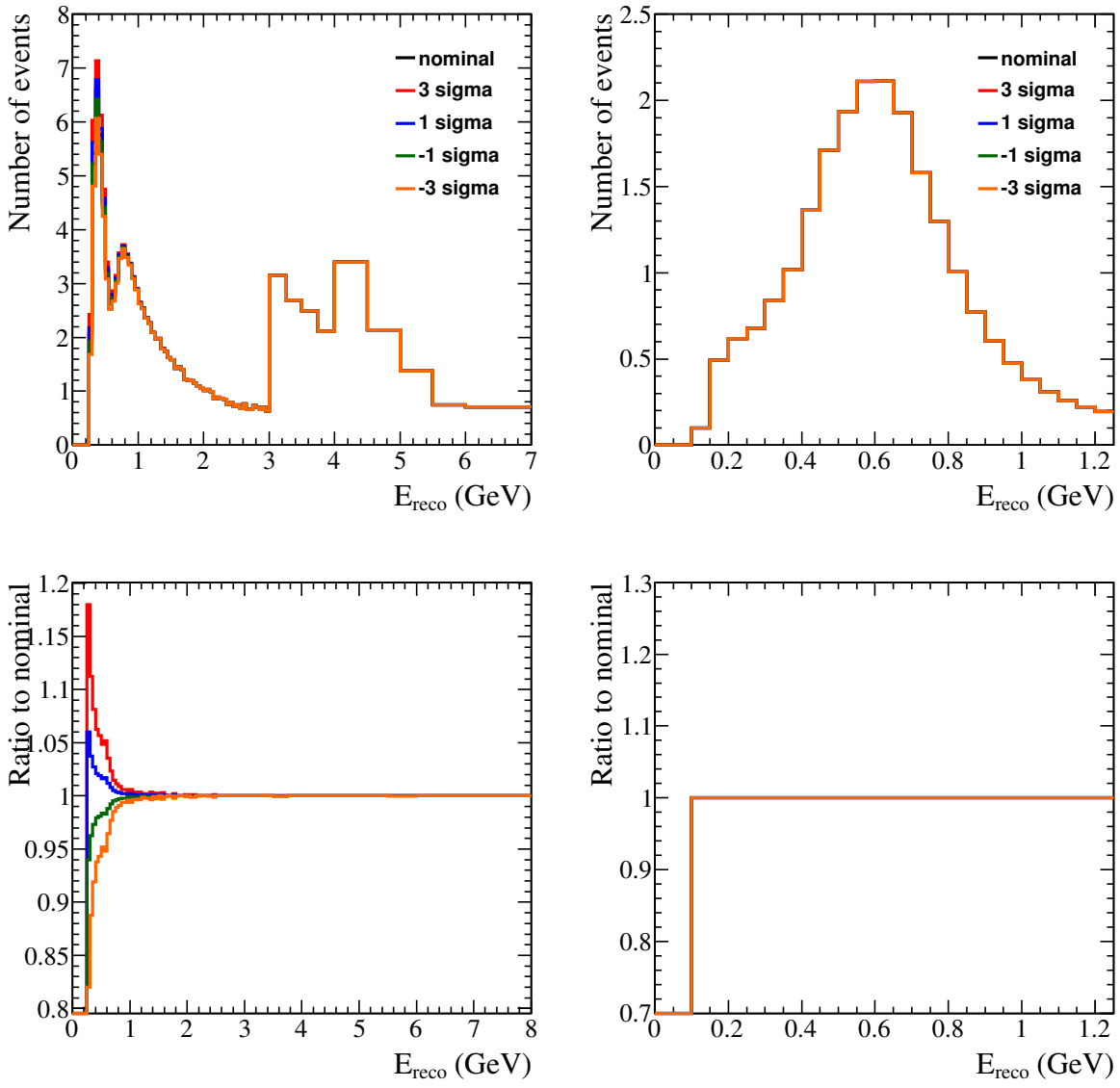


Figure 41: Effect of Norm, NC 1piC with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.05**	**0.2381**	**21.95**	**-0.2381**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9897	4.893	0.8973	-4.893
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05933	8.738	0.04979	-8.738
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.128	0.0395	3.126	-0.0395
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1525	0.1469	0.1521	-0.1469
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 83: Effect of Norm, NC Coh with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.05**	**0.2381**	**21.95**	**-0.2381**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.2001	30	0.1077	-30
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.02066	30	0.01112	-30
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.005353	30	0.002882	-30
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0009693	30	0.000522	-30
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 84: Effect of Norm, NC Coh with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

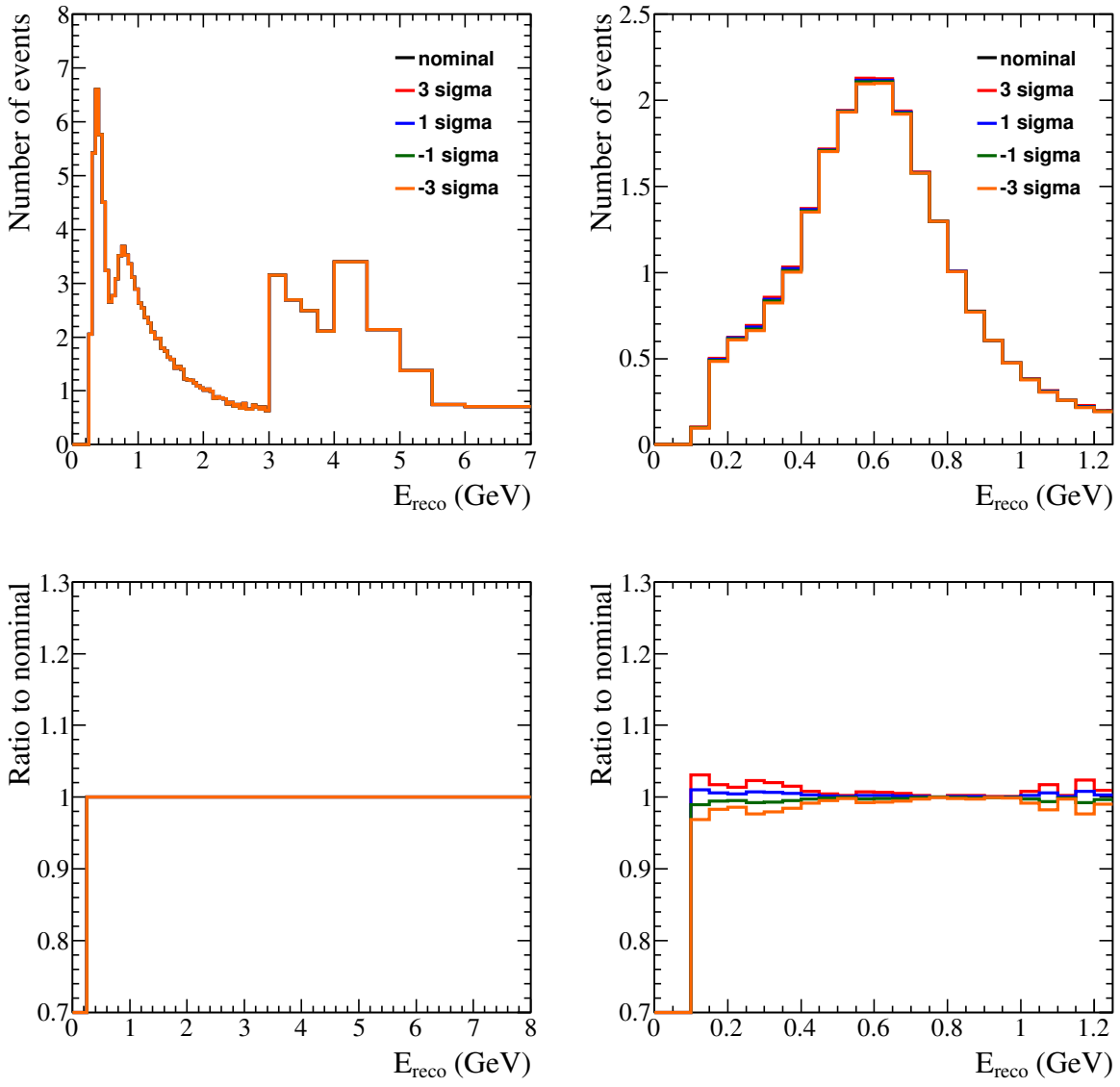


Figure 42: Effect of Norm, NC Coh with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.8**	**0.8505**	**123.7**	**-0.8505**	**22**	**22.11**	**0.4908**	**21.89**	**-0.4908**
ν_μ	116.2	117.2	0.8334	115.3	-0.8334	0.9435	1.043	10.54	0.8441	-10.54
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.856	0.6731	7.751	-0.6731	0.05456	0.05982	9.631	0.04931	-9.631
ν_e	0.2619	0.2983	13.89	0.2255	-13.89	3.127	3.13	0.09694	3.124	-0.09694
$\bar{\nu_e}$	0.0242	0.02783	15	0.02057	-15	0.1523	0.1526	0.1906	0.152	-0.1906
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 85: Effect of Norm, NC Oth with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.8**	**0.8505**	**123.7**	**-0.8505**	**22**	**22.11**	**0.4908**	**21.89**	**-0.4908**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	4.198	30	2.26	-30	0.3314	0.4308	30	0.232	-30
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.2276	30	0.1226	-30	0.01752	0.02277	30	0.01226	-30
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1577	30	0.08489	-30	0.0101	0.01314	30	0.007073	-30
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.01573	30	0.00847	-30	0.0009675	0.001258	30	0.0006773	-30
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 86: Effect of Norm, NC Oth with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

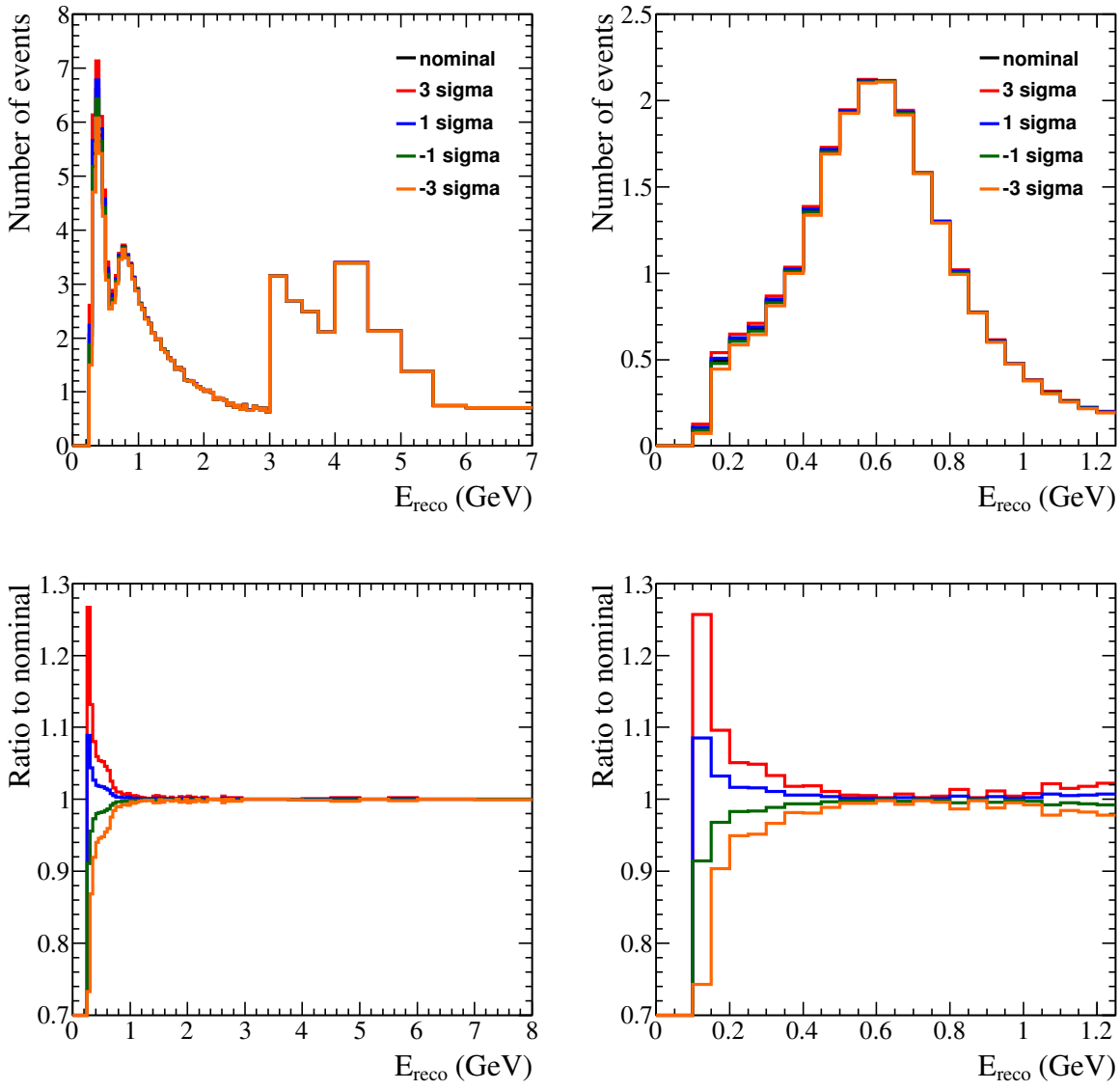


Figure 43: Effect of Norm, NC Oth with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.008339**	**124.8**	**-0.008339**	**22**	**22.63**	**2.86**	**21.37**	**-2.86**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2639	0.7506	0.26	-0.7506	3.127	3.22	2.976	3.034	-2.976
$\bar{\nu_e}$	0.0242	0.02432	0.4795	0.02409	-0.4795	0.1523	0.1568	2.943	0.1478	-2.943
Osc.$\bar{\nu_e}$	0.003176	0.003271	0	0.003081	0	0.2339	0.2409	0	0.2269	0
$Osc.\nu_e$	0.2742	0.2825	3	0.266	-3	17.49	18.01	3	16.97	-3

Table 87: Effect of Norm, ν_e To ν_μ with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0.008339**	**124.8**	**-0.008339**	**22**	**22.63**	**2.86**	**21.37**	**-2.86**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03612	3	0.03402	-3	2.264	2.332	3	2.196	-3
ν_e CC 1π	0.02275	0.02344	3	0.02207	-3	0.7705	0.7936	3	0.7473	-3
ν_e CC coherent	0.001091	0.001123	3	0.001058	-3	0.009237	0.009515	3	0.00896	-3
ν_e CC other	0.00662	0.006819	3	0.006422	-3	0.0581	0.05984	3	0.05635	-3
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002086	3	0.001965	-3	0.0973	0.1002	3	0.09438	-3
$\bar{\nu_e}$ CC 1π	0.001296	0.001335	3	0.001258	-3	0.04178	0.04303	3	0.04053	-3
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001357	3	0.0001278	-3	0.006662	0.006862	3	0.006462	-3
$\bar{\nu_e}$ CC other	0.0004146	0.000427	3	0.0004021	-3	0.003647	0.003756	3	0.003538	-3
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002236	3	0.002106	-3	0.1808	0.1862	3	0.1754	-3
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008758	3	0.0008248	-3	0.04064	0.04186	3	0.03942	-3
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001171	3	0.0001103	-3	0.01087	0.0112	3	0.01055	-3
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.201e-05	3	3.956e-05	-3	0.001548	0.001595	3	0.001502	-3
Osc. ν_e CCQE	0.1948	0.2006	3	0.1889	-3	14.79	15.23	3	14.35	-3
Osc. ν_e CC 1π	0.07266	0.07484	3	0.07048	-3	2.624	2.703	3	2.546	-3
Osc. ν_e CC coherent	0.005219	0.005376	3	0.005063	-3	0.04383	0.04514	3	0.04251	-3
Osc. ν_e CC other	0.001592	0.00164	3	0.001545	-3	0.03175	0.03271	3	0.0308	-3

Table 88: Effect of Norm, ν_e To ν_μ with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

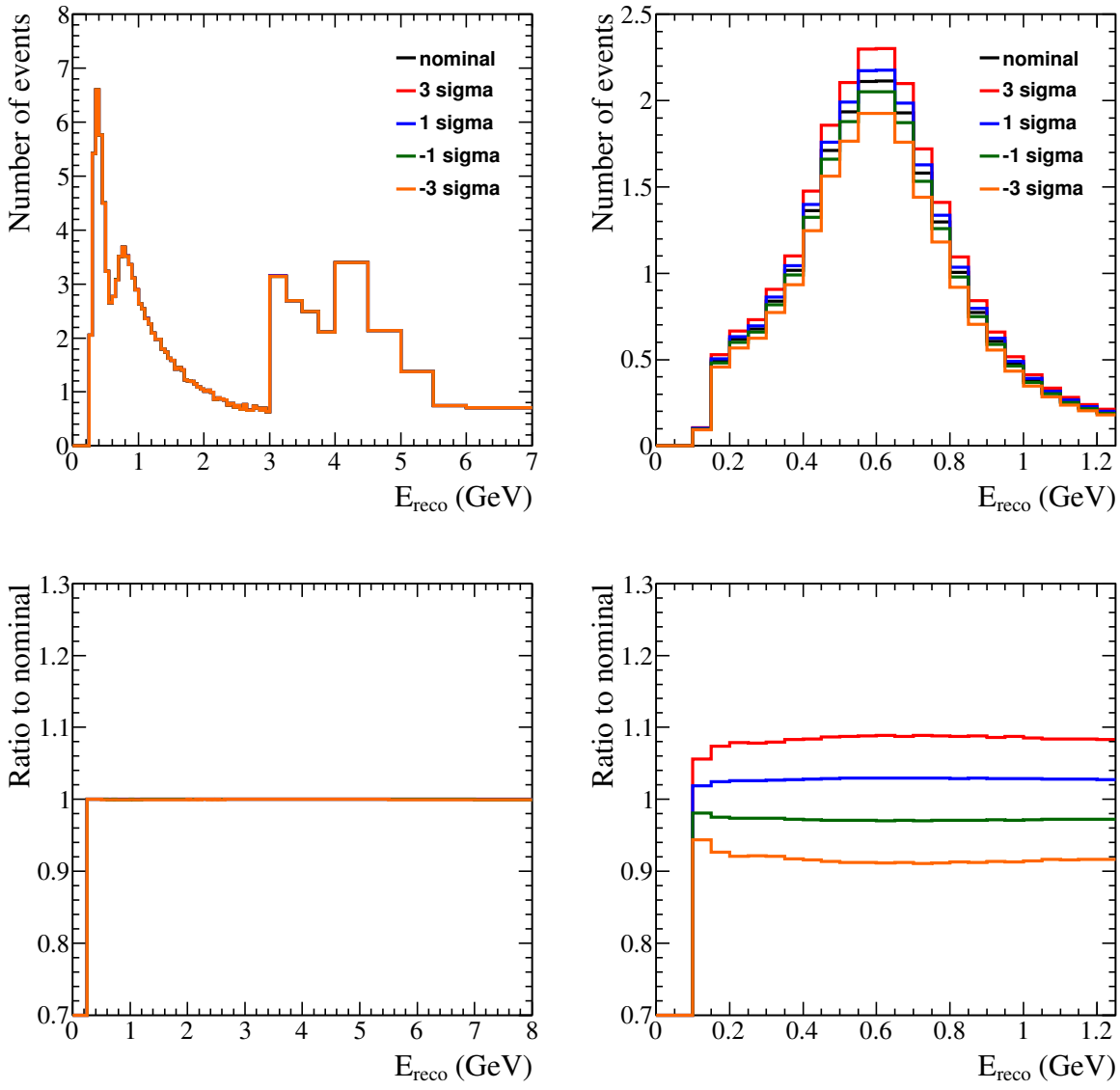


Figure 44: Effect of Norm, ν_e To ν_μ with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.3**	**1.201**	**123.3**	**-1.201**	**22**	**22.08**	**0.3495**	**21.92**	**-0.3495**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	9.301	19.18	6.307	-19.18	0.05456	0.05481	0.4589	0.05431	-0.4589
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.02498	3.197	0.02343	-3.197	0.1523	0.1822	19.62	0.1224	-19.62
Osc.$\bar{\nu_e}$	0.003176	0.003811	0	0.002541	0	0.2339	0.2806	0	0.1871	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 89: Effect of Norm, $\bar{\nu}$ To ν with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**126.3**	**1.201**	**123.3**	**-1.201**	**22**	**22.08**	**0.3495**	**21.92**	**-0.3495**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	5.543	20	3.695	-20	0.0008752	0.00105	20	0.0007002	-20
$\bar{\nu_\mu}$ CC 1π	2.132	2.559	20	1.706	-20	0.0002486	0.0002983	20	0.0001988	-20
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2978	20	0.1985	-20	4.264e-06	5.117e-06	20	3.412e-06	-20
$\bar{\nu_\mu}$ CC other	0.4855	0.5826	20	0.3884	-20	0.000124	0.0001488	20	9.917e-05	-20
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002431	20	0.001621	-20	0.0973	0.1168	20	0.07784	-20
$\bar{\nu_e}$ CC 1π	0.001296	0.001556	20	0.001037	-20	0.04178	0.05014	20	0.03342	-20
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001581	20	0.0001054	-20	0.006662	0.007995	20	0.00533	-20
$\bar{\nu_e}$ CC other	0.0004146	0.0004975	20	0.0003317	-20	0.003647	0.004376	20	0.002918	-20
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002605	20	0.001737	-20	0.1808	0.217	20	0.1446	-20
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.00102	20	0.0006802	-20	0.04064	0.04877	20	0.03251	-20
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001364	20	9.095e-05	-20	0.01087	0.01305	20	0.0087	-20
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.894e-05	20	3.263e-05	-20	0.001548	0.001858	20	0.001238	-20
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 90: Effect of Norm, $\bar{\nu}$ To ν with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

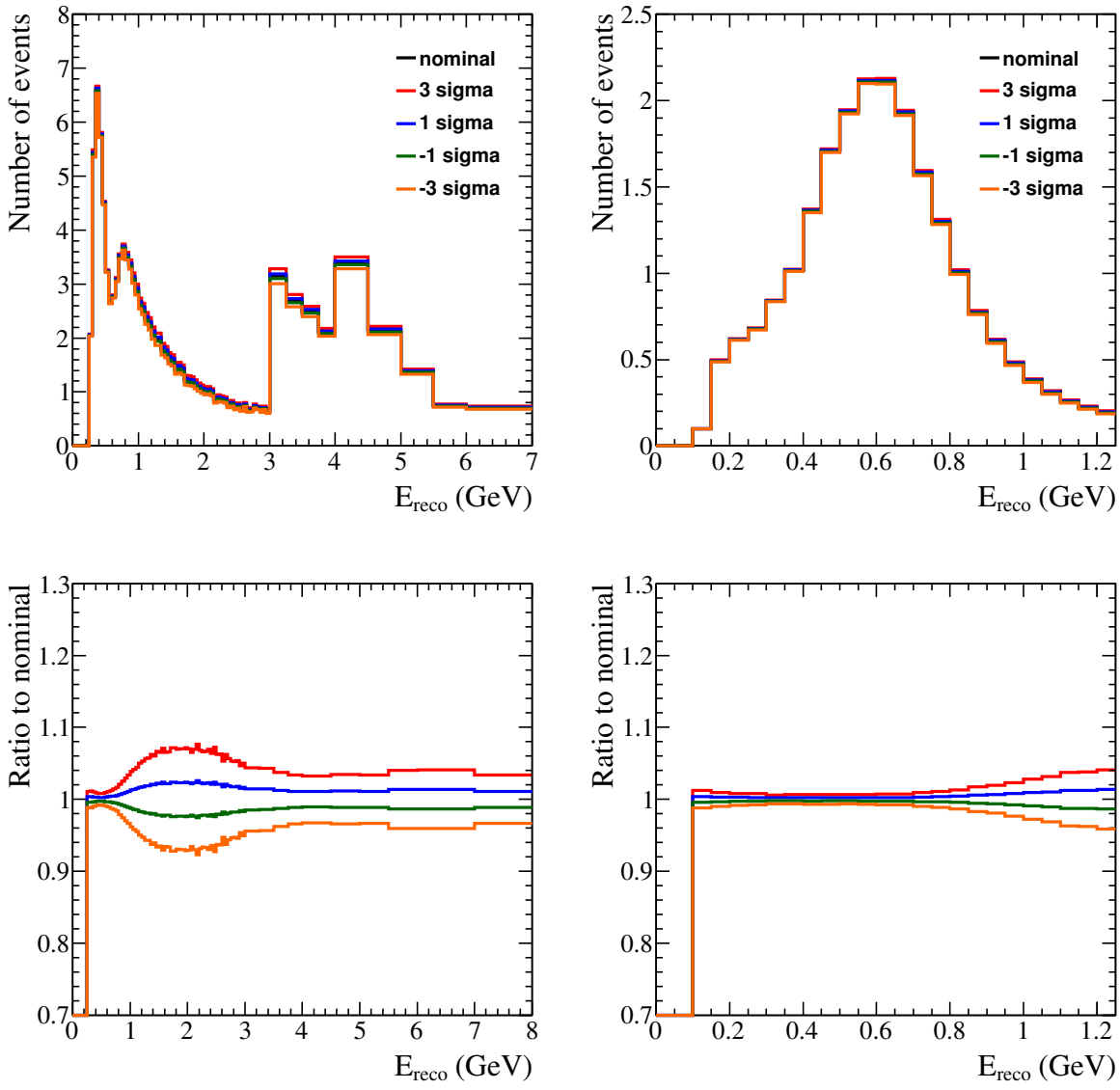


Figure 45: Effect of Norm, $\bar{\nu}$ To ν with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**2.278e-14**	**124.8**	**1.139e-14**	**22**	**22**	**-1.615e-14**	**22**	**1.615e-14**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 91: Effect of SK energy scale with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**2.278e-14**	**124.8**	**1.139e-14**	**22**	**22**	**-1.615e-14**	**22**	**1.615e-14**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 92: Effect of SK energy scale with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

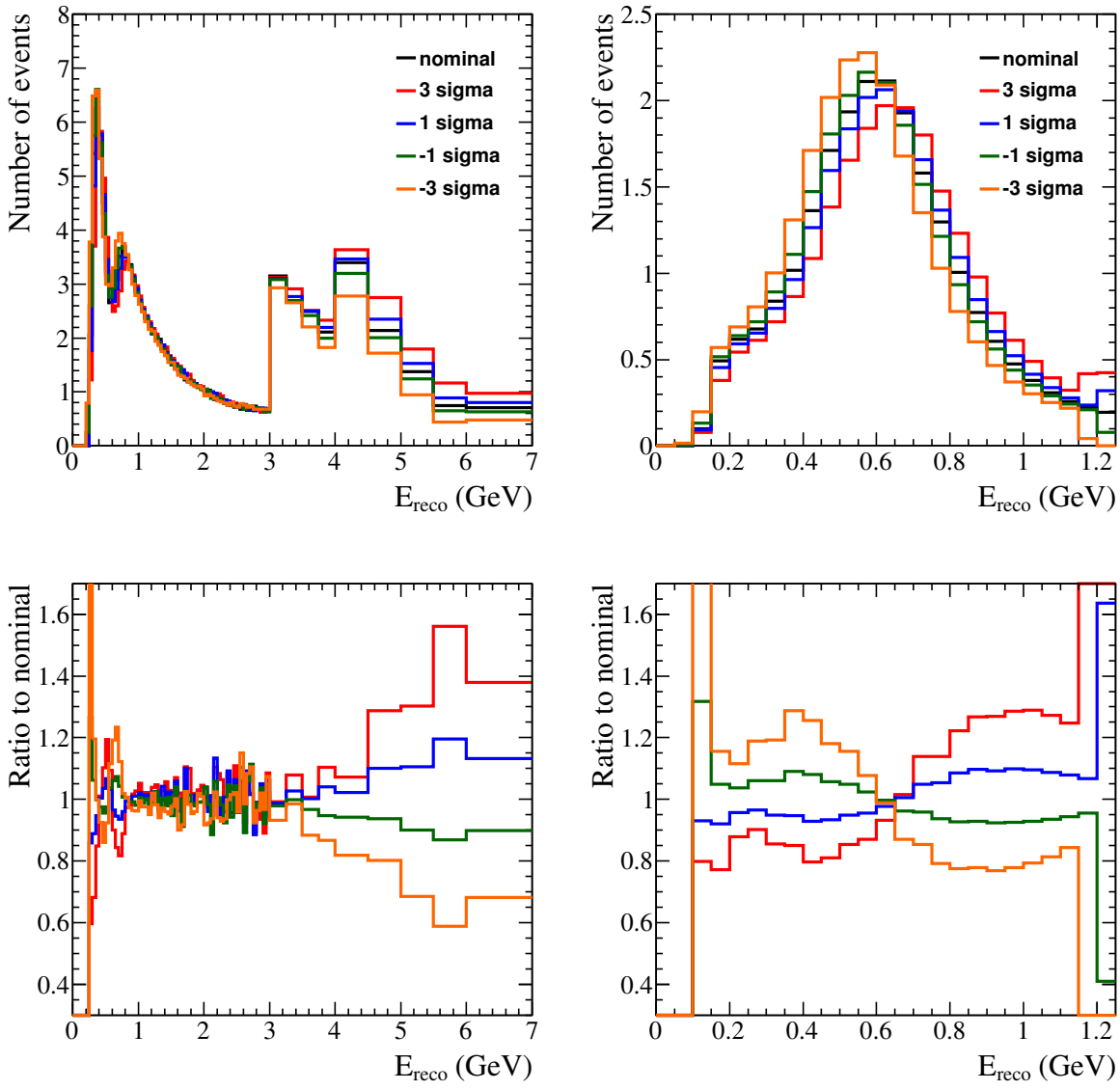


Figure 46: Effect of SK energy scale with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2427**	**124.5**	**-0.2427**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	116.5	0.2537	115.9	-0.2537	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1658	0.06261	0.1656	-0.06261	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.812	0.09995	7.796	-0.09995	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 93: Effect of SKDet + FSI/SI 0; Ereco range 0.00 - 0.40 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2427**	**124.5**	**-0.2427**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	71.09	0.4165	70.5	-0.4165	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1306	0.07949	0.1304	-0.07949	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.627	0.1689	4.611	-0.1689	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 94: Effect of SKDet + FSI/SI 0; Ereco range 0.00 - 0.40 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

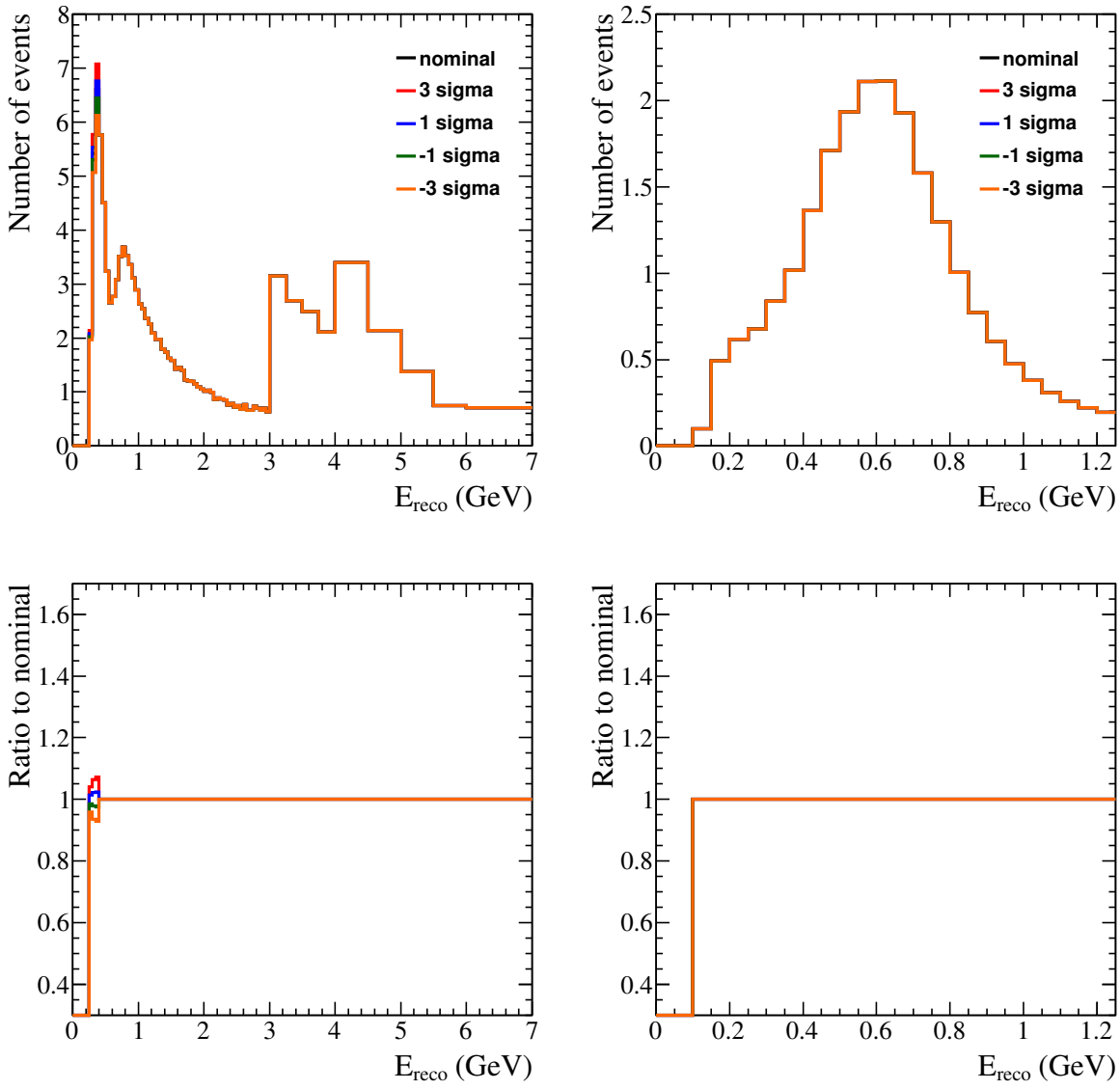


Figure 47: Effect of SKDet + FSI/SI 0; Ereco range 0.00 - 0.40 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.6**	**0.6583**	**123.9**	**-0.6583**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	117	0.6866	115.4	-0.6866	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1682	1.488	0.1632	-1.488	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.824	0.2653	7.783	-0.2653	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 95: Effect of SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.6**	**0.6583**	**123.9**	**-0.6583**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	71.59	1.127	70	-1.127	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.133	1.888	0.1281	-1.888	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.64	0.4482	4.598	-0.4482	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 96: Effect of SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

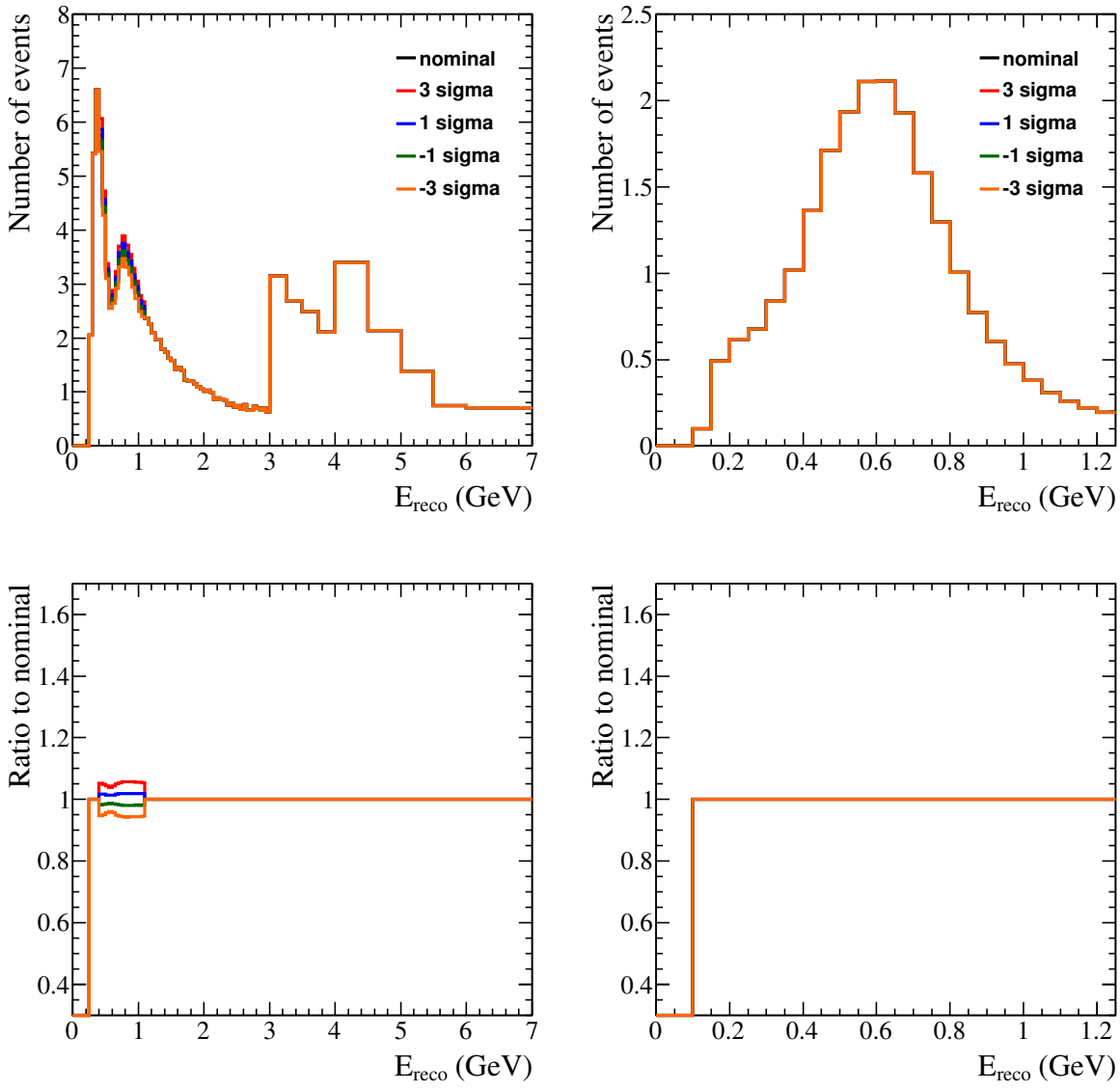


Figure 48: Effect of SKDet + FSI/SI 1; Ereco range 0.40 - 1.10 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.7**	**0.7373**	**123.9**	**-0.7373**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	117.1	0.7084	115.4	-0.7084	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1665	0.4858	0.1649	-0.4858	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.899	1.225	7.708	-1.225	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 97: Effect of SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; ν_μ/$\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.7**	**0.7373**	**123.9**	**-0.7373**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	71.62	1.163	69.97	-1.163	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1313	0.6168	0.1297	-0.6168	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.715	2.069	4.524	-2.069	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 98: Effect of SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; ν_μ/$\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

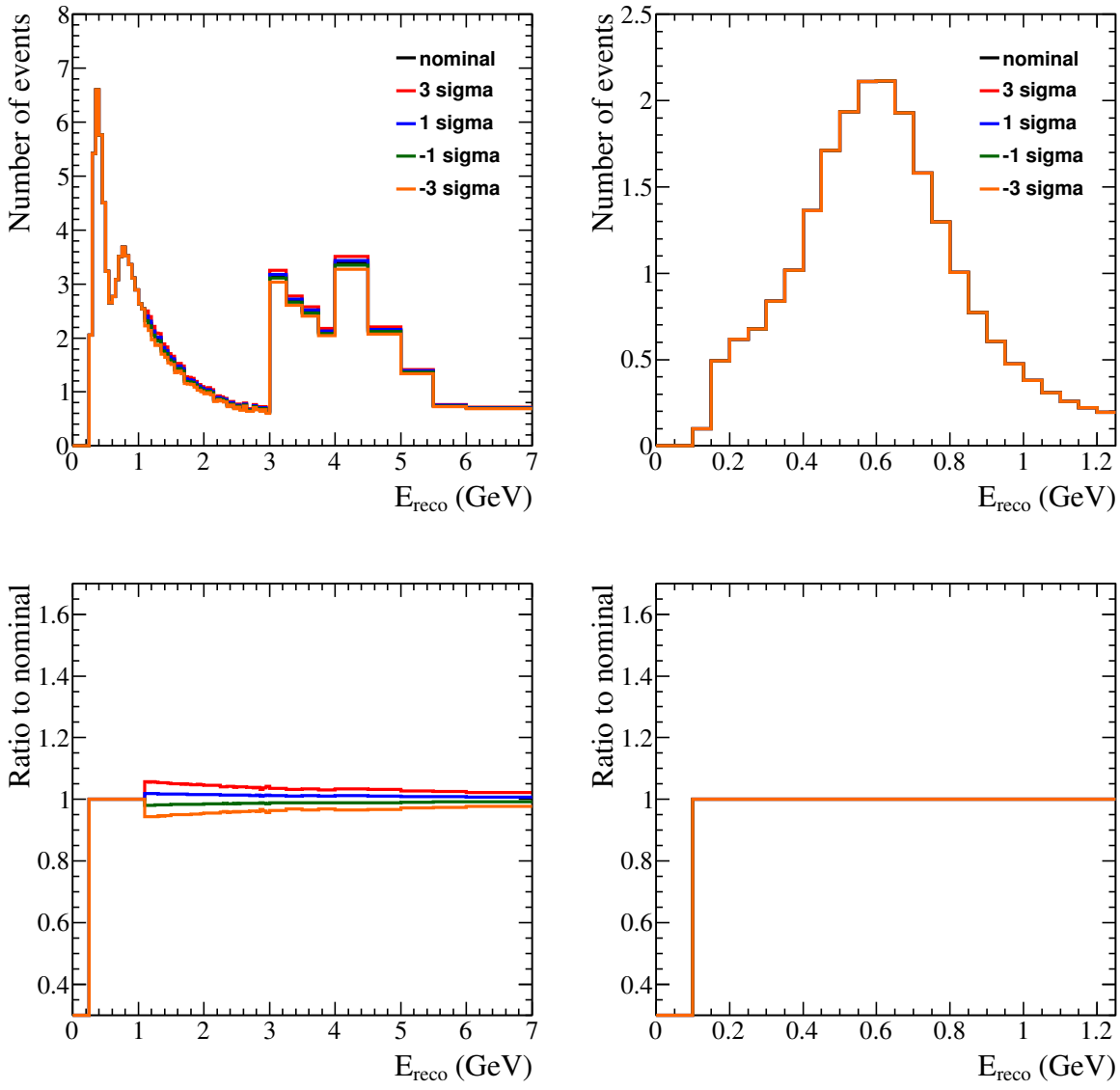


Figure 49: Effect of SKDet + FSI/SI 2; Ereco range 1.10 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.5**	**2.974**	**121.1**	**-2.974**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	119.7	2.972	112.8	-2.972	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1688	1.867	0.1626	-1.867	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	8.056	3.231	7.552	-3.231	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 99: Effect of SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCnQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.5**	**2.974**	**121.1**	**-2.974**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	32.39	8.796	27.16	-8.796	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9872	8.796	0.8276	-8.796	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	9.354	8.796	7.841	-8.796	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03433	8.796	0.02878	-8.796	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.001015	8.796	0.0008512	-8.796	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.00292	8.796	0.002448	-8.796	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.32	8.796	1.945	-8.796	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.27	8.796	0.2263	-8.796	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.5282	8.796	0.4428	-8.796	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 100: Effect of SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCnQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

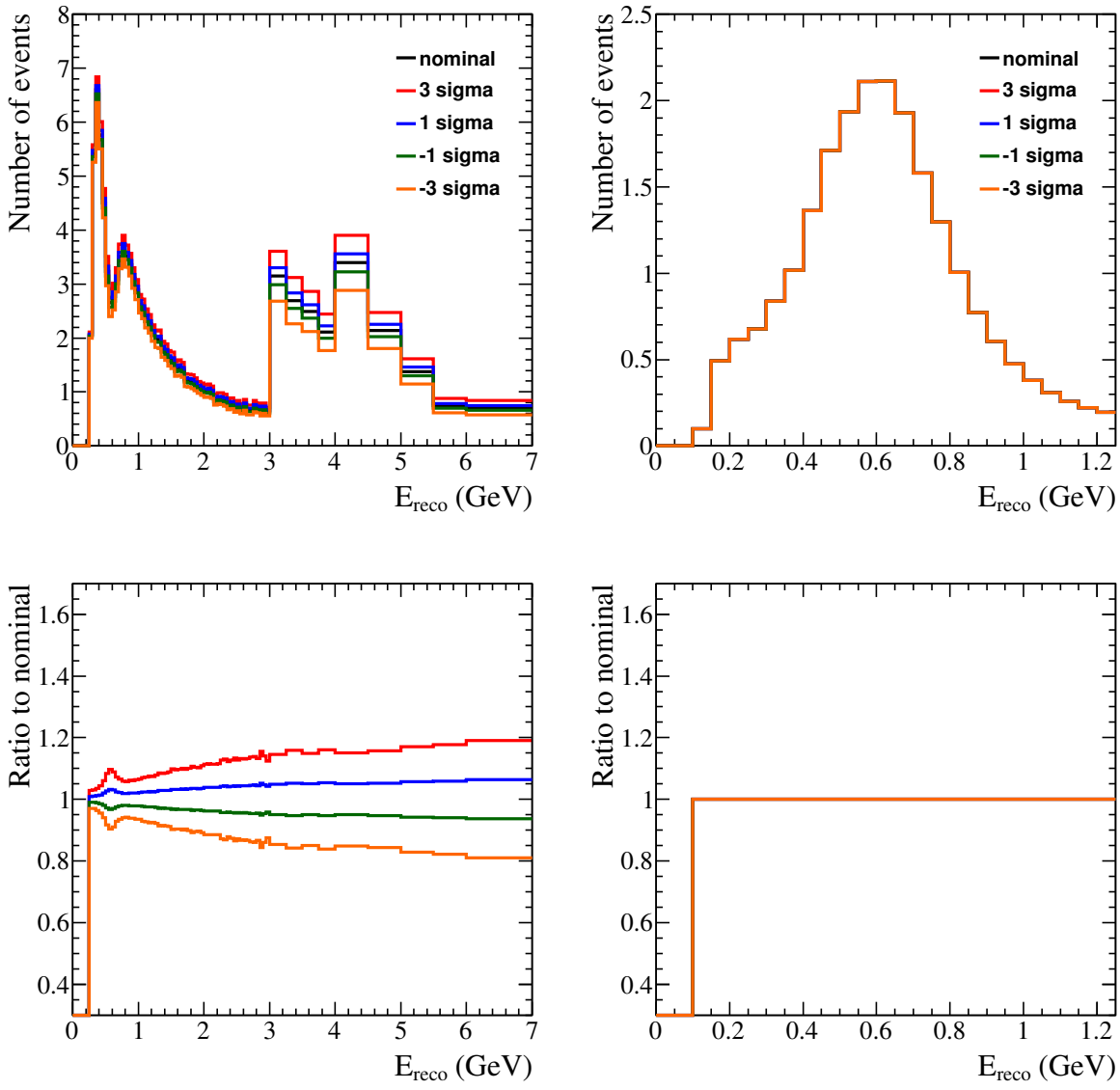


Figure 50: Effect of SKDet + FSI/SI 3; Ereco range 0.00 - 30.00 GeV; $\nu_\mu/\bar{\nu_\mu}$ CCnQE ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2771**	**124.4**	**-0.2771**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.3273	24.95	0.1966	-24.95	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.02806	15.94	0.02035	-15.94	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.006343	99.71	9.225e-06	-99.71	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.5477	99.71	0.0007965	-99.71	17.49	17.49	0	17.49	0

Table 101: Effect of SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; ν_e/$\bar{\nu_e}$/sigν_e CC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**125.1**	**0.2771**	**124.4**	**-0.2771**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}$/$\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.07004	99.71	0.0001019	-99.71	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.04544	99.71	6.609e-05	-99.71	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.002178	99.71	3.168e-06	-99.71	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.01322	99.71	1.923e-05	-99.71	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.004046	99.71	5.884e-06	-99.71	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.002589	99.71	3.766e-06	-99.71	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0002631	99.71	3.826e-07	-99.71	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.000828	99.71	1.204e-06	-99.71	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.004336	99.71	6.306e-06	-99.71	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.001698	99.71	2.47e-06	-99.71	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.000227	99.71	3.302e-07	-99.71	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	8.145e-05	99.71	1.185e-07	-99.71	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.3889	99.71	0.0005657	-99.71	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.1451	99.71	0.000211	-99.71	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.01042	99.71	1.516e-05	-99.71	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.00318	99.71	4.625e-06	-99.71	0.03175	0.03175	0	0.03175	0

Table 102: Effect of SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; ν_e/$\bar{\nu_e}$/sigν_e CC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

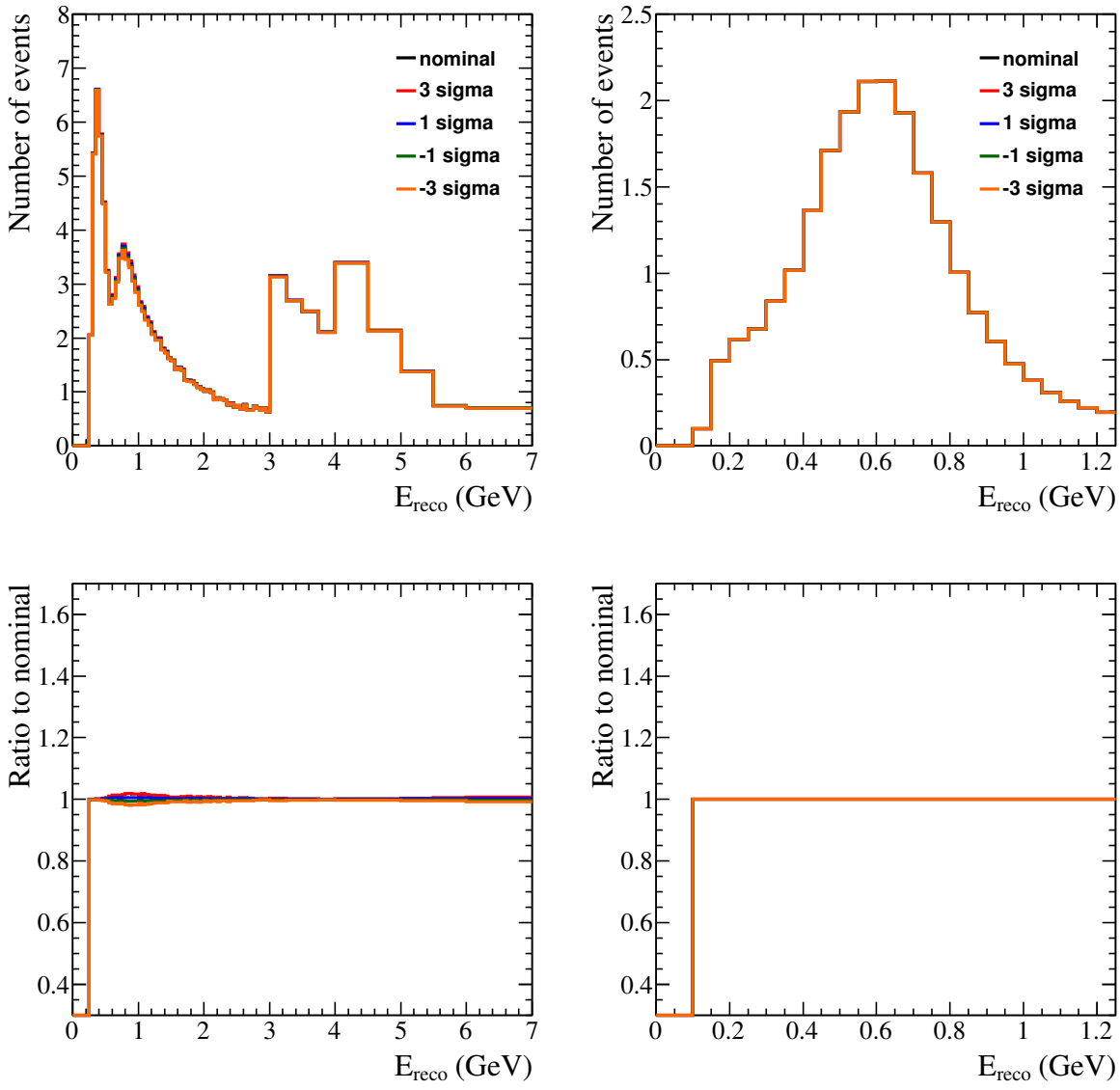


Figure 51: Effect of SKDet + FSI/SI 4; Ereco range 0.00 - 30.00 GeV; $\nu_e/\bar{\nu_e}$/sigν_e CC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.8**	**3.192**	**120.8**	**-3.192**	**22**	**22**	**0**	**22**	**0**
ν_μ	116.2	119.9	3.153	112.6	-3.153	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.993	2.427	7.614	-2.427	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.3787	44.58	0.1452	-44.58	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0363	49.96	0.01211	-49.96	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 103: Effect of SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**128.8**	**3.192**	**120.8**	**-3.192**	**22**	**22**	**0**	**22**	**0**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	4.68	59.46	1.19	-59.46		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	5.149	59.46	1.309	-59.46	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.2287	59.46	0.05815	-59.46		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.2792	59.46	0.07098	-59.46	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.1198	59.46	0.03045	-59.46		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1934	59.46	0.04916	-59.46	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.01313	59.46	0.003339	-59.46		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.01929	59.46	0.004905	-59.46	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 104: Effect of SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

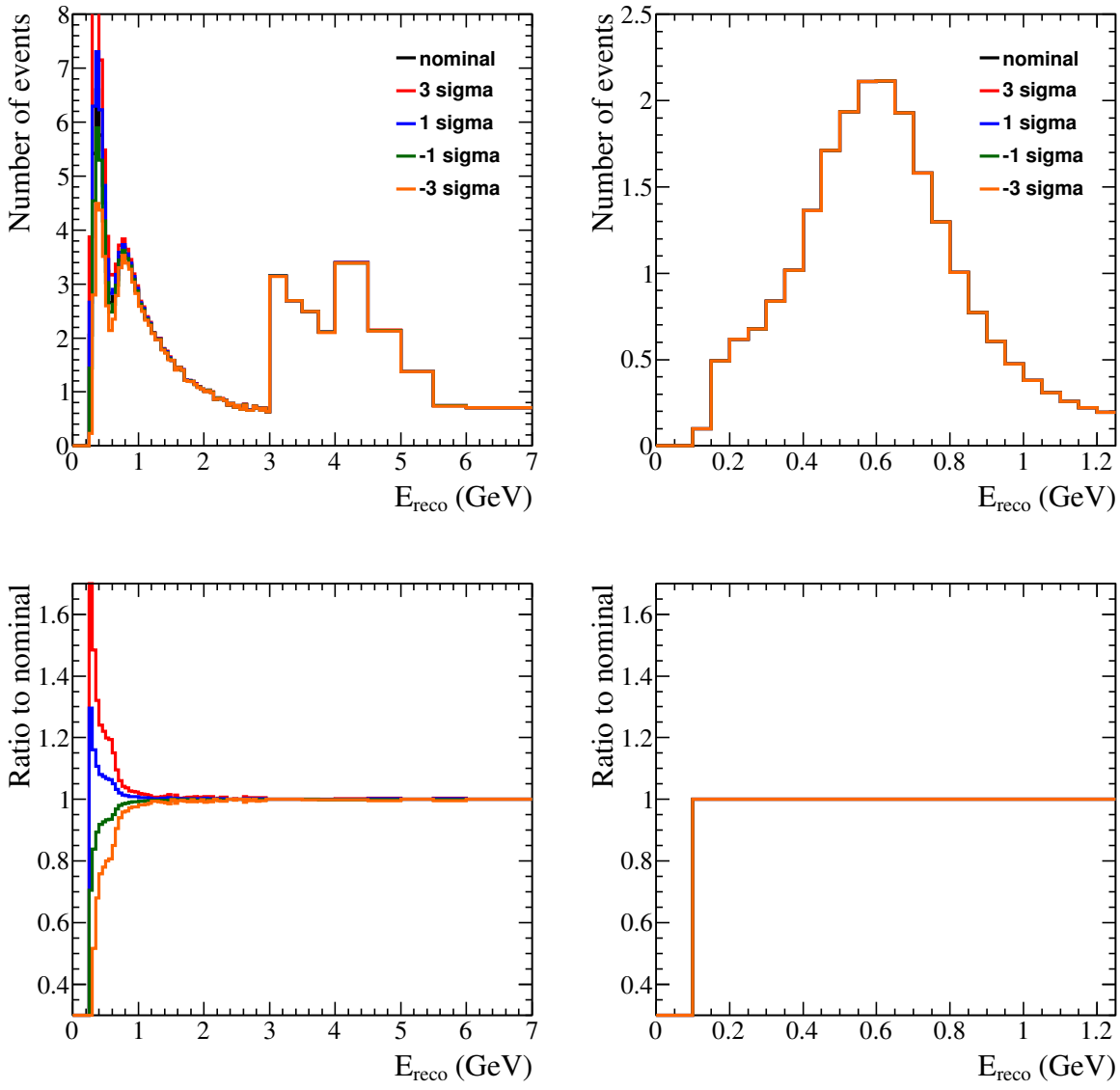


Figure 52: Effect of SKDet + FSI/SI 5; Ereco range 0.00 - 30.00 GeV; all NC ($1R_\mu$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.22**	**1.012**	**21.78**	**-1.012**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.71	1.273	17.27	-1.273

Table 105: Effect of SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.22**	**1.012**	**21.78**	**-1.012**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.84	0.3422	14.74	-0.3422
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.793	6.43	2.456	-6.43
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04547	3.755	0.04218	-3.755
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.0334	5.177	0.03011	-5.177

Table 106: Effect of SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

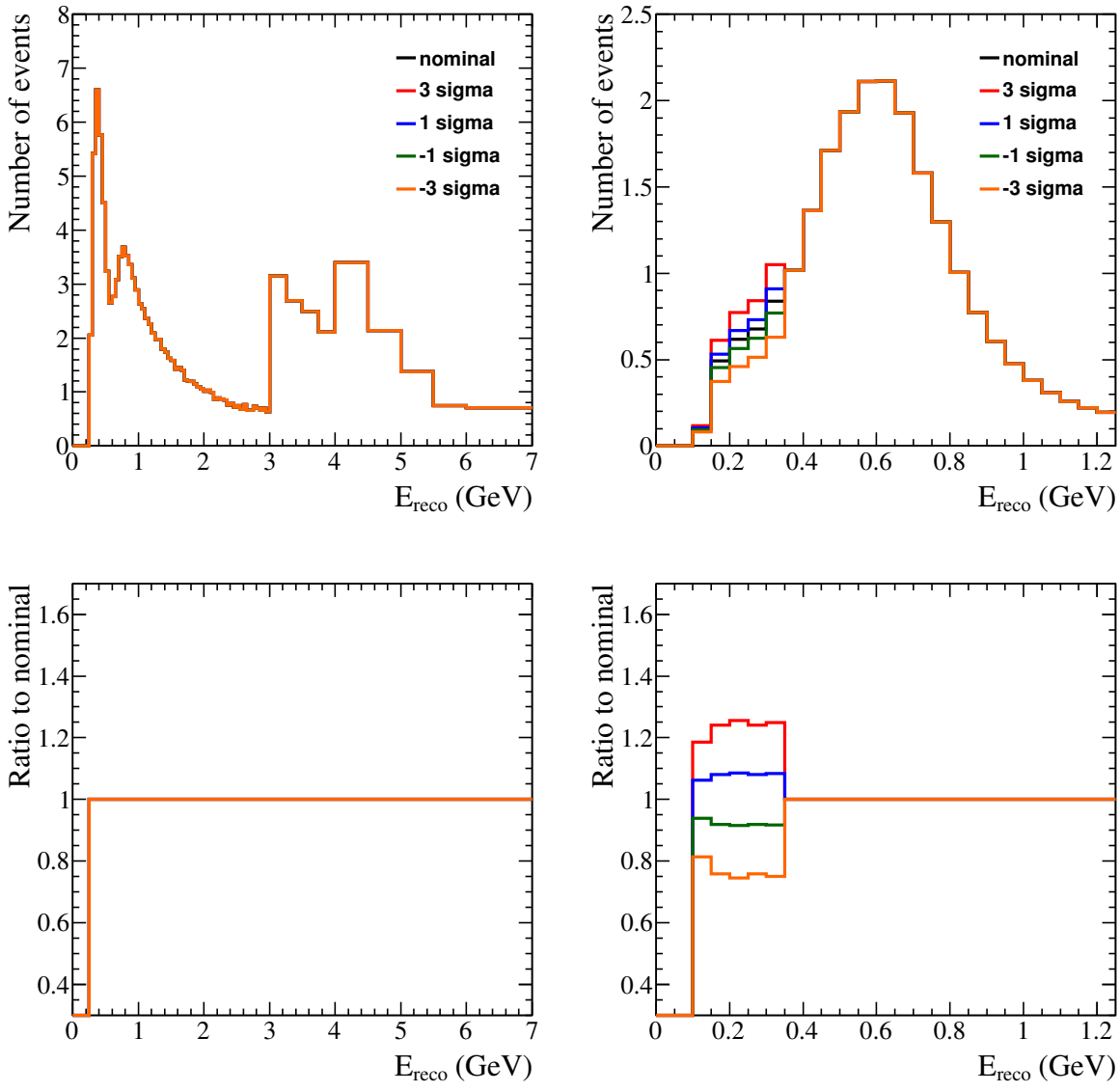


Figure 53: Effect of SKDet + FSI/SI 6; Ereco range 0.00 - 0.35 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.49**	**2.199**	**21.52**	**-2.199**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.97	2.767	17.01	-2.767

Table 107: Effect of SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.49**	**2.199**	**21.52**	**-2.199**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	15.23	2.986	14.35	-2.986
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.665	1.553	2.584	-1.553
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04488	2.403	0.04278	-2.403
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03228	1.645	0.03123	-1.645

Table 108: Effect of SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

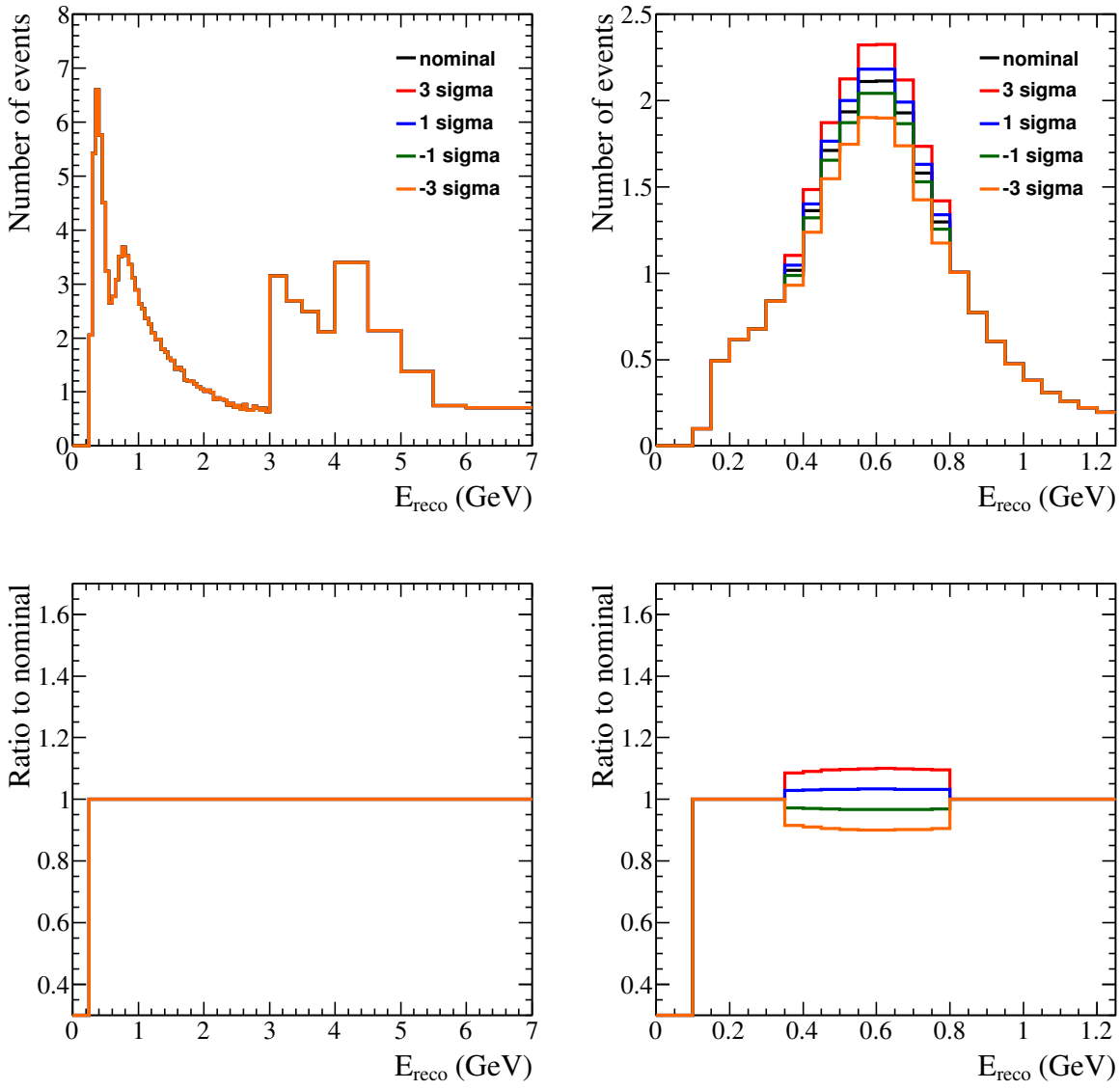


Figure 54: Effect of SKDet + FSI/SI 7; Ereco range 0.35 - 0.80 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.12**	**0.5575**	**21.88**	**-0.5575**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.61	0.7013	17.37	-0.7013

Table 109: Effect of SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.12**	**0.5575**	**21.88**	**-0.5575**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.91	0.7923	14.67	-0.7923
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.63	0.1988	2.619	-0.1988
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04392	0.209	0.04374	-0.209
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03193	0.5502	0.03158	-0.5502

Table 110: Effect of SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

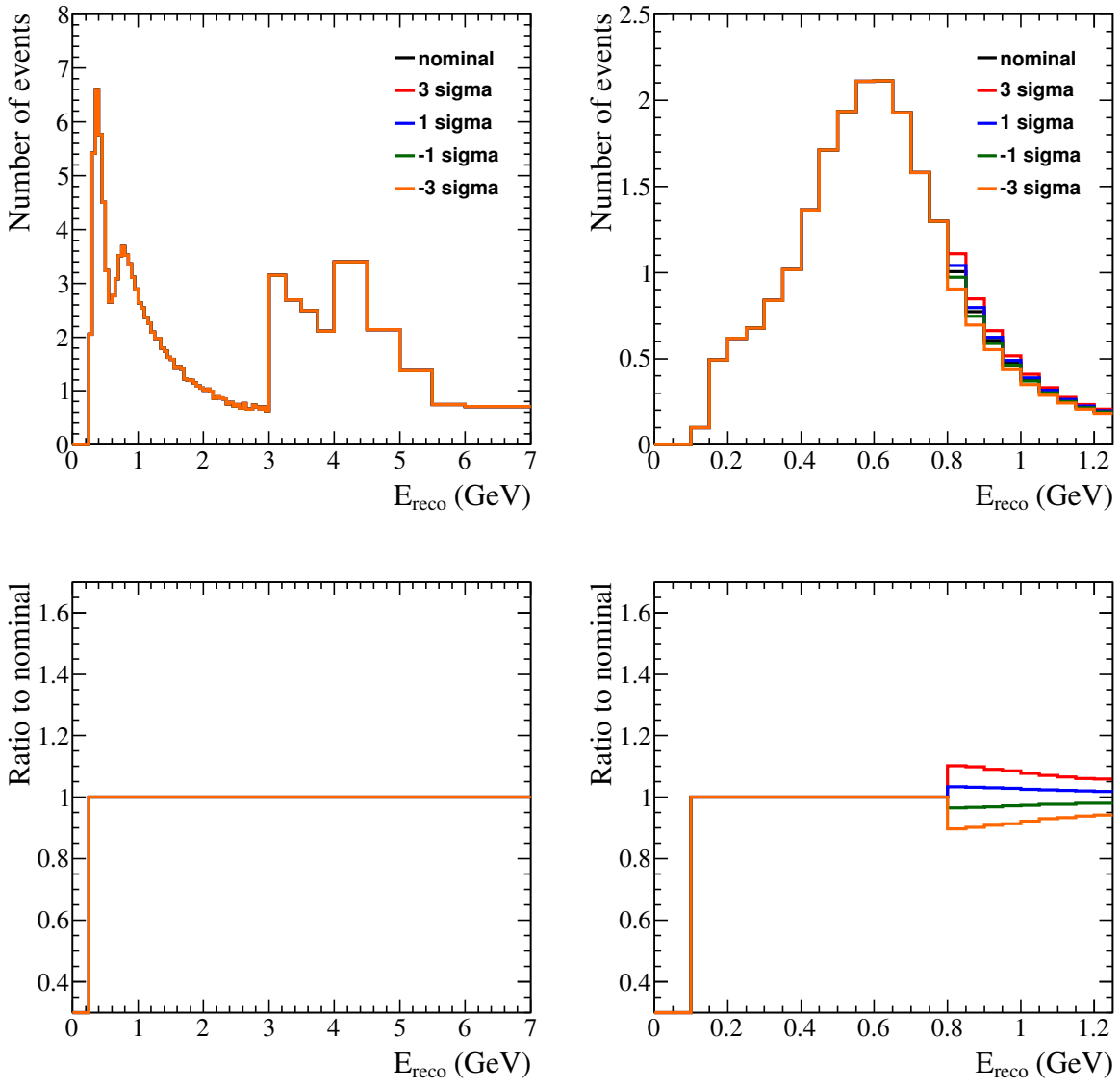


Figure 55: Effect of SKDet + FSI/SI 8; Ereco range 0.80 - 1.25 GeV; Osc. ν_e CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.05**	**0.206**	**21.96**	**-0.206**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9882	4.74	0.8988	-4.74
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.000308	74.28	4.546e-05	-74.28
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05503	0.8618	0.05409	-0.8618
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 111: Effect of SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.05**	**0.206**	**21.96**	**-0.206**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.0805	68.5	0.01505	-68.5
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.02907	70.16	0.005099	-70.16
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	9.821e-06	124.9	-1.088e-06	-124.9
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0002368	72.21	3.822e-05	-72.21
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	6.749e-05	84.67	5.601e-06	-84.67
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	1.836e-06	124.9	-2.034e-07	-124.9
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.001284	46.76	0.000466	-46.76
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.000251	0.9754	0.0002461	-0.9754
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	9.592e-06	124.9	-1.063e-06	-124.9
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.0001771	42.9	7.079e-05	-42.9
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 112: Effect of SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

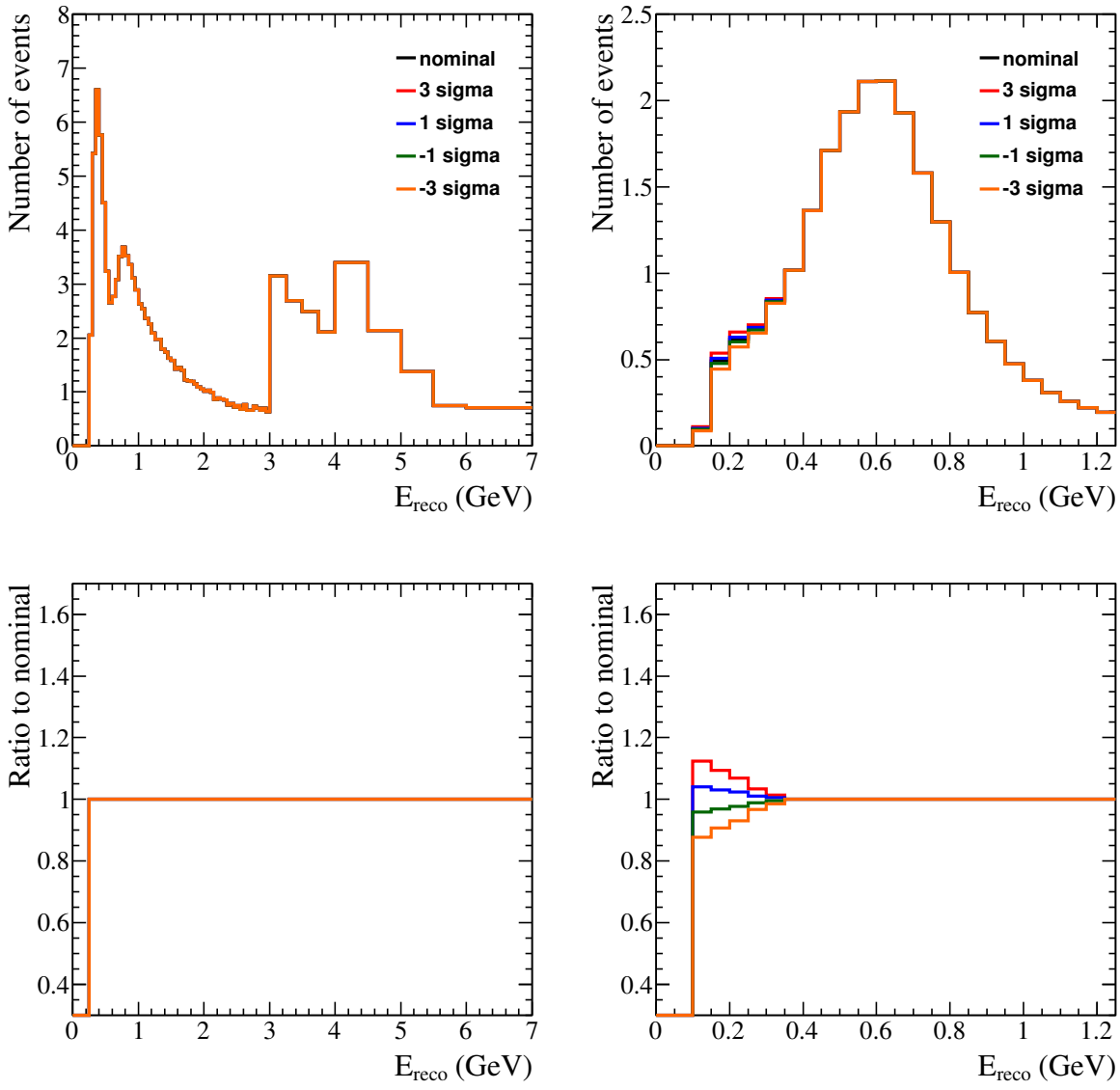


Figure 56: Effect of SKDet + FSI/SI 9; Ereco range 0.00 - 0.35 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.03**	**0.1203**	**21.97**	**-0.1203**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9693	2.733	0.9177	-2.733
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0002437	37.87	0.0001098	-37.87
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05518	1.13	0.05394	-1.13
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 113: Effect of SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.03**	**0.1203**	**21.97**	**-0.1203**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.06364	33.2	0.03192	-33.2
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.02585	51.27	0.008326	-51.27
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.002118	122.1	-0.0002104	-122.1
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001887	37.2	8.636e-05	-37.2
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	5.006e-05	36.98	2.303e-05	-36.98
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	4.102e-06	122.1	-4.075e-07	-122.1
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.00116	32.59	0.00059	-32.59
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0004804	93.29	1.668e-05	-93.29
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.0002233	80.14	2.462e-05	-80.14
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 114: Effect of SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

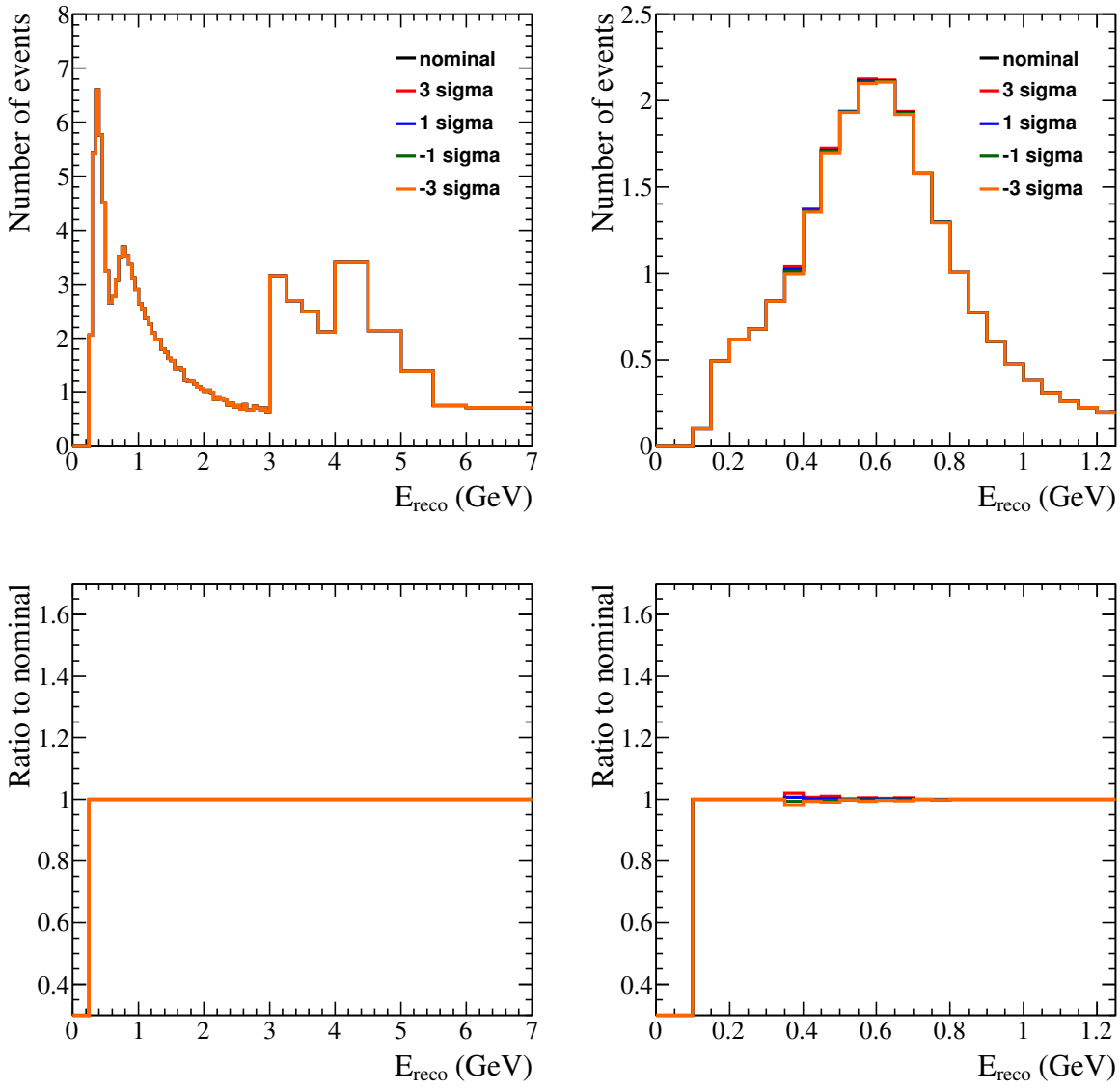


Figure 57: Effect of SKDet + FSI/SI 10; Ereco range 0.35 - 0.80 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.01**	**0.05206**	**21.99**	**-0.05206**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9545	1.163	0.9325	-1.163
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001975	11.74	0.000156	-11.74
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05502	0.838	0.0541	-0.838
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.127	0	3.127	0
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1523	0	0.1523	0
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 115: Effect of SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.01**	**0.05206**	**21.99**	**-0.05206**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.05837	22.17	0.03719	-22.17
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01747	2.258	0.0167	-2.258
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001574	14.45	0.0001176	-14.45
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.741e-05	2.365	3.568e-05	-2.365
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.001263	44.25	0.0004879	-44.25
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0003184	28.12	0.0001787	-28.12
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 116: Effect of SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

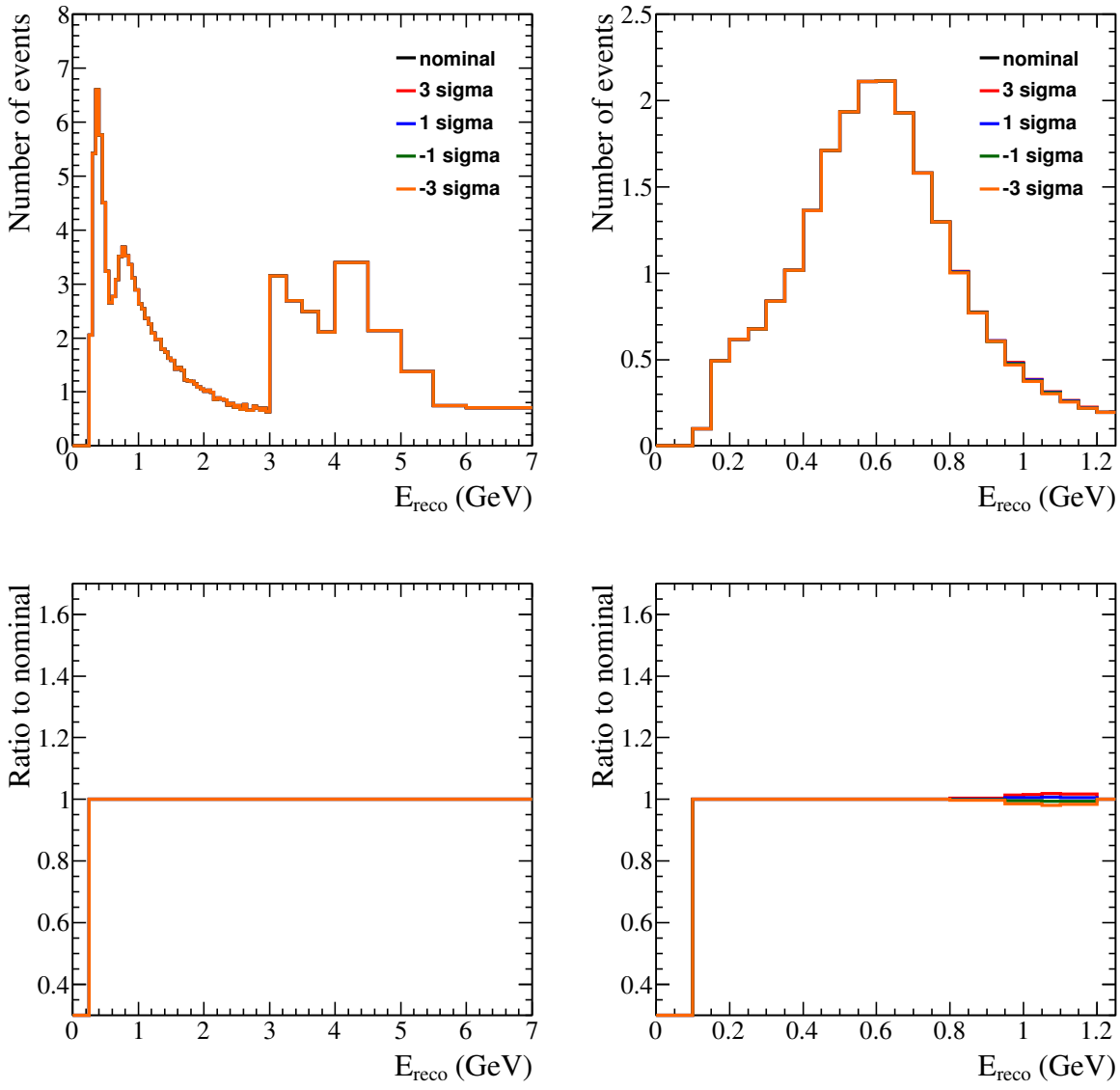


Figure 58: Effect of SKDet + FSI/SI 11; Ereco range 0.80 - 1.25 GeV; $\nu_\mu/\bar{\nu_\mu}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.04**	**0.163**	**21.97**	**-0.163**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.16	1.056	3.094	-1.056
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1533	0.629	0.1513	-0.629
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2358	0.8026	0.232	-0.8026
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 117: Effect of SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.04**	**0.163**	**21.97**	**-0.163**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.281	0.7513	2.247	-0.7513
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7856	1.959	0.7554	-1.959
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009342	1.132	0.009133	-1.132
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.05892	1.417	0.05727	-1.417
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.09775	0.4675	0.09684	-0.4675
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04219	0.9758	0.04137	-0.9758
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006729	1.005	0.006595	-1.005
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003676	0.7815	0.003619	-0.7815
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1818	0.5407	0.1798	-0.5407
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04139	1.852	0.03989	-1.852
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.011	1.127	0.01075	-1.127
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001572	1.568	0.001524	-1.568
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 118: Effect of SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

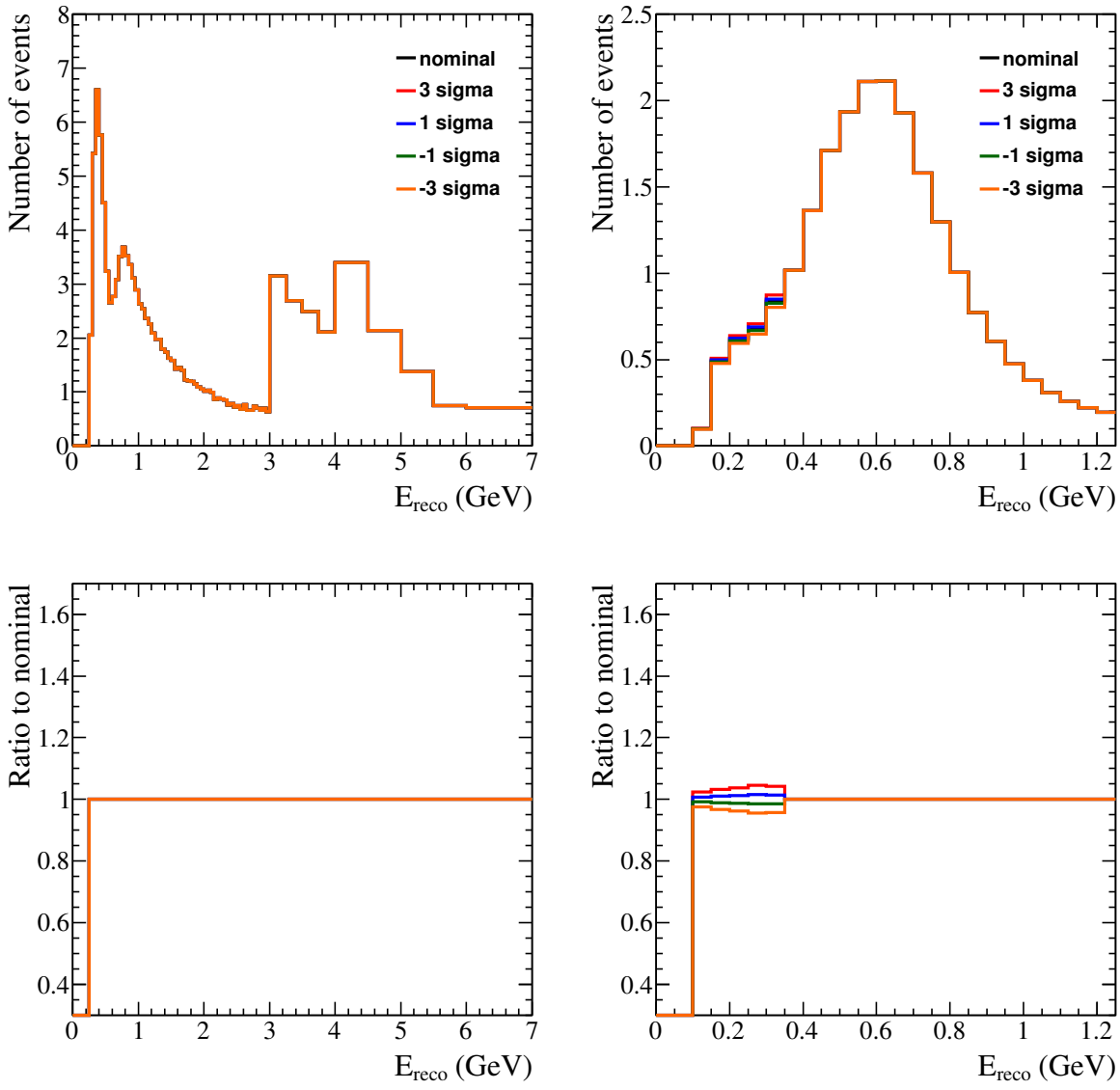


Figure 59: Effect of SKDet + FSI/SI 12; Ereco range 0.00 - 0.35 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.09**	**0.3992**	**21.91**	**-0.3992**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.206	2.52	3.048	-2.52
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1553	1.993	0.1493	-1.993
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2399	2.565	0.2279	-2.565
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 119: Effect of SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.09**	**0.3992**	**21.91**	**-0.3992**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.324	2.637	2.205	-2.637
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7881	2.284	0.7529	-2.284
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009454	2.346	0.009021	-2.346
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.05938	2.204	0.05682	-2.204
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.09928	2.032	0.09532	-2.032
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04262	2.002	0.04094	-2.002
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006809	2.199	0.006516	-2.199
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003722	2.061	0.003572	-2.061
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1854	2.549	0.1762	-2.549
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04166	2.512	0.03962	-2.512
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.0112	3.027	0.01055	-3.027
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001587	2.532	0.001509	-2.532
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 120: Effect of SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

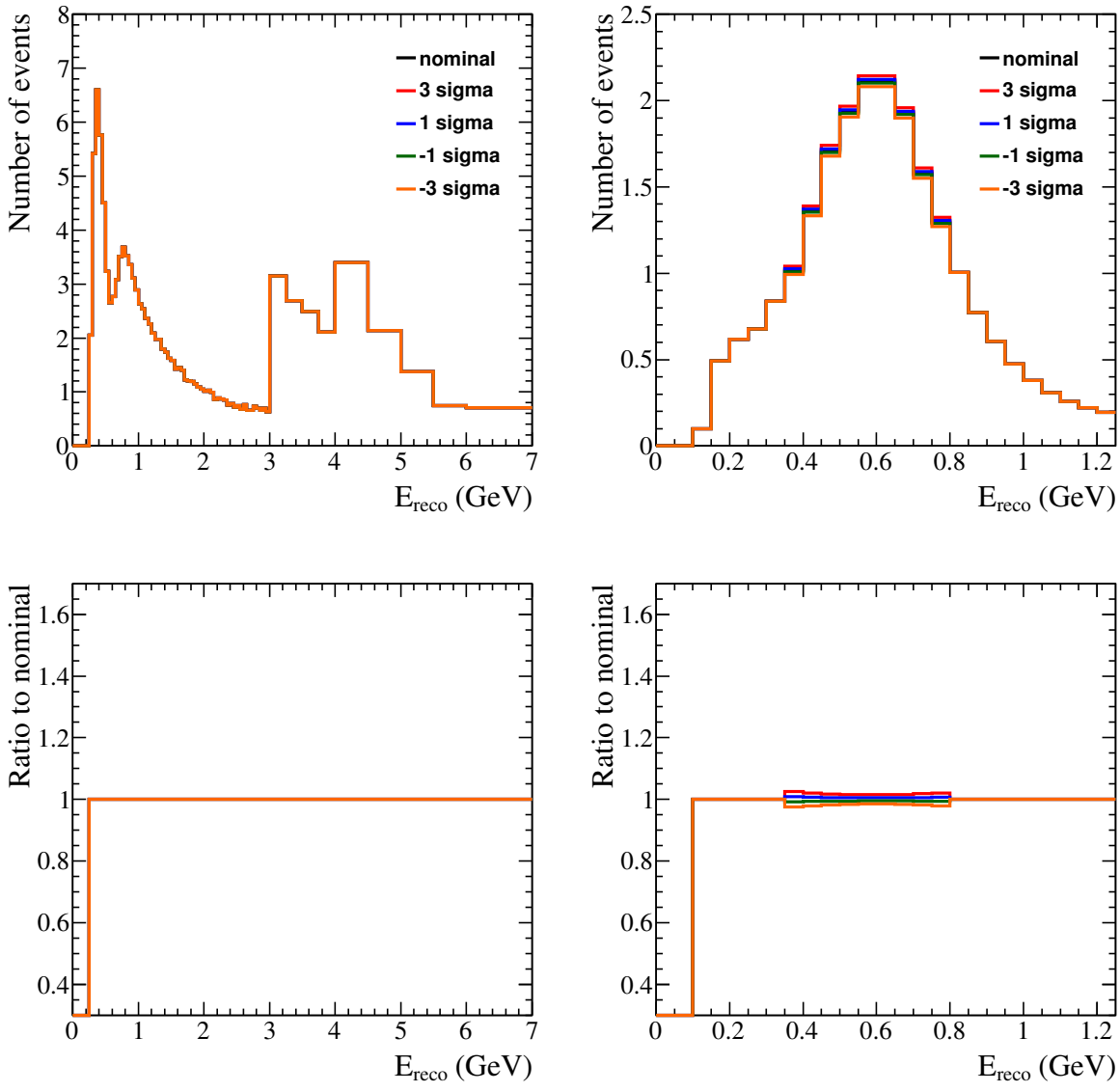


Figure 60: Effect of SKDet + FSI/SI 13; Ereco range 0.35 - 0.80 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.08**	**0.3482**	**21.92**	**-0.3482**
ν_μ	116.2	116.2	0	116.2	0	0.9435	0.9435	0	0.9435	0
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.05456	0	0.05456	0
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.193	2.125	3.061	-2.125
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.157	3.102	0.1476	-3.102
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2393	2.32	0.2284	-2.32
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 121: Effect of SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.08**	**0.3482**	**21.92**	**-0.3482**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.3924	0	0.3924	0
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1539	0	0.1539	0
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3314	0	0.3314	0
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.0199	0	0.0199	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01589	0	0.01589	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01752	0	0.01752	0
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.316	2.264	2.213	-2.264
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7841	1.769	0.7568	-1.769
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009454	2.349	0.00902	-2.349
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.05944	2.311	0.05676	-2.311
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.0107	0	0.0107	0
ν_e NC coherent		-		–		0.004117	0.004117	0	0.004117	0
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.0101	0	0.0101	0
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.1005	3.301	0.09409	-3.301
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04301	2.933	0.04056	-2.933
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006839	2.647	0.006486	-2.647
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003757	3.011	0.003537	-3.011
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001191	0	0.001191	0
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0007456	0	0.0007456	0
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.0009675	0	0.0009675	0
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1854	2.551	0.1762	-2.551
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04127	1.55	0.04001	-1.55
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01103	1.442	0.01072	-1.442
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001575	1.751	0.001521	-1.751
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 122: Effect of SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

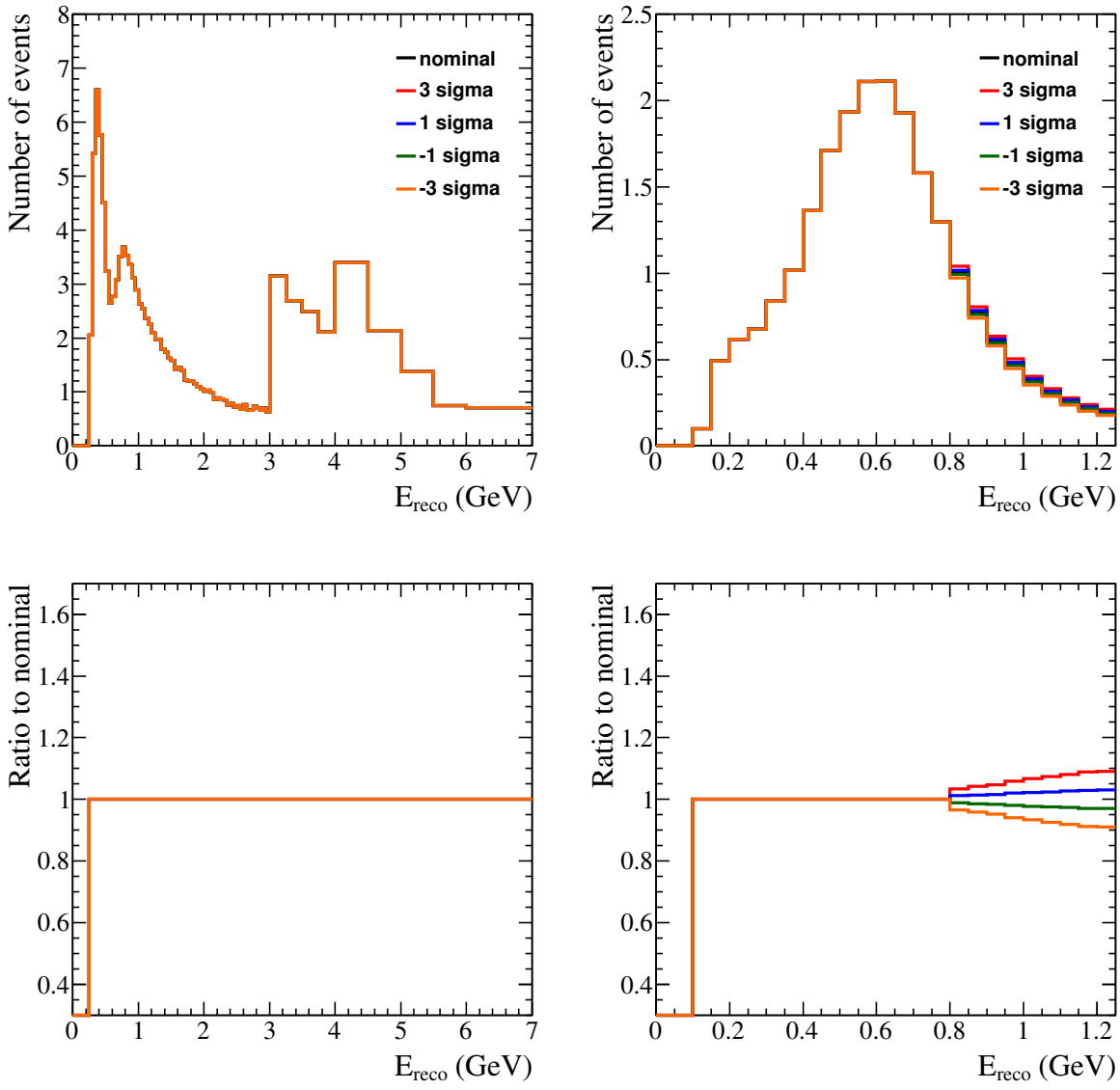


Figure 61: Effect of SKDet + FSI/SI 14; Ereco range 0.80 - 1.25 GeV; $\nu_e/\bar{\nu_e}$ CC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.19**	**0.8472**	**21.81**	**-0.8472**
ν_μ	116.2	116.2	0	116.2	0	0.9435	1.117	18.39	0.77	-18.39
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.06297	15.42	0.04615	-15.42
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.131	0.128	3.123	-0.128
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1527	0.2968	0.1518	-0.2968
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 123: Effect of SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.19**	**0.8472**	**21.81**	**-0.8472**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC 1$\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC 1π^0		-		–		0.3924	0.4477	14.11	0.337	-14.11
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1818	18.15	0.126	-18.15
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.4216	27.23	0.2411	-27.23
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1$\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC 1π^0		-		–		0.0199	0.02238	12.45	0.01742	-12.45
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01835	15.43	0.01344	-15.43
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.021	19.88	0.01403	-19.88
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC 1$\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC 1π^0		-		–		0.0107	0.01184	10.62	0.009569	-10.62
ν_e NC coherent		-		–		0.004117	0.004618	12.17	0.003616	-12.17
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.01247	23.42	0.007738	-23.42
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC 1$\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC 1π^0		-		–		0.001191	0.001333	11.93	0.001049	-11.93
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0008191	9.855	0.0006722	-9.855
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.001204	24.44	0.000731	-24.44
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 124: Effect of SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

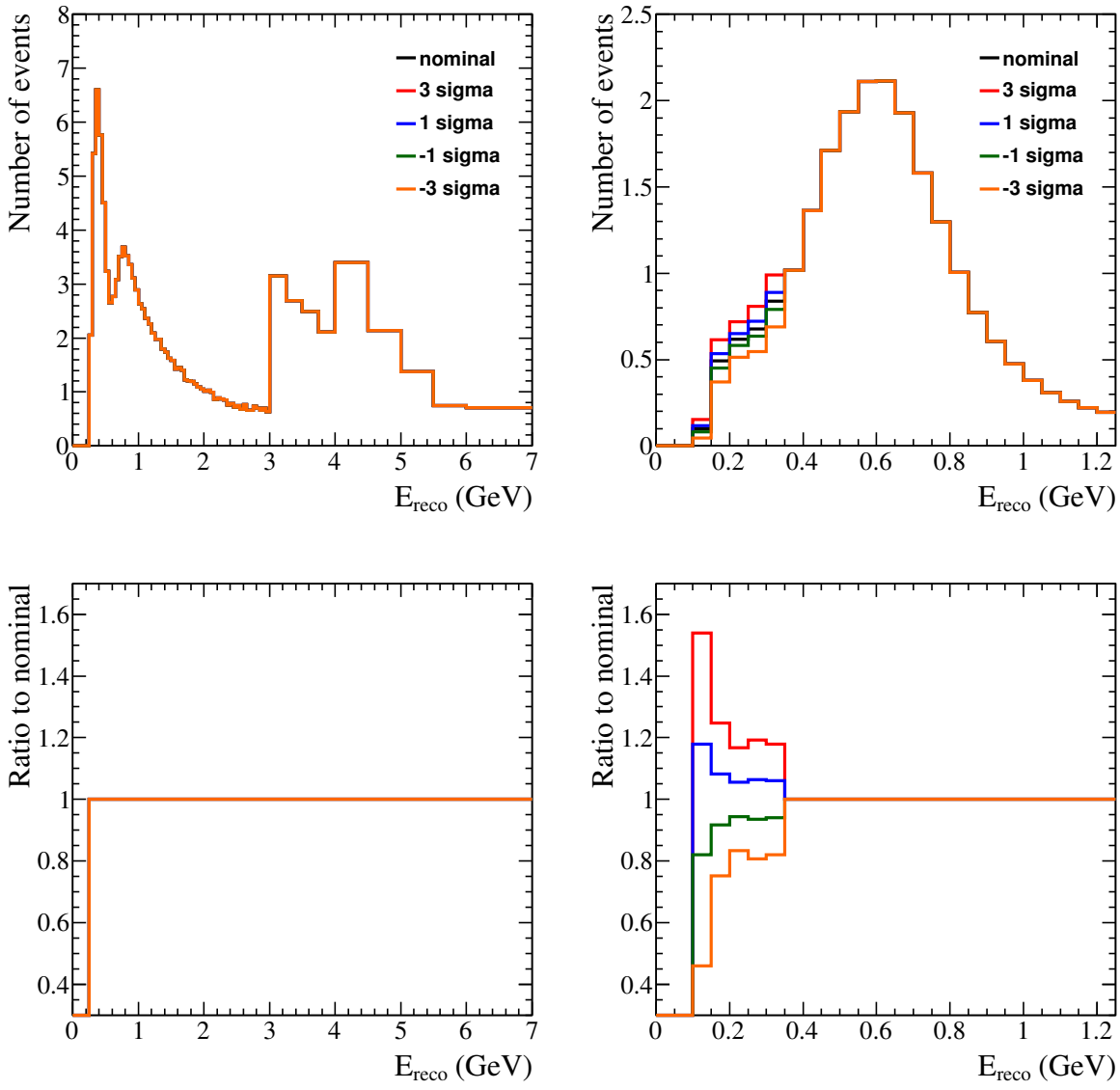


Figure 62: Effect of SKDet + FSI/SI 15; Ereco range 0.00 - 0.35 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.11**	**0.4881**	**21.89**	**-0.4881**
ν_μ	116.2	116.2	0	116.2	0	0.9435	1.042	10.44	0.845	-10.44
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.0605	10.88	0.04862	-10.88
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.13	0.08524	3.124	-0.08524
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1526	0.1977	0.152	-0.1977
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 125: Effect of SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.11**	**0.4881**	**21.89**	**-0.4881**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.4429	12.88	0.3419	-12.88
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1738	12.94	0.134	-12.94
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3594	8.458	0.3033	-8.458
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.02207	10.93	0.01773	-10.93
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01794	12.9	0.01384	-12.9
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01923	9.78	0.0158	-9.78
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01196	11.71	0.009452	-11.71
ν_e NC coherent		-		–		0.004117	0.004667	13.35	0.003568	-13.35
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.01097	8.534	0.009243	-8.534
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001319	10.73	0.001063	-10.73
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0008324	11.63	0.0006589	-11.63
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.001054	8.944	0.000881	-8.944
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 126: Effect of SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

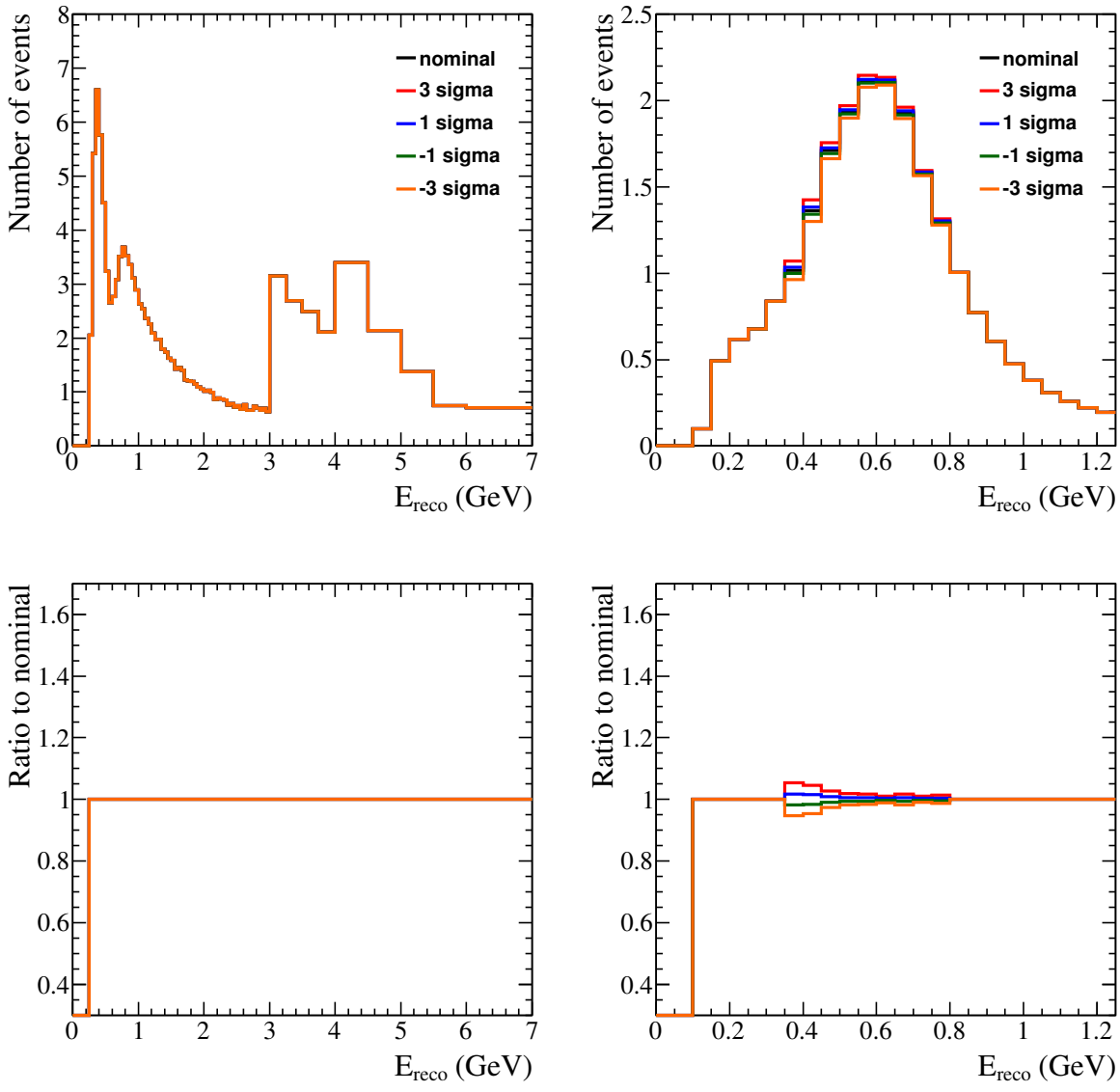


Figure 63: Effect of SKDet + FSI/SI 16; Ereco range 0.35 - 0.80 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.1**	**0.4559**	**21.9**	**-0.4559**
ν_μ	116.2	116.2	0	116.2	0	0.9435	1.032	9.331	0.8555	-9.331
$Osc.\nu_\mu$	0.1657	0.1657	0	0.1657	0	0.0001767	0.0001767	0	0.0001767	0
$\bar{\nu_\mu}$	7.804	7.804	0	7.804	0	0.05456	0.06246	14.47	0.04667	-14.47
ν_e	0.2619	0.2619	0	0.2619	0	3.127	3.131	0.1242	3.123	-0.1242
$\bar{\nu_e}$	0.0242	0.0242	0	0.0242	0	0.1523	0.1528	0.3232	0.1518	-0.3232
Osc.$\bar{\nu_e}$	0.003176	0.003176	0	0.003176	0	0.2339	0.2339	0	0.2339	0
$Osc.\nu_e$	0.2742	0.2742	0	0.2742	0	17.49	17.49	0	17.49	0

Table 127: Effect of SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

	$1R_\mu$					$1R_e$				
Component	**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**		**BANFF tuned**	**BANFF tuned +1σ**		**BANFF tuned -1σ**	
	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)	N_{SK}	N_{SK}	change (%)	N_{SK}	change (%)
Total	**124.8**	**124.8**	**0**	**124.8**	**0**	**22**	**22.1**	**0.4559**	**21.9**	**-0.4559**
ν_μ CCQE	70.79	70.79	0	70.79	0	0.04778	0.04778	0	0.04778	0
ν_μ CC 1π	29.78	29.78	0	29.78	0	0.01709	0.01709	0	0.01709	0
ν_μ CC coherent	0.9074	0.9074	0	0.9074	0	4.366e-06	4.366e-06	0	4.366e-06	0
ν_μ CC other	8.597	8.597	0	8.597	0	0.0009536	0.0009536	0	0.0009536	0
ν_μ/ν_τ NC $1\pi^{+/-}$	2.935	2.935	0	2.935	0		-		–	
ν_μ/ν_τ NC $1\pi^0$		-		–		0.3924	0.4412	12.45	0.3436	-12.45
ν_μ/ν_τ NC coherent		-		–		0.1539	0.1657	7.657	0.1421	-7.657
ν_μ/ν_τ NC other	3.229	3.229	0	3.229	0	0.3314	0.3588	8.271	0.304	-8.271
Osc. ν_μ CCQE	0.1305	0.1305	0	0.1305	0	0.0001375	0.0001375	0	0.0001375	0
Osc. ν_μ CC 1π	0.03155	0.03155	0	0.03155	0	3.655e-05	3.655e-05	0	3.655e-05	0
Osc. ν_μ CC coherent	0.0009333	0.0009333	0	0.0009333	0	8.165e-07	8.165e-07	0	8.165e-07	0
Osc. ν_μ CC other	0.002684	0.002684	0	0.002684	0	1.847e-06	1.847e-06	0	1.847e-06	0
$\bar{\nu_\mu}$ CCQE	4.619	4.619	0	4.619	0	0.0008752	0.0008752	0	0.0008752	0
$\bar{\nu_\mu}$ CC 1π	2.132	2.132	0	2.132	0	0.0002486	0.0002486	0	0.0002486	0
$\bar{\nu_\mu}$ CC coherent	0.2481	0.2481	0	0.2481	0	4.264e-06	4.264e-06	0	4.264e-06	0
$\bar{\nu_\mu}$ CC other	0.4855	0.4855	0	0.4855	0	0.000124	0.000124	0	0.000124	0
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^{+/-}$	0.1434	0.1434	0	0.1434	0		-		–	
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC $1\pi^0$		-		–		0.0199	0.02371	19.14	0.01609	-19.14
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC coherent		-		–		0.01589	0.01762	10.89	0.01416	-10.89
$\bar{\nu_\mu}/\bar{\nu_\tau}$ NC other	0.1751	0.1751	0	0.1751	0	0.01752	0.01987	13.44	0.01516	-13.44
ν_e CCQE	0.03507	0.03507	0	0.03507	0	2.264	2.264	0	2.264	0
ν_e CC 1π	0.02275	0.02275	0	0.02275	0	0.7705	0.7705	0	0.7705	0
ν_e CC coherent	0.001091	0.001091	0	0.001091	0	0.009237	0.009237	0	0.009237	0
ν_e CC other	0.00662	0.00662	0	0.00662	0	0.0581	0.0581	0	0.0581	0
ν_e NC $1\pi^{+/-}$	0.07512	0.07512	0	0.07512	0		-		–	
ν_e NC $1\pi^0$		-		–		0.0107	0.01277	19.32	0.008637	-19.32
ν_e NC coherent		-		–		0.004117	0.004674	13.52	0.003561	-13.52
ν_e NC other	0.1213	0.1213	0	0.1213	0	0.0101	0.01136	12.45	0.008846	-12.45
$\bar{\nu_e}$ CCQE	0.002026	0.002026	0	0.002026	0	0.0973	0.0973	0	0.0973	0
$\bar{\nu_e}$ CC 1π	0.001296	0.001296	0	0.001296	0	0.04178	0.04178	0	0.04178	0
$\bar{\nu_e}$ CC coherent	0.0001317	0.0001317	0	0.0001317	0	0.006662	0.006662	0	0.006662	0
$\bar{\nu_e}$ CC other	0.0004146	0.0004146	0	0.0004146	0	0.003647	0.003647	0	0.003647	0
$\bar{\nu_e}$ NC $1\pi^{+/-}$	0.008237	0.008237	0	0.008237	0		-		–	
$\bar{\nu_e}$ NC $1\pi^0$		-		–		0.001191	0.001432	20.23	0.0009504	-20.23
$\bar{\nu_e}$ NC coherent		-		–		0.0007456	0.0008976	20.38	0.0005937	-20.38
$\bar{\nu_e}$ NC other	0.0121	0.0121	0	0.0121	0	0.0009675	0.001067	10.27	0.0008681	-10.27
Osc. $\bar{\nu_e}$ CCQE	0.002171	0.002171	0	0.002171	0	0.1808	0.1808	0	0.1808	0
Osc. $\bar{\nu_e}$ CC 1π	0.0008503	0.0008503	0	0.0008503	0	0.04064	0.04064	0	0.04064	0
Osc. $\bar{\nu_e}$ CC coherent	0.0001137	0.0001137	0	0.0001137	0	0.01087	0.01087	0	0.01087	0
Osc. $\bar{\nu_e}$ CC other	4.078e-05	4.078e-05	0	4.078e-05	0	0.001548	0.001548	0	0.001548	0
Osc. ν_e CCQE	0.1948	0.1948	0	0.1948	0	14.79	14.79	0	14.79	0
Osc. ν_e CC 1π	0.07266	0.07266	0	0.07266	0	2.624	2.624	0	2.624	0
Osc. ν_e CC coherent	0.005219	0.005219	0	0.005219	0	0.04383	0.04383	0	0.04383	0
Osc. ν_e CC other	0.001592	0.001592	0	0.001592	0	0.03175	0.03175	0	0.03175	0

Table 128: Effect of SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.

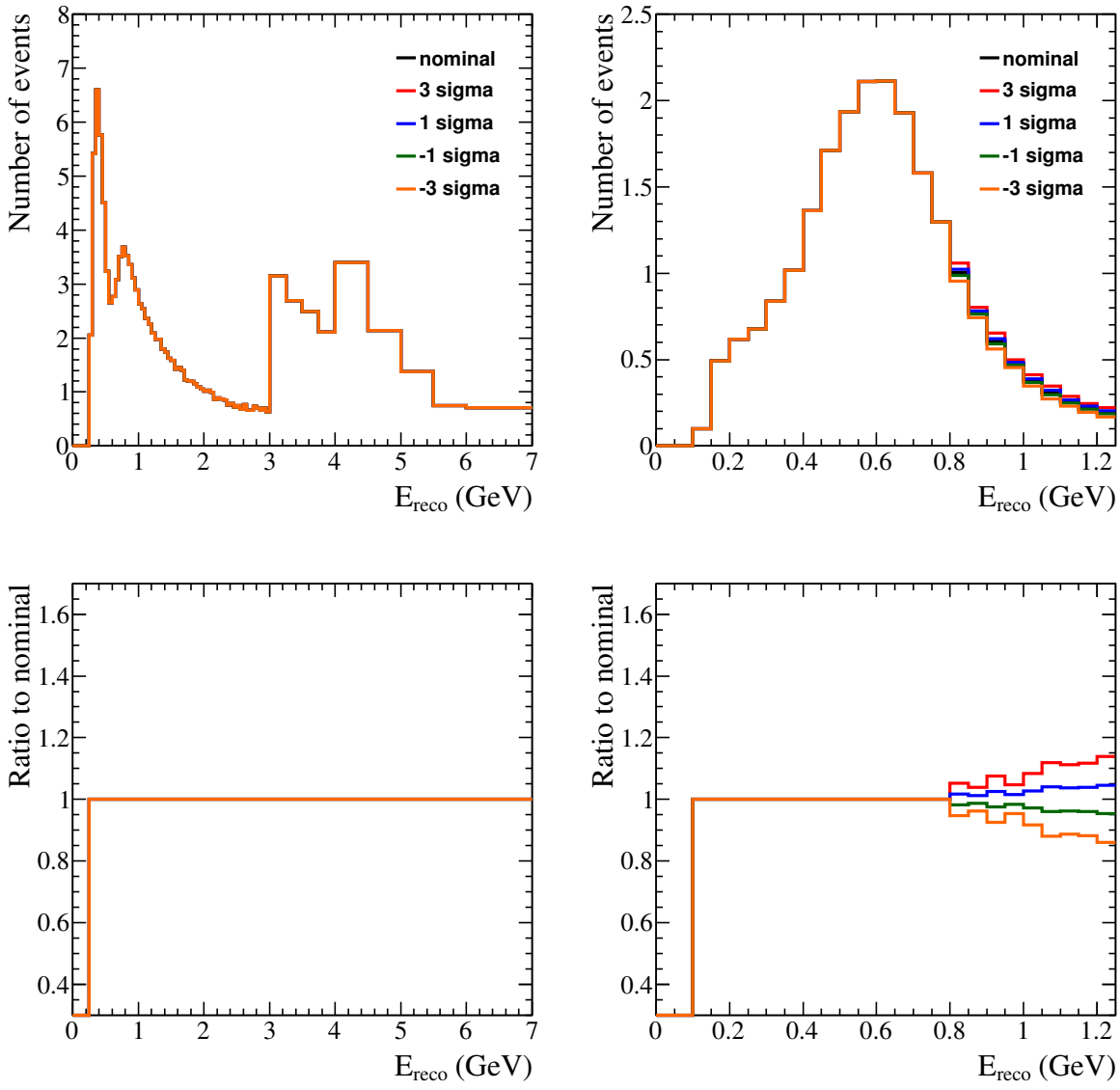


Figure 64: Effect of SKDet + FSI/SI 17; Ereco range 0.80 - 1.25 GeV; all NC ($1R_e$) with Δm^2 = 0.002437 , δm^2 = 7.5e-05 , $sin^2\theta_{12}$ = 0.306 , $sin^2\theta_{13}$ = 0.0251 , $sin^2\theta_{23}$ = 0.513 , δ_{CP} = 0 and normal hierarchy.